

Uncovering Insights of Compound Flooding with Data-Driven AI

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Abstract

Compound flooding, driven by nonlinear interactions between multiple hydrometeorological factors, poses a significant challenge to hazard prevention. Existing forecasting approaches, whether physics-based or data-driven, often emphasize temporal patterns while underexploring how multiple interacting factors jointly shape flood dynamics. To address this problem, we conduct a large-scale data-driven analysis of compound flooding in South Florida, a typical area for compound flooding, by integrating tidal conditions, rainfall, groundwater stage, and human water management activities. Our analysis reveals three key findings: (i) models that capture temporal dynamics alone fail to represent multi-factor interactions during compound events; (ii) subsurface saturation, as reflected by groundwater levels, emerges as a dominant predictor of flood severity, often outweighing immediate rainfall intensity in this porous coastal region; and (iii) the spatial state of surrounding monitoring stations within a finite effective radius provides critical causal context for flooding, while extending temporal history yields diminishing returns during extreme events. These findings suggest that compound flooding is governed more by spatially coupled system states than by long-term temporal dependencies, challenging rain-centric and sequence-dominated forecasting paradigms. By framing data-driven models as tools for scientific inquiry rather than prediction alone, this study offers new insights into the mechanisms of compound flooding and informs

the design of more physically grounded early-warning systems for coastal environments. Our dataset and code are publicly available at <https://github.com/AslanDing/SFBench>.

Keywords

Compound Flood, Machine Learning, Data-Driven Modeling

1 Introduction

Floods are among the most frequent and destructive natural hazards, causing severe environmental damage [77], catastrophic loss of life [30], and economic disruption [8], as shown in Figure 1. Compared with single-driver flood events, such as stream overflows or flash floods [21], compound floods arise from the concurrence of multiple interacting drivers across meteorological, hydrological, and oceanographic processes [59]. According to previous research [6], compound flooding is projected to increase globally by over 25% by 2100 relative to present under a high-emissions scenario. These events are particularly difficult to predict, as nonlinear interactions among drivers can amplify flood severity beyond what any single factor would suggest. Recent studies indicate that both the frequency and spatial extent of compound floods are increasing under climate change [25, 62, 68], underscoring the urgency of understanding their underlying mechanisms.

Despite their importance, the causal structure of compound flooding remains poorly understood. Classical physics-based methods simulate flood dynamics by solving partial differential equations that encode hydrodynamic processes [51, 78], such as the Hydrologic Engineering Center's River Analysis System (HEC-RAS) [9]. Despite their accuracy and explainability, their application is constrained by extensive data requirements, including high-resolution terrain data, canal networks, and river geometries [58, 80]. As a

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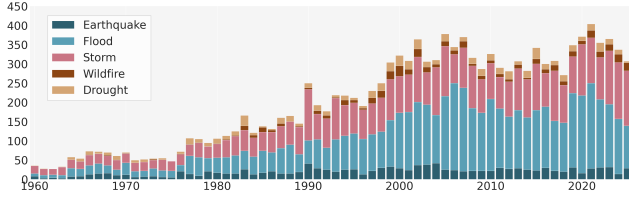


Figure 1: Global historical statistics for 5 types of natural disaster events. Data source: EM-DAT [17]. Flood is the most common disaster.

result, physics-based models are often difficult to deploy consistently across regions, limiting their utility for large-scale comparative analysis of compound flood drivers. In parallel, data-driven approaches based on machine learning have emerged as an alternative for flood forecasting, leveraging advances in deep learning to model complex, nonlinear relationships directly from observations. Prior studies have applied Convolutional Neural Networks (CNN) [37], Recurrent Neural Networks (RNN) [26], Graph Neural Networks (GNN) [32], and Transformers [61] to capture temporal and spatial dependencies in flood processes. However, most existing models emphasize a single driving force and treat the compound flooding forecasting task as a temporal analysis task, overlooking the interaction of various factors. For example, Google researchers use Long Short-Term Memory networks (LSTMs) [26] to forecast extreme floods with watershed streamflow data [47]. In [35], the authors build a model to capture the relationship between rainfall and discharge of catchments. These methods lead to oversimplified scientific insights, compressing the complex interplay among multiple driving factors into a sequential dependency.

A fundamental obstacle to resolving these questions is the lack of integrated observational frameworks that support systematic, multi-factor analysis of compound flooding. Existing datasets [31, 34, 56, 60] are often limited in geographic scope or omit key drivers such as groundwater dynamics and human water management operations. Consequently, previous studies have largely examined compound floods through isolated case studies or restricted regional settings [73], making it difficult to unravel the relative importance of interacting factors or to assess how spatial context influences the causality of floods.

In this study, to bridge the gap between existing methods and real-world scenarios, we leverage data-driven models not merely as forecasting tools but as analytical probes to investigate the mechanisms underlying compound flooding. We select South Florida as a typical area for the analysis, a low-lying coastal region that serves as a natural laboratory for compound flood analysis due to its porous geology, dense water management infrastructure, and frequent exposure to hurricanes and extreme precipitation [27]. Flooding in this region is shaped by the interaction of multiple drivers, including rainfall, tide, groundwater saturation, and human-controlled hydraulic structures, providing a uniquely rich setting for studying multi-factor flood dynamics, as shown in Figure 2. To enable this analysis, we assemble a comprehensive observational dataset spanning 1985 to 2024, integrating water stage, tide, subsurface water(groundwater) stage, rainfall, and operational data from gates and pumps across 2,452 monitoring stations. Rather than treating this dataset as a benchmark alone, we use it to support controlled

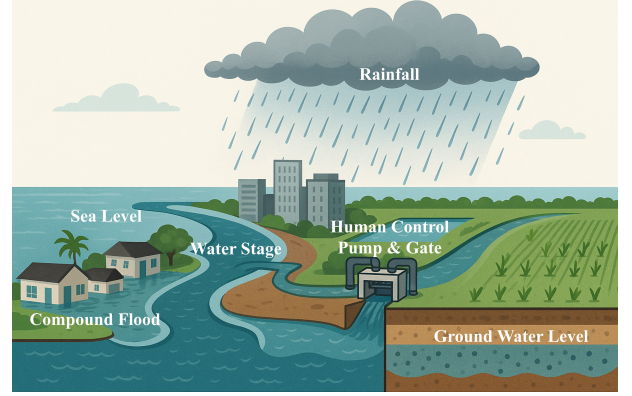


Figure 2: Schematic diagram of multiple factors.

analyses of how different modeling paradigms respond to variations in temporal history, spatial context, and driver availability.

With a diverse set of learning architectures, including Multilayer Perceptrons (MLPs), RNNs, CNNs, GNNs, Transformers, and Large Language Models (LLM)-based approaches, we systematically examine model behavior under feature ablation, spatial perturbation, and extreme-event evaluation. The analysis yields three central findings: (i) models that depend on temporal patterns struggle to represent critical multi-factor interactions during compound flood events; (ii) subsurface saturation consistently emerges as a dominant driver of flood severity in South Florida, frequently exceeding the influence of short-term rainfall; and (iii) spatially localized system states within a finite effective radius provide essential causal context for flooding, while extending temporal history offers limited benefit during extremes.

While demonstrated in South Florida, these results reveal fundamental insights that compound flood dynamics emerge not from isolated temporal sequences but from the spatial entanglement of multi-hazard drivers. This work reframes data-driven models as generalizable tools for scientific discovery, offering a universal blueprint for understanding compound flood mechanisms and developing flood forecasting systems, especially for coastal regions.

2 Related Work

Flood Datasets. Monitoring floods presents a significant challenge due to unpredictable nature and potentially devastating consequences. Existing flood datasets can be broadly categorized into satellite image datasets [7, 18, 46, 52, 55, 74] and time series monitoring datasets [2, 12, 19, 31, 34, 56, 73]. Satellite image datasets utilize remote sensing to capture the extent of surface water [4]. While effective in delineating flood-affected areas, this type of dataset [7, 74] often lacks temporal dynamics and information on the underlying hydrological and meteorological factors driving flood formation. Time series monitoring datasets typically utilize fixed monitoring stations to record hydrological-related data such as soil moisture and temperature. A prominent example within this category is the CAMELS-x family of datasets [1, 3, 12, 16, 19]. For instance, CAMELS-BR [12] encompasses data from 3,679 gauges across Brazil. LamaH-CE [34] provides daily and hourly time series data from 882 gauges, including runoff and catchment attributes. In [33], LamaH-CE is used as a benchmark for flood forecasting, primarily focusing on water stream data across temporal and spatial aspects. In

summary, these datasets primarily focus on general hydrological modeling. Analyzing compound floods, as highlighted in [27], necessitates detailed data on rainfall, water levels, and groundwater, which are often limited in existing time series datasets.

Data-Driven Methods for Forecasting. High computational costs often constrain traditional physical models [9, 51, 78]. In contrast, data-driven AI excels at extracting complex, non-linear patterns directly from high-dimensional data, providing a scalable framework that can efficiently decode the spatial entanglement of multi-hazard drivers across diverse regions. Deep learning methodologies have shown promise in this domain. Based on their core architectural designs, these methods applied to time series forecasting can be categorized as follows: *MLP-based models* [14, 38, 39, 66, 81] leverage the capabilities of multilayer perceptrons for analyzing temporal sequences. *RNN-based methods* [28, 36, 53, 57, 64] are widely adopted in forecasting due to their inherent ability to model temporal dependencies within sequential data. *CNN-based methods* [15, 45, 63, 65, 69] employ convolutional operations to extract hierarchical features from time series data, enabling effective learning of underlying patterns and trends. *GNN-based methods* [10, 11, 42, 71, 72, 76] utilize graph structures to model intricate relationships between different time series variables, enhancing forecasting accuracy. *Transformer-based methods* [40, 41, 48, 67, 70, 82, 83] have demonstrated remarkable performance in capturing long-range dependencies and complex temporal dynamics within time series data. *LLM-based methods* [22, 29, 50, 84] explore the application of prompting and reprogramming techniques to align time series data with text embeddings for forecasting tasks.

3 Methods

To enable systematic analysis of these interactions, we first construct an observational data framework, SF²Bench, that integrates heterogeneous data sources with explicit physical meaning into a coherent spatiotemporal representation. Then, we perform spatiotemporal alignment to construct a coherent representation. After that, we apply data-driven AI analysis to reveal the underlying dependencies among various processes.

3.1 Data Sources and Physical Interpretation

SF²Bench is an observational time series dataset that characterizes the evolving state of a managed coastal hydrological system in South Florida. It aggregates measurements from 2,452 monitoring stations operated by the South Florida Water Management District (SFWMD)¹, covering a spatial extent of 67,349 km² and a temporal range from 1985 to 2024. Because station availability varies over time, the dataset is organized into eight temporal splits that reflect the effective coverage period of individual monitoring locations.

The dataset integrates multiple physical and anthropogenic variables that jointly govern the emergence of compound flooding events [27]. These variables include surface water stage, rainfall, groundwater stage, coastal water levels, and records of human-controlled hydraulic operations. Together, they define the instantaneous state of the regional flood system by capturing external forcings, boundary conditions, internal storage, and active system controls. Rainfall serves as the primary external forcing driving

surface runoff and water accumulation. Groundwater stage represents antecedent subsurface storage and reflects the land’s capacity to absorb incoming precipitation, which critically modulates flood severity during extreme rainfall events. Coastal water levels, including tidal influences, act as dynamic boundary conditions for inland drainage. At several monitoring stations, these coastal effects are inherently embedded in the observed water stage due to direct hydraulic connectivity with the ocean. In addition to natural drivers, SF²Bench explicitly incorporates records of human-controlled hydraulic operations, including pumps and gates. Pumps enable water transfer from lower to higher elevations, while gates regulate downstream flow from higher to lower elevations. In South Florida, where an extensive network of canals, lakes, storage areas, and control structures actively manages water distribution, such operational interventions directly alter system dynamics and flood propagation pathways. Representing these controls is therefore essential for capturing the coupled human–natural processes underlying compound flood behavior.

A comparison between SF²Bench and existing hydrological datasets is provided in Table 1. Compared to CAMELS-x [1, 3, 12, 16, 19], which emphasizes large-sample hydrology across diverse regions, SF²Bench focuses on a flood-prone coastal area where compound flooding is a dominant hazard. In contrast to BangladeshFlood [56], SF²Bench incorporates a broader set of interacting drivers, capturing both temporal evolution and spatial heterogeneity relevant to compound flood processes.

3.2 Spatiotemporal Alignment

To facilitate the construction of AI-ready datasets, we provide detailed information on aligning heterogeneous observational records.

The Observational Network. The raw observations underlying SF²Bench are sourced from DBHYDRO², an environmental data repository maintained by SFWMD. The raw data stream, originally encompassing 3,731 sensors, underwent a rigorous spatiotemporal density filtration protocol. We prioritized stations with long-term data continuity and high relevance to cross-boundary hydraulic interactions, resulting in a finalized network of 2,452 high-fidelity nodes. This multi-modal array comprises water stage ($n = 993$), precipitation ($n = 349$), and groundwater levels ($n = 582$), integrated with the active control signals of 99 pump stations and 429 hydraulic gates. By synthesizing these disparate sensing modalities, the repository provides a comprehensive digital pulse of the region’s hydrological state. Each station provides localized observations of distinct system components. Together, these measurements form a distributed sensing network that captures external forcings, internal storage states, boundary conditions, and human-controlled interventions within the regional flood system. In our framework, water stage, rather than streamflow, is used here because streamflow approaches zero and is hard to measure in many catchments, such as lakes and wetlands. A summary of data types and their physical interpretations is provided in Table 2.

Temporal Alignment. Raw observations in the DBHYDRO are recorded at native breakpoint frequencies, which vary substantially across stations and variables, ranging from sub-hourly to irregular event-based recordings. While this high-resolution logging

¹<https://www.sfwmd.gov/>

²<https://apps.sfwmd.gov/dbhydroInsights/%23/homepage>

Table 1: Comparison with other datasets for flood forecasting. N/A indicates Not Available. * marks datasets with Other Attributions. Climatic indices capture climate statistics (e.g., evapotranspiration). Land cover attributes describe surface materials. Soil attributes include properties such as porosity, while geological attributes refer to subsurface features. Anthropogenic influences cover human activities (e.g., discharges). Other catchment attributes include location, area, and topography.

Dataset	Time Span	Interval	Type	Gauges	Area(km ²)	Public	Other Attributions
DarlingFlood	1900-2018	Daily	Flow	12	3.5×10^4	No	Rainfall
SekongFlood	1981-2013	Daily	Flow	8	2.8×10^4	No	Rainfall
BangladeshFlood	1979-2013	Daily	Stage	24	1.5×10^5	No	N/A
Qi River	1979-2020	Hour	Flow	7	7.1×10^3	No	Rainfall
Tunxi basins	1981-2007	Hour	Flow	12	N/A	No	Rainfall
CAMELS*	1989-2009	Daily	Flow	671	1.0×10^4	Yes	Climatic Indices
CAMELS-CL*	1913-2018	Daily	Flow	516	N/A	Yes	Land Cover Attributes
CAMELS-GB*	1970-2015	Daily	Flow	671	2.1×10^5	Yes	Soil Attributes
CAMELS-BR*	1925-2024	Daily	Flow	4,025	N/A	Yes	Geological Attributes
CAMELS-AUS*	1951-2014	Daily	Flow	107	6.9×10^5	Yes	Anthropogenic Influences
LamaH-CE*	1951-2014	Daily & Hour	Flow	859	1.7×10^5	Yes	Other Catchment Attributes
SF ² Bench	1985-2024	Hour	Stage	2,452	6.7×10^4	Yes	Rainfall, Groundwater, Human Control

preserves fine-scale dynamics, it results in heterogeneous and asynchronous time series that are not directly comparable across space or variables. Moreover, observational gaps and uneven station lifespans introduce additional temporal inconsistency. To construct a unified system-level representation, we align all observations to an hourly time grid. For each station and variable, measurements within each hourly interval are aggregated using their mean value, yielding a temporally consistent snapshot. This hourly resolution balances the need to capture flood-relevant dynamics with the requirement for long-term temporal coverage across stations. Due to heterogeneous data availability, the aligned time series are organized into eight contiguous temporal splits³, each corresponding to a contiguous period with sufficient station coverage.

Handling Missing Observations. Temporal alignment inevitably introduces missing values where observations are unavailable. To ensure the integrity of the physical signals during this transition, we applied differentiated interpolation strategies based on the variables' nature: continuous field and event-based sparse representation. Surface water stage and groundwater level are treated as continuous state variables that evolve smoothly over time, and missing values are filled using linear interpolation. In contrast, rainfall and hydraulic control records (pumps and gates) represent stochastic external events; missing entries are therefore interpreted as the absence of events and filled with zero.

3.3 Data-Driven AI Analysis

We investigate the extent to which data-driven models can capture the coupled dynamics of a human-regulated hydrological system using large-scale observational data. By reframing water stage forecasting from a purely predictive task into a system identification problem, we aim to reveal the underlying spatiotemporal dependencies linking surface water dynamics, subsurface processes, meteorological forcing, and human interventions.

Scientific Objective. We examine whether AI models can learn hydrological dynamics and forecast water stages from multi-factor

Table 2: The summary of the data type in SF²Bench.

Type	Stations	Unit	Description
Water	993	Feet	Water Stage
Groundwater	582	Feet	Stage of Groundwater
Rainfall	349	Inches	Rainfall
Pump	99	RPM	Rotational Speed
Gate	429	Feet	Opening Level

observational data. Specifically, we aim to understand how surface water levels evolve in response to meteorological forcing, subsurface groundwater interactions, and human-operated hydraulic controls, such as pumps and gates.

We formulate the hydrological system as a spatio-temporal coupled network driven by both natural processes and human control signals. Given water stage time series data $X \in \mathbb{R}^{N \times L}$ from N water monitoring stations, the forecasting objective is to predict the water stage values for the next T time steps, denoted as $Y \in \mathbb{R}^{N \times T}$, using a fixed look-back window of length L . In addition to the surface water stage, the system also has access to additional time series information, including groundwater levels $X_w \in \mathbb{R}^{N_w \times L}$ (from N_w stations), rainfall $X_r \in \mathbb{R}^{N_r \times L}$ (from N_r stations), pump control data $X_p \in \mathbb{R}^{N_p \times L}$ (from N_p stations), gate control data $X_g \in \mathbb{R}^{N_g \times L}$ (from N_g stations), and location information for the monitoring stations. From the data-driven model perspective, this formulation treats water stage evolution as the outcome of interacting latent processes operating across space and time. For the primary benchmark experiments aimed at fair comparison across different modeling approaches, we focus on water stage forecasting as the supervised prediction target. The additional modalities are regarded as external drivers that can potentially enrich system understanding and improve predictive performance, and their integration is explored in extended experimental settings.

Data-Driven Models. To cover a broad range of inductive biases, we benchmark six categories of time series forecasting architectures, including MLP, CNN, RNN, GNN, Transformer, and LLM-based models. We select two advanced methods as representative examples for each of these categories. Additionally, for the

³Detailed split information is reported in Appendix Table 5.

first four classical architectures, we also implement a basic foundational architecture. The specific advanced methods we evaluate are: 1. MLP-Based Models: NLinear [81], TSMixer [14], 2. CNN-Based Models: ModernTCN [45], TimesNet [69], 3. RNN-Based Models: DeepAR [57], DilatedRNN [13], 4. GNN-Based Models: FourierGNN [76], StemGNN [11], 5. Transformer-Based Models: PatchTST [48], iTransformer [41], 6. LLM-Based Models: GPT4TS [84], AutoTimes [43]. We follow the source code of NeuralForecast⁴ for the implementation of these approaches⁵.

3.4 Experimental Design

We design a series of controlled experiments to examine whether existing machine learning models can capture the spatiotemporal and multi-factor characteristics of compound flooding systems. Rather than optimizing for a single benchmark score, our experimental design aims to answer three scientific questions: (i) *whether current AI models can effectively learn spatiotemporal dependencies under multi-factor factors*, (ii) *which input factors are most informative for compound flooding prediction*, and (iii) *the relative importance of temporal versus spatial context in forecasting performance*.

Spatial Scope and Computational Constraints. Due to the substantial memory and training costs associated with large-scale models, particularly LLM-based approaches, we adopt a two-level evaluation strategy. First, we conduct comprehensive benchmarking within three hydrologically relevant regions selected based on observed flood occurrences. The three areas are selected based on the flood observation data presented in Figure 3a, which visualizes these flood locations alongside the selected areas. These regions represent areas where compound flooding effects are most pronounced. Second, for the full dataset, we report results for four foundational methods to assess large-scale generalization while maintaining computational feasibility.

Factor and Context Ablations. To isolate the contribution of different factors, we conduct targeted ablation studies. These include removing or selectively incorporating groundwater, rainfall, and human control signals, as well as varying the spatial extent of input stations and the temporal length of historical context. By systematically controlling these factors, we assess how different models leverage temporal accumulation, spatial coupling, and cross-modal information in learning compound flooding dynamics.

Temporal Splits and Forecasting Horizons. To evaluate generalization across hydrological regimes, we apply temporally disjoint data splits, using the last year for testing, the second-to-last year for validation, and all preceding data for training. A fixed look-back window of two days is used for all models, while prediction horizons of one, three, five, and seven days are evaluated to assess short- and medium-term forecasting capability.

3.5 Evaluation Criteria

The evaluation is designed to reflect both overall predictive fidelity and sensitivity to hydrologically critical extreme events. To quantify general predictive accuracy, we follow standard time series forecasting practices to report the Mean Absolute Error (MAE) and

Mean Squared Error (MSE), which measure deviations between predicted and observed water stage values across all time steps.

In addition, to explicitly evaluate performance on extreme flood-related events, we also employ the Symmetric Extremal Dependence Index (SEDI) [23, 75], following prior work in flood and climate forecasting [24]. The formulation of SEDI is as follows:

$$\text{SEDI}(p) = \frac{|\hat{Y} < V_{1-\frac{p}{2}} \& Y < V_{1-\frac{p}{2}}| + |\hat{Y} > V_{\frac{p}{2}} \& Y > V_{\frac{p}{2}}|}{|Y < V_{1-\frac{p}{2}}| + |Y > V_{\frac{p}{2}}|}, \quad (1)$$

where $|\cdot|$ means the number of the true values satisfying the condition, $\hat{Y}$ is the forecasting results, p is the quantile of the threshold, and $V_{1-\frac{p}{2}}, V_{\frac{p}{2}}$ are the lower and upper threshold of top and worst $\frac{p}{2}$ percent, respectively. SEDI focuses on the co-occurrence of extreme predictions and observations, thereby emphasizing the model's responsiveness to tail events. Specifically, extreme conditions are defined using quantile-based thresholds (e.g., the 95th and 5th percentiles of the observed values), and each time step is classified as either normal or extreme. A higher SEDI score indicates stronger dependence between predicted and observed extremes, reflecting improved capability in capturing flood-relevant dynamics.

3.6 Experimental Setup

In this paper, the default look-back window size is two days, and the forecast window length is a list of [1, 3, 5, 7] days. For the SEDI metric, we report the average performance across four forecast settings. Except for Table 3, MAE and MSE results are reported based on the average of four forecast window lengths.

4 Results

4.1 Dataset Analysis & Visualization

To establish system-level intuition and verify the physical consistency of Sf²Bench, we conduct a comprehensive series of visualization-based analyses from both spatial and temporal perspectives. These analyses serve as sanity checks to examine whether the observed patterns are coherent across space and time and consistent with established hydrological processes.

Spatial Distribution. Figure 3a presents the geographic distribution of all monitoring stations, which are predominantly deployed along the complex river network of South Florida and radiate outward from Lake Okeechobee. The staggered spatial layout of hydrological, groundwater, and rainfall stations enables joint spatial reasoning across different components of the hydrological system. For instance, water level variations at a given site are expected to be influenced by nearby precipitation patterns as well as local groundwater conditions, which reflect subsurface water storage capacity. To further contextualize the dataset, we overlay observed flood events recorded in 2008 and from 2020 to 2023 in Figure 3b, based on reports from the South Florida Water Management District (SFWMD) [49]. Notably, these flood events are primarily concentrated in urbanized regions, highlighting the interaction between natural hydrological processes and human-modified landscapes.

Temporal Pattern. Figure 3 illustrates the average annual temporal patterns across all monitoring stations, using data from split S5 and S7 as a representative example. From these patterns, we observe a strong correlation between groundwater level and water

⁴<https://github.com/Nixtla/neuralforecast>

⁵The summary of these methods can be found in Appendix D.

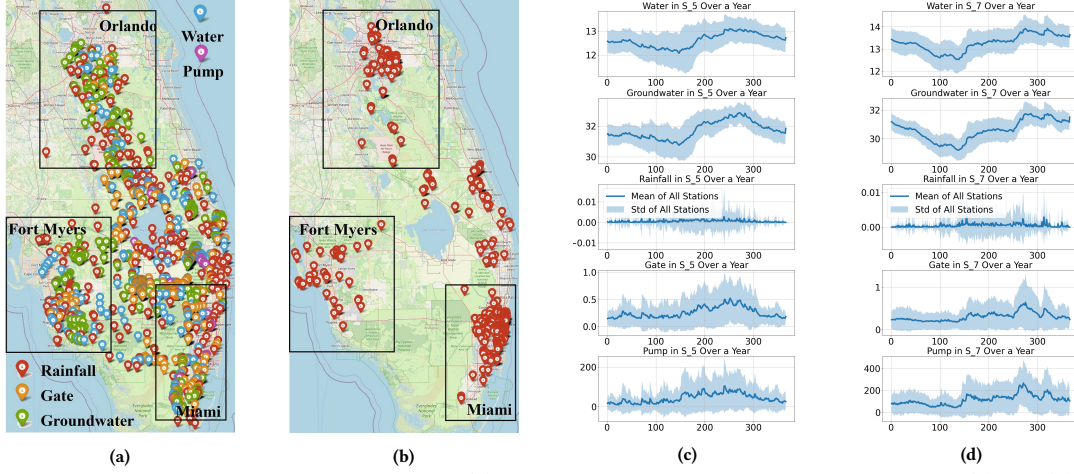


Figure 3: (a). Location distribution of monitor stations. (b). Flood observation location distribution. In (a) and (b), we highlight three interest parts: Orlando, Fort Myers, and Miami. (c) and (d). Temporal patterns of key features over a year. The x-axis is the number of days in one year.

level data. The water stage generally rises from approximately the 150th to the 300th day of the year, followed by a gradual decrease. According to the National Weather Service (NWS), Florida’s climate is characterized by distinct dry and rainy seasons, with the latter typically spanning from May to October. This aligns well with the observed data patterns in our dataset. Taking rainfall as a reference, we note that increases in water level tend to correspond with rainfall events, such as the rainfall peak after the 300th day and the subsequent rise in water level. In addition, human control activities on hydraulic structures appear to influence water level changes in response to rainfall. We can infer that human intervention has played a role in mitigating potential flooding.

4.2 Observations & Discovery

We organize our observations around the core scientific questions concerning the modeling of compound flooding systems.

Q1: Can existing machine learning models effectively capture multi-factor spatiotemporal dynamics?

Across all benchmark experiments, we observe that current machine learning models exhibit uneven capability in capturing the multi-factor and spatiotemporal structure of the hydrological system. Table 3 reports the average MSE and MAE results across three regions and eight data splits. The top-performing methods include PatchTST, iTransformer, MLP, NLinear, and AutoTimes. PatchTST and iTransformer achieve the best MAE, while MLP, NLinear, and AutoTimes perform best under MSE. In general, a small MAE but large MSE suggests high accuracy for most points with occasional large errors, whereas the opposite indicates greater stability but lower pointwise accuracy. Table 3 also shows model performance on extreme cases; in contrast, ModernTCN and NLinear demonstrate competitive performance on extreme-event metrics. We also observe similar results for performance across all stations in Appendix F.1.

From our observations, the MLP achieves near-best performance across almost all experiments, despite its relatively concise model structure. This suggests that existing methods that emphasize temporal dependency may fail to adequately capture the interactions

among multiple driving factors. Furthermore, these results reveal a trade-off in model biases that certain architectures favor overall system averages, whereas others are specifically tuned to the patterns of correlated extremes.

Q2: Do average accuracy metrics reflect performance on extreme events (e.g., floods)?

We find that improvements in MAE and MSE do not fully correspond to improved extreme-event prediction. Models such as ModernTCN achieve strong SEDI performance despite weaker average accuracy, indicating a decoupling between mean behavior and tail dynamics. This highlights the necessity of evaluating compound flooding models using metrics that explicitly target extreme events, as average error measures alone may obscure critical system-level behaviors relevant to flood risk assessment.

Q3: Which factors are most informative for compound flooding prediction?

To quantify the contribution of diverse drivers to compound flooding predictability, we conducted a systematic ablation study across five representative architectures. These models are categorized into two paradigms: channel-independent (NLinear, PatchTST) and channel-dependent (iTransformer, TimesNet, TSMixer). While the former treats all variables as a unified input pool, the latter variants designate water stage as the primary supervised target, treating auxiliary factors as exogenous covariates.

By systematically isolating factors including rainfall (R), anthropogenic control signals (C), and groundwater (G), we reveal a non-uniform contribution of auxiliary inputs that is deeply tied to model inductive biases. As shown in Table 4, a distinct bifurcation emerges that, while NLinear and TSMixer reach a performance plateau using water stage data alone, architectures like iTransformer and PatchTST demonstrate a superior capacity for multimodal synergy, showing substantial error reductions when the input space is enriched. This suggests that transformer-based architectures are more adept at capturing the cross-channel dependencies essential for simulating complex hydrological responses. Beyond architectural differences, the ablation reveals a critical physical insight regarding the hierarchy of environmental drivers. Across the most responsive

Table 3: Benchmark of average MAE, MSE, and SEDI results on three interest areas across 8 splits. The best and second results are shown in bold font and underlined, respectively. The best performances are in **bold, and the second-best method is underlined.**

Methods	MAE↓					MSE↓					SEDI↑		
	1D	3D	5D	7D	Avg.	1D	3D	5D	7D	Avg.	10%	5%	1%
MLP-Based Models													
MLP	0.0788	0.1351	0.1764	0.2063	0.1492	<u>0.0521</u>	0.1029	0.1432	0.1707	0.1172	0.5302	0.3952	0.1721
TSMixer	0.0928	0.1442	0.1816	0.2126	0.1578	0.0648	0.1132	0.1566	0.1903	0.1283	0.5927	0.4639	0.2274
NLinear	0.0817	0.1373	0.1769	0.2082	0.1510	0.0556	<u>0.1105</u>	0.1547	0.1841	<u>0.1262</u>	0.6097	0.4825	0.2513
CNN-Based Models													
TCN	0.1792	0.2281	0.2697	0.3088	0.2465	0.3722	0.4647	0.4173	0.5386	0.4482	0.2020	0.1449	0.0983
ModernTCN	0.0798	0.1441	0.1891	0.2248	0.1594	0.0681	0.2443	0.2808	0.2723	0.2164	<u>0.6279</u>	0.5130	0.2828
TimesNet	0.0983	0.1528	0.1927	0.2267	0.1676	0.0704	0.1358	0.1829	0.2241	0.1533	0.6079	0.4880	0.2509
RNN-Based Models													
LSTM	0.1182	0.1821	0.2232	0.2576	0.1953	0.1339	0.1985	0.2348	0.2873	0.2136	0.4427	0.3062	0.0867
DeepAR	0.1178	0.1837	0.2247	0.2596	0.1964	0.1230	0.1954	0.2389	0.2691	0.2066	0.4271	0.2852	0.0663
DilatedRNN	0.0919	0.1573	0.2022	0.2374	0.1722	0.1033	0.1672	0.2341	0.2591	0.1909	0.5536	0.4179	0.1730
GNN-Based Models													
GCN	0.1696	0.2006	0.2504	0.2799	0.2251	7.2299	1.5645	2.1341	1.0374	2.9915	0.3662	0.2582	0.1210
StemGNN	0.1332	0.2181	0.3153	0.3570	0.2559	0.1632	0.2543	0.4189	0.4716	0.3270	0.4350	0.2993	0.0985
FourierGNN	0.0921	0.1503	0.1930	0.2280	0.1658	0.0768	0.1416	0.2071	0.2125	0.1595	0.4657	0.3647	0.1645
Transformer-Based Models													
PatchTST	0.0741	<u>0.1316</u>	0.1719	<u>0.2044</u>	0.1455	0.0531	0.1132	0.1566	0.1903	0.1283	0.6254	0.5043	0.2711
iTransformer	<u>0.0756</u>	0.1314	<u>0.1722</u>	0.2041	<u>0.1458</u>	0.0253	0.1112	0.1583	0.1911	0.1284	0.6292	<u>0.5077</u>	<u>0.2705</u>
LLM-Based Models													
GPT4TS	0.1256	0.1521	0.1911	0.2247	0.1734	0.0966	0.1410	0.1847	0.2245	0.1617	0.6163	0.4924	0.2566
AutoTimes	0.0846	0.1362	0.1752	0.2062	0.1505	0.0584	0.1125	<u>0.1522</u>	<u>0.1823</u>	0.1263	0.6054	0.4794	0.2476

Table 4: Input factors ablation study on S6. All, G, R, and C represent all factors: groundwater, rainfall, and human control(pump and gate). The best performances are in **bold, and the second-best method is underlined.**

	Models	w/ All	w/o G	w/o R	w/o C
MAE	iTransformer	0.1406	0.1407	0.1406	<u>0.1405</u>
	PatchTST	0.1376	0.1391	0.1378	<u>0.1377</u>
	TSMixer	0.1596	<u>0.1419</u>	0.2523	0.2111
	NLinear	0.1546	0.1577	0.1483	0.1540
	TimesNet	0.1642	0.1599	0.1662	0.1650
MSE	iTransformer	<u>0.0953</u>	0.0960	0.0949	0.0954
	PatchTST	<u>0.0917</u>	0.0932	0.0927	0.0916
	TSMixer	0.1080	0.0946	0.4061	0.2698
	NLinear	0.0992	0.1011	0.0966	0.0984
	TimesNet	0.1188	0.1154	0.1226	0.1202
	Models	w/o GR	w/o RC	w/o GC	w/o GRC
MAE	iTransformer	0.1411	0.1410	0.1402	0.1411
	PatchTST	0.1398	0.1396	0.1385	0.1421
	TSMixer	0.1418	0.2807	0.1423	0.1422
	NLinear	0.1485	<u>0.1450</u>	0.1579	0.1435
	TimesNet	0.1592	0.1656	0.1578	<u>0.1580</u>
MSE	iTransformer	0.0957	0.0956	0.0949	0.0962
	PatchTST	0.0950	0.0938	0.0923	0.0971
	TSMixer	0.0946	0.6697	<u>0.0943</u>	0.0941
	NLinear	0.0973	<u>0.0954</u>	0.1006	0.0947
	TimesNet	0.1145	0.1237	0.1111	<u>0.1123</u>

models, groundwater consistently emerges as the most informative auxiliary factor, far outweighing the predictive contribution of immediate rainfall or control data. The exclusion of groundwater (denoted as “w/o G” or “w/o GR”) leads to a significantly higher degradation in MSE and MAE compared to the removal of other variables, such as “w/o R” or “w/o C”. Notably, for iTransformer and TimesNet, providing groundwater as the sole auxiliary input obtains the highest performance gains among all ablation settings. This suggests that subsurface water storage serves as a dominant latent state that governs surface water dynamics during compound events, effectively acting as a long-term memory component of the hydrological cycle.

Q4: Is temporal context or spatial context more critical for accurate forecasting?

To characterize the boundary of predictability in compound flooding, we investigated the relative importance of spatial coverage and temporal receptivity across different modeling paradigms. By defining an anchor area of radius R as the forecasting target and incrementally expanding the interest area using scale factors from 1.0 to 1.8 (as illustrated in Figure 4a), we quantified the impact of regional spatial information. As shown in Figure 4b, models such as iTransformer, PatchTST, and TSMixer exhibit consistent error reductions as more surrounding monitoring stations are integrated. This performance gain underscores the value of capturing upstream-downstream hydraulic correlations and regional meteorological trends. However, these spatial benefits tend to saturate as the radius increases, suggesting that the marginal utility of geographic data

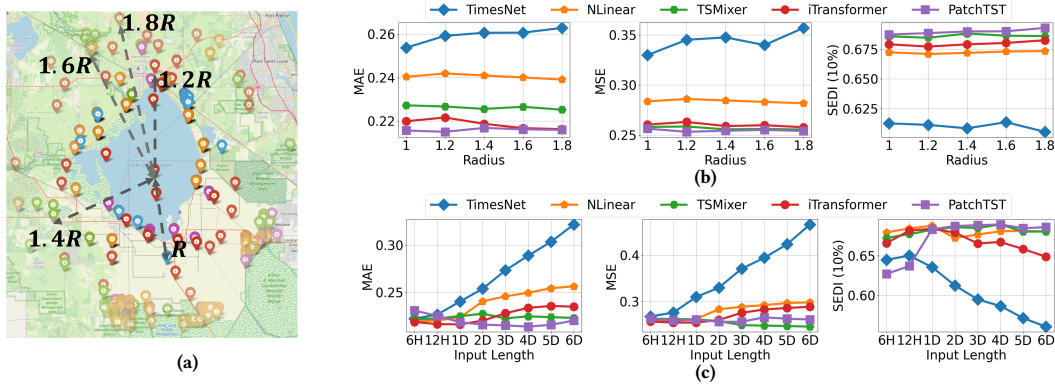


Figure 4: (a).The different interest areas with a radius for the ablation study on spatial information. (b). Study on spatial information. (c). Study on temporal information.

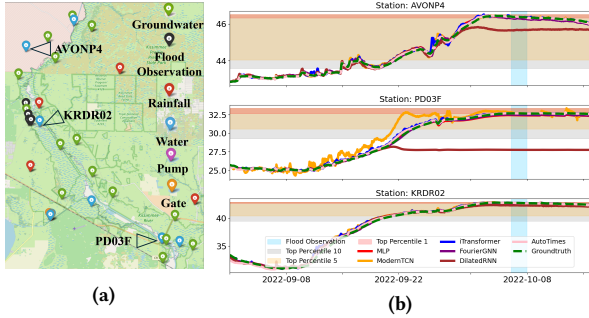


Figure 5: (a) The interest area of the case study. (b) The comparison among different methods. We mark the flood observation on the x-axis and the top percentile 10/5/1 with colors. diminishes once the immediate hydrological catchment or local forcing zone is fully represented.

In parallel, we examined the influence of temporal context by varying the input look-back window from 6 hours to 6 days. The results in Figure 4c reveal that temporal extension generally brings in more substantial performance improvements than spatial expansion, particularly for PatchTST and TSMixer, which achieved significant MSE reductions with longer histories. This dominance of the temporal dimension suggests that the sequential evolution of water stages contains more critical predictive information than instantaneous spatial snapshots, reflecting cumulative rainfall effects and tidal phase transitions. We also observed a performance inflection point for the iTransformer, where accuracy began to degrade beyond a one-day input window. This phenomenon reflects a trade-off between the depth of the temporal context and the data density required to resolve long-range dependencies without overfitting.

In summary, our analysis indicates that while spatial information provides necessary geographical constraints, the temporal context is the primary driver of accuracy, though its benefits are ultimately constrained by the complexity of the underlying physical patterns and the limits of the available training data.

4.3 Compound Flooding Event Study

To complement the statistical evaluation, we provide an event-level case study on Hurricane Ian, which struck Florida in September 2022 and triggered widespread compound flooding. This event provides a natural stress test for AI models, as it involves intense rainfall,

elevated groundwater levels, and rapid surface water response under human regulation. Evaluating model behavior during such an extreme, non-stationary period allows us to assess whether learned dynamics remain physically plausible beyond average conditions.

We focus on the middle reach of the Kissimmee River, where multiple water monitoring stations are located in close proximity to documented flood zones (Figure 5a). This region provides a representative setting for evaluating compound flooding dynamics, with high-quality observations available for key driving factors, including surface water level, groundwater level, rainfall, and human management activities. All models are trained using data before the event and tasked with forecasting water stage trajectories over the subsequent 24-hour period, enabling a direct comparison of model behavior under extreme, non-stationary conditions.

Q5. How do different AI models respond to rapidly evolving compound flood dynamics under extreme conditions? Figure 5b summarizes the 24-hour water level forecasting results at three representative monitoring stations along the Kissimmee River. Across all stations, most methods successfully capture the overall rising trend of water levels driven by sustained rainfall. However, clear differences emerge in their ability to track the timing and magnitude of flood peaks. ModernTCN and DilatedRNN exhibit delayed responses and reduced peak predictions, suggesting limited capacity to adapt to unexpected regime shifts. In contrast, transformer-based models demonstrate improved responsiveness to rapid water level increases, although iTransformer shows noticeable overestimation and vibrant behavior at certain stations, indicating potential sensitivity to noise amplification under highly non-stationary inputs. These qualitative observations are consistent with the quantitative evaluation reported in Table 3, reinforcing that accurate compound flood forecasting requires not only trend-following capability but also robust temporal alignment and stability under extreme hydrological forcing.

5 Conclusion

We investigate the use of AI-based forecasting models for compound floods forecasting. By combining standard accuracy metrics with SEDI, we provide a comprehensive assessment of model performance. Our results reveal that while AI models can capture the overall flooding trend, their reliability in predicting extreme water level responses varies substantially. The case study demonstrates

that model failures often emerge during rapid regime shifts driven by the interaction of multiple factors. These findings reveal the shortcomings of existing data-driven AI methods and explore the driving factors behind compound flooding events. The future work will focus on developing more effective data-driven models capable of integrating multiple driving factors from spatiotemporal information, as well as constructing interpretable frameworks to support the prevention and management of compound floods. In particular, such frameworks should help elucidate how models balance and weigh competing drivers, such as storm surge and peak river discharge, in shaping compound flood dynamics.

GenAI Disclosure

In this paper, we leverage LLMs, including ChatGPT and Gemini, to refine sentence-level writing.

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A Time Split of SF²Bench

We provide the detailed information about the number of monitor stations in each split in Table 5. In each split, the number of water monitor stations is more than other kinds of features, which means the water level feature is the main information, and others are additional parts.

Table 5: The summary of the all splits in SF²Bench

Splits	Time Span	Interval	Water	Groundwater	Rainfall	Pump	Gate
S0	1985-1990	1 Hour	159	40	143	17	82
S1	1990-1995	1 Hour	227	36	139	18	104
S2	1995-2000	1 Hour	332	44	170	26	94
S3	2000-2005	1 Hour	402	178	227	31	107
S4	2005-2010	1 Hour	518	296	254	48	172
S5	2010-2015	1 Hour	585	333	216	65	256
S6	2015-2020	1 Hour	670	317	186	85	300
S7	2020-2024	1 Hour	716	352	194	89	329

We also provide the geographical distribution information of monitor stations in different splits in Figure 10. As time goes by, the number of monitoring stations gradually increases, and in terms of spatial distribution, the locations of monitoring stations remain consistent.

In addition, we visualize the temporal pattern of features over a year in each split. As shown in Figure 6, 7, 8, and 9, the temporal patterns in 8 splits are similar and consistent. From the view of the climate, the dry and rainy seasons are highly consistent but the intensity of rainfall in different splits is different. These patterns that are highly aligned with the actual situation demonstrate the quality of the dataset.

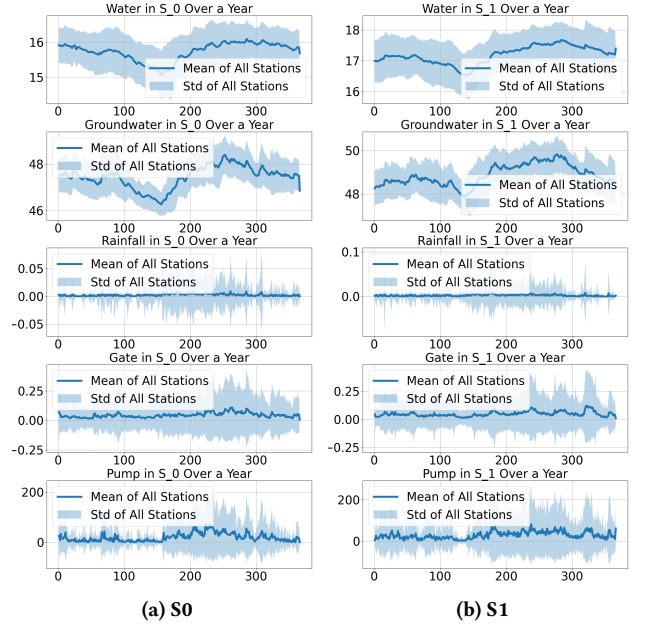


Figure 6: The temporal pattern of features of S0 and S1.

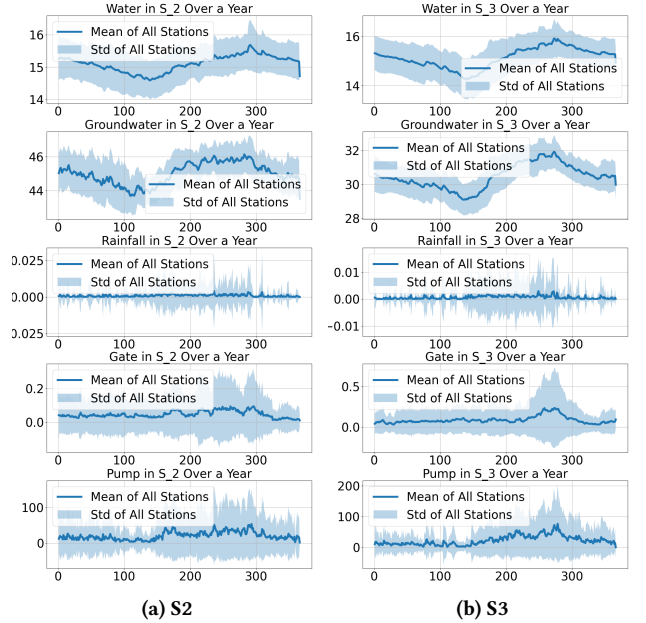


Figure 7: The temporal pattern of features of S2 and S3.

B Experimental Details

C Data Preprocessing

Normalization. To make the data have a zero mean and unit variance, we follow [20, 79, 82] using z-score to normalize the time series data. For the time series, whose variance is less than $1E-4$, we regard its variance as one to avoid the variable overflow (inf or

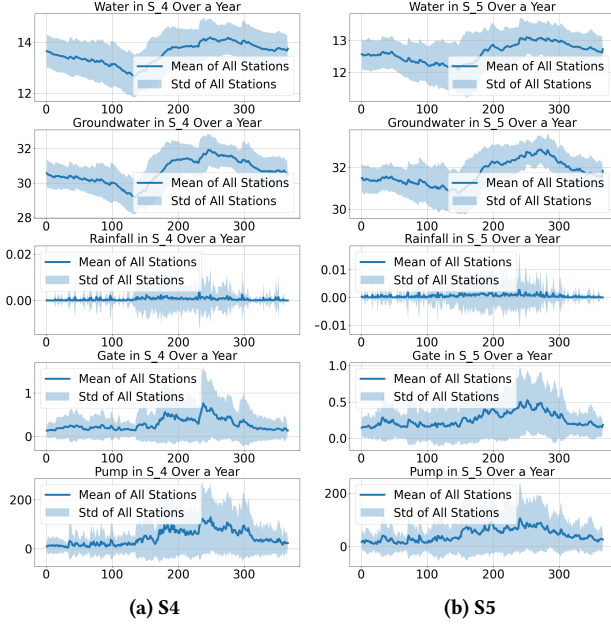


Figure 8: The temporal pattern of features of S4 and S5.

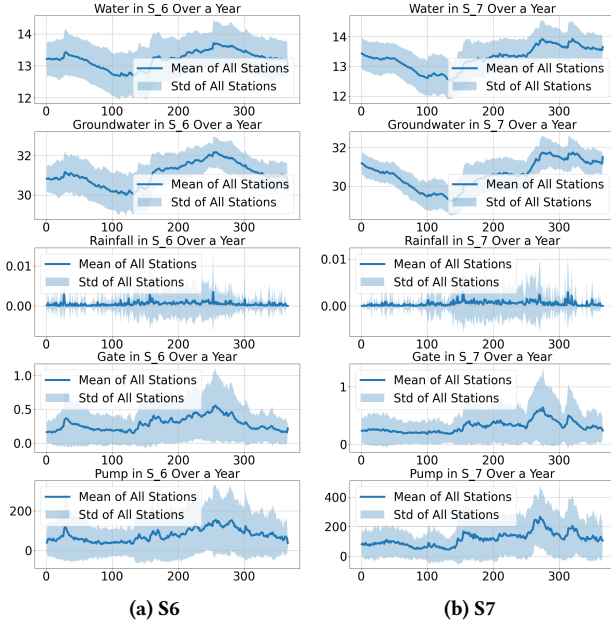


Figure 9: The temporal pattern of features of S6 and S7.

NaN). For forecasting tasks, all the report metrics are based on the normalized data.

Timestamp Features. Because of some methods, such as NLinear, we do not consider extracting the timestamp feature as part of the input. For those methods that are capable of handling the

timestamp feature, we ignore this part. In our code, we also provide timestamp features extracted by following [79, 82] for further works.

D Methods

MLP. We implement a classical three-layer MLP with ReLU as the activation function. The input layer dimension is determined by the input sequence length, and the output layer dimension corresponds to the forecast horizon. The hidden layer dimension is 64. For training, the learning rate is 1×10^{-4} , weight decay is 1×10^{-6} , and the batch size is 64. The model is trained for 15 epochs.

NLinear [81]. This is a simple linear model that treats each time series independently, modeling future values using a linear transformation of the most recent input values. Implementation is based on the NeuralForecast library⁶. The learning rate is 1×10^{-4} , weight decay is 1×10^{-6} , batch size is 64, and training is performed for 50 epochs.

TSMixer [14]. Inspired by MLP-Mixer models from vision tasks, TSMixer is a neural network architecture for time series forecasting. It alternately applies MLPs along the time and feature axes, learning dependencies across both dimensions without requiring attention mechanisms or complex sequence modeling. Implementation follows the NeuralForecast default settings. The architecture includes 2 mixing layers, and the second feed-forward layer has 64 units. The learning rate is 1×10^{-4} , batch size is 32, and the model is trained for 10 epochs.

TCN [5]. It incorporates causal convolutions, ensuring predictions depend only on current and past inputs, thus preserving temporal order. They also utilize dilated convolutions to efficiently capture long-range dependencies by expanding the receptive field without significantly increasing layers. Our implementation follows the official code⁷ using the popular channel-wise setting. It employs a three-layer backbone with a kernel size of 3 and a fixed dilation of 1. The learning rate is 1×10^{-3} , weight decay is 1×10^{-7} , batch size is 256, and training is performed for 50 epochs.

ModernTCN [45]. ModernTCN introduces a streamlined, fully convolutional architecture that aims to simplify design while enhancing performance. It incorporates components like depth-wise separable convolutions and Gated Linear Units (GLUs) to efficiently capture local and long-range temporal dependencies. Implementation follows the long-term forecasting settings from the source code⁸, using the Weather dataset hyperparameters as defaults. The learning rate is 1×10^{-4} , batch size is 256, and the model is trained for 100 epochs.

TimesNet [69]. TimesNet models temporal variations in a two-dimensional space by reshaping time series data into a pseudo-image format and applying 2D convolutional techniques. This enables it to capture both short-term dynamics and long-term dependencies. Implementation uses the Time-Series-Library⁹, with default hyperparameters from the long-term forecasting setting for the Weather dataset. The learning rate is 1×10^{-4} , batch size is 32, and training is performed for 10 epochs.

LSTM [26]. Our Long Short-Term Memory (LSTM) implementation

⁶<https://github.com/Nixtla/neuralforecast>

⁷<https://github.com/locuslab/TCN/tree/master>

⁸<https://github.com/luodh/ModernTCN>

⁹<https://github.com/thuml/Time-Series-Library>

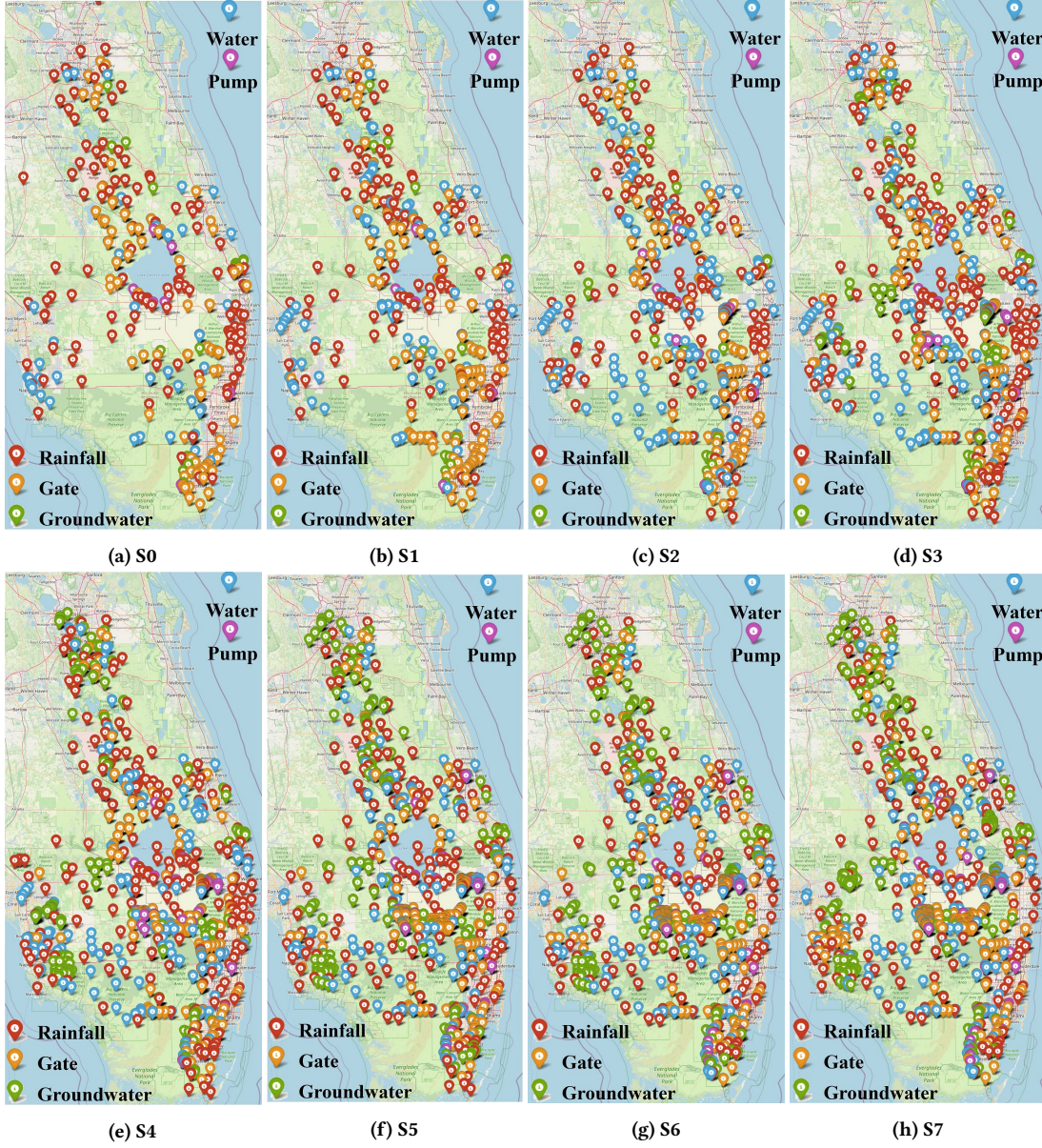


Figure 10: The location distribution in each split.

is a two-layer model with a hidden dimension of 32. The learning rate is 1×10^{-3} , weight decay is 1×10^{-6} , batch size is 64, and the model is trained for 50 epochs.

DeepAR [57]. DeepAR is a global model trained on multiple related time series, which aids generalization, especially for series with limited history. It employs an RNN architecture to predict future values by modeling the conditional distribution of the next value given past observations. Implementation is based on the NeuralForecast library. The learning rate is 1×10^{-3} , batch size is 64, and training is performed for 20 epochs.

DilatedRNN [13]. Dilated RNNs exponentially expand their receptive field by stacking layers with different dilation factors, allowing

efficient capture of short- and long-range patterns without a drastic increase in parameters. This makes them suitable for forecasting tasks with wide-ranging temporal dependencies. Implementation is based on the NeuralForecast library. The learning rate is 1×10^{-3} , batch size is 64, and training is performed for 40 epochs.

GCN [32]. The GCN architecture consists of two layers with a hidden dimension of 32. The graph topology is derived from location information using Delaunay triangulation¹⁰. The learning rate is 1×10^{-4} , weight decay is 1×10^{-5} , batch size is 32, and the model is trained for 50 epochs.

FourierGNN [76]. FourierGNN leverages graph neural networks

¹⁰<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.Delaunay.html>

and Fourier transforms to capture temporal and inter-variable dependencies. Time series variables are treated as graph nodes, with edges representing their relationships. Fourier transforms project data into the frequency domain to model periodic and long-range dependencies. Implementation follows the source code¹¹. The learning rate is 1×10^{-5} , batch size is 32, and training is performed for 100 epochs.

StemGNN [11]. StemGNN is designed to capture both temporal (via temporal convolutions) and spatial (via spectral graph convolutions) dependencies in time-series data, learning smooth representations over the graph structure and dynamic patterns. Implementation follows the source code¹². The learning rate is 1×10^{-4} , batch size is 32, and the model is trained for 50 epochs.

iTransformer [41]. The iTransformer uses an encoder-decoder structure where the encoder processes the sequence in reverse order. This allows the decoder to predict future values based on this processed representation, enhancing focus on relevant temporal sequences while mitigating the computational cost of traditional transformers. Implementation is based on the NeuralForecast library. The learning rate is 1×10^{-4} , batch size is 32, and training is performed for 10 epochs.

PatchTST [48]. PatchTST is a Transformer-based architecture employing patching and channel independence. Time series are divided into patches, which are transformed into tokens and processed by a transformer model to capture local and global dependencies via self-attention. This is particularly useful for long-term forecasting. Implementation is based on the NeuralForecast library. The learning rate is 1×10^{-4} , batch size is 128, and training is performed for 100 epochs.

GPT4TS [84]. GPT4TS treats time series as a language, leveraging pretrained language models (LLMs) to learn temporal patterns. Time series data is tokenized for LLM processing, enabling zero-shot or few-shot generalization. Implementation follows the source code¹³, using long-term forecasting settings for the Weather dataset as default hyperparameters. The default language model is GPT-2 [54]. The learning rate is 1×10^{-4} , batch size is 64, and training is performed for 10 epochs.

AutoTimes [43]. AutoTimes projects time series segments into the embedding space of language tokens, leveraging the autoregressive capabilities of LLMs for forecasting. By training the model to predict subsequent time series segments given preceding ones, AutoTimes generates multi-step forecasts. Implementation follows the source code¹⁴, with GPT-2 [54] as the language model. The learning rate is 5×10^{-4} , batch size is 64, and training is performed for 10 epochs.

For all benchmark experiments, AdamW [44] is used as the optimizer, and the loss function is Mean Squared Error (MSE).

E Platform

All experiments are conducted on two Linux machines, one with 8 NVIDIA A100 GPUs, each with 40GB of memory, and another with 4 RTX 4090 GPUs. We used Python 3.12.9 and Pytorch 2.6.0 to construct our project.

¹¹<https://github.com/aikunyi/FourierGNN>

¹²<https://github.com/microsoft/StemGNN>

¹³<https://github.com/DAMO-DI-ML/NeurIPS2023-One-Fits-All>

¹⁴<https://github.com/thuml/AutoTimes/tree/main>

Table 6: Average results of basic methods on the whole dataset. The best and second results are shown in bold font and underlined, respectively. The best performances are in bold, and the second-best method is underlined.

Metric	MLP	LSTM	TCN	GCN
↓MAE	0.2328	0.3302	0.3491	<u>0.3053</u>
↓MSE	0.3902	2.2772	<u>0.6807</u>	1.0673
↑SEDI(10%)	0.4670	0.3367	0.1689	<u>0.4136</u>
↑SEDI(5%)	0.3549	0.2121	0.1221	<u>0.3118</u>
↑SEDI(1%)	0.1718	0.0545	0.0571	<u>0.1477</u>

F Detailed Results

F.1 Results on All Stations

For the entire dataset, we only report the results of the basic foundational techniques. We provide the results in Table 6. As the results show, MLP achieves the best performance, and GCN is the second-best, Consistent with previous results in Section 4.2.

F.2 Detailed Benchmark Results

In this section, we report the detailed results, including MSE, MAE, and three SEDI values on different prediction lengths, on three interest areas (Orlando, Miami, and Fort Myers) from Table 7 to 36. The best average results are presented in bold font, while the second-best are underlined. Furthermore, from Table 37 to 41, we report the detailed results of basic methods on the whole dataset. To provide qualitative analysis, from Figure 11 to 16, we demonstrate the case visualization of each method on the S7 split in the Orlando area.

F.3 Detailed Other Results

To provide more information about the ablation study, we report the detailed results of the factor ablation study in Table 42 to 44. We report the average results on three interest areas on the S6 split. For the spatial and temporal information ablation study, the results are available from Table 45 to 48.

G Limitations

While SF²Bench provides five key factors, it currently lacks explicit topological linkage information between monitoring stations due to the intricate nature of South Florida’s water system. Although we provide some flood observation data, the locations of these observations may not directly correspond to our monitoring stations. Therefore, this information is provided in a separate file rather than being integrated into the time series data.

Table 7: Benchmark(MAE) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.0915	0.1229	0.0679	0.0810	0.0716	0.0691	0.0598	0.0846	0.0810
	3D	0.1424	0.1928	0.1167	0.1437	0.1174	0.1154	0.1086	0.1314	0.1335
	5D	0.1746	0.2472	0.1607	0.1844	0.1503	0.1529	0.1418	0.1718	0.1730
	7D	0.1919	0.2899	0.1848	0.2244	0.1750	0.1793	0.1753	0.1970	0.2022
	Avg.	0.1501	0.2132	0.1325	0.1584	0.1286	0.1292	0.1214	0.1462	0.1474
MLP	1D	0.1046	0.1355	0.0812	0.0846	0.0767	0.0778	0.1184	0.1015	0.0975
	3D	0.1494	0.2140	0.1252	0.1378	0.1185	0.1219	0.1500	0.1458	0.1453
	5D	0.1769	0.2673	0.1595	0.1783	0.1496	0.1547	0.1751	0.1798	0.1802
	7D	0.1938	0.3093	0.1892	0.2120	0.1765	0.1819	0.2098	0.2096	0.2103
	Avg.	0.1562	0.2315	0.1388	0.1532	0.1303	0.1341	0.1633	0.1592	0.1583
MLP	1D	0.0980	0.1279	0.0777	0.0820	0.0771	0.0741	0.0640	0.0929	0.0867
	3D	0.1475	0.2109	0.1234	0.1390	0.1207	0.1211	0.1079	0.1415	0.1390
	5D	0.1835	0.2663	0.1584	0.1814	0.1527	0.1541	0.1422	0.1763	0.1769
	7D	0.1964	0.3085	0.1891	0.2161	0.1803	0.1814	0.1711	0.2065	0.2062
	Avg.	0.1563	0.2284	0.1371	0.1546	0.1327	0.1327	0.1213	0.1543	0.1522
TCN	1D	0.1887	0.1417	0.2145	0.3830	0.1322	0.0889	0.1164	0.1913	0.1821
	3D	0.1588	0.2302	0.1857	0.4058	0.1968	0.1359	0.1471	0.2127	0.2091
	5D	0.2110	0.3028	0.2139	0.4483	0.2128	0.1928	0.2024	0.2579	0.2552
	7D	0.2135	0.3379	0.2605	0.5715	0.2686	0.2052	0.2102	0.2695	0.2921
	Avg.	0.1930	0.2532	0.2186	0.4522	0.2026	0.1557	0.1690	0.2329	0.2346
CNN	1D	0.0893	0.1238	0.0721	0.0760	0.0704	0.0661	0.0588	0.0802	0.0796
	3D	0.1486	0.2113	0.1366	0.1450	0.1212	0.1162	0.1073	0.1324	0.1398
	5D	0.1886	0.2685	0.1798	0.1887	0.1632	0.1537	0.1426	0.1697	0.1818
	7D	0.2140	0.3228	0.2240	0.2310	0.1936	0.1833	0.1728	0.2002	0.2177
	Avg.	0.1601	0.2316	0.1531	0.1602	0.1371	0.1298	0.1204	0.1456	0.1547
CNN	1D	0.1068	0.1436	0.0805	0.0909	0.0919	0.0925	0.0787	0.1051	0.0987
	3D	0.1558	0.2190	0.1252	0.1515	0.1382	0.1402	0.1254	0.1582	0.1517
	5D	0.1813	0.2745	0.1637	0.1933	0.1735	0.1766	0.1578	0.1958	0.1896
	7D	0.2034	0.3222	0.2025	0.2309	0.2102	0.2146	0.1886	0.2263	0.2248
	Avg.	0.1618	0.2398	0.1430	0.1666	0.1534	0.1560	0.1376	0.1714	0.1662
Transformer	1D	0.0865	0.1227	0.0645	0.0730	0.0710	0.0700	0.0629	0.0889	0.0799
	3D	0.1395	0.1963	0.1103	0.1313	0.1193	0.1195	0.1058	0.1376	0.1325
	5D	0.1717	0.2546	0.1471	0.1723	0.1534	0.1528	0.1401	0.1775	0.1712
	7D	0.1889	0.2962	0.1801	0.2068	0.1833	0.1830	0.1708	0.2048	0.2017
	Avg.	0.1466	0.2175	0.1255	0.1459	0.1317	0.1313	0.1199	0.1522	0.1463
Transformer	1D	0.0843	0.1148	0.0638	0.0678	0.0711	0.0693	0.0617	0.0858	0.0773
	3D	0.1365	0.1922	0.1113	0.1269	0.1149	0.1199	0.1046	0.1362	0.1303
	5D	0.1677	0.2451	0.1471	0.1728	0.1489	0.1569	0.1409	0.1718	0.1689
	7D	0.1866	0.2927	0.1787	0.2054	0.1786	0.1859	0.1679	0.2019	0.1997
	Avg.	0.1438	0.2112	0.1252	0.1432	0.1284	0.1330	0.1188	0.1489	0.1441

Table 8: Benchmark(MAE) on part 1(Orlando area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.1275	0.1790	0.0935	0.1714	0.1044	0.1032	0.0865	0.1211	0.1233
	3D	0.1652	0.2751	0.1431	0.2458	0.1518	0.1528	0.1481	0.1647	0.1808
	5D	0.1972	0.3297	0.1801	0.2977	0.1871	0.1891	0.1784	0.1985	0.2197
	7D	0.2139	0.3517	0.2277	0.3351	0.2198	0.2143	0.1978	0.2266	0.2484
	Avg.	0.1759	0.2839	0.1611	0.2625	0.1658	0.1648	0.1527	0.1777	0.1931
RNN DeepAR	1D	0.1209	0.1855	0.0946	0.1726	0.1024	0.0971	0.0881	0.1171	0.1223
	3D	0.1667	0.2677	0.1488	0.2465	0.1534	0.1620	0.1421	0.1736	0.1826
	5D	0.1942	0.3110	0.1907	0.3237	0.1889	0.1828	0.1770	0.2045	0.2216
	7D	0.2146	0.3522	0.2210	0.3619	0.2122	0.2197	0.2123	0.2317	0.2532
	Avg.	0.1741	0.2791	0.1638	0.2762	0.1642	0.1654	0.1548	0.1817	0.1949
DilatedRNN	1D	0.0939	0.1317	0.0700	0.1552	0.0836	0.0748	0.0672	0.1006	0.0971
	3D	0.1499	0.2318	0.1249	0.2031	0.1339	0.1318	0.1224	0.1457	0.1554
	5D	0.1794	0.2793	0.1627	0.2897	0.1678	0.1635	0.1520	0.1809	0.1969
	7D	0.2110	0.3329	0.1959	0.3061	0.1990	0.1945	0.1931	0.1478	0.2225
	Avg.	0.1586	0.2439	0.1384	0.2385	0.1461	0.1411	0.1337	0.1437	0.1680
GCN	1D	0.1003	0.2139	0.1034	0.3710	0.1030	0.1005	0.0775	0.0962	0.1457
	3D	0.1543	0.3137	0.1411	0.3384	0.1524	0.1517	0.1208	0.1392	0.1889
	5D	0.1862	0.3741	0.1665	0.4935	0.1893	0.1818	0.1610	0.1732	0.2407
	7D	0.2034	0.4156	0.1912	0.5967	0.2153	0.2073	0.1927	0.2024	0.2781
	Avg.	0.1610	0.3293	0.1505	0.4499	0.1650	0.1603	0.1380	0.1528	0.2134
GNN FourierGNN	1D	0.0985	0.1324	0.0796	0.1337	0.0863	0.0929	0.0672	0.0927	0.0979
	3D	0.1445	0.2034	0.1178	0.2733	0.1295	0.1246	0.1125	0.1392	0.1556
	5D	0.1746	0.2644	0.1524	0.3672	0.1712	0.1640	0.1513	0.1719	0.2021
	7D	0.1918	0.3028	0.1936	0.4107	0.1992	0.2019	0.1871	0.2068	0.2367
	Avg.	0.1523	0.2257	0.1358	0.2962	0.1466	0.1459	0.1295	0.1527	0.1731
StemGNN	1D	0.1270	0.1782	0.0786	0.6896	0.1005	0.0833	0.0741	0.0969	0.1785
	3D	0.1717	0.3602	0.1536	0.6742	0.1778	0.1720	0.1498	0.1715	0.2539
	5D	0.2081	0.4895	0.2141	1.4591	0.2144	0.2729	0.2258	0.2412	0.4156
	7D	0.2477	0.5122	0.2577	1.4988	0.2552	0.3307	0.2486	0.2419	0.4491
	Avg.	0.1886	0.3850	0.1760	1.0804	0.1870	0.2147	0.1746	0.1879	0.3243
LLM GPT4TS	1D	0.1050	0.1372	0.0833	0.0934	0.0905	0.0898	0.0778	0.1106	0.0984
	3D	0.1530	0.2169	0.1221	0.1502	0.1335	0.1380	0.1222	0.1547	0.1488
	5D	0.1817	0.2650	0.1565	0.1928	0.1666	0.1723	0.1559	0.1948	0.1857
	7D	0.1967	0.3105	0.1930	0.2294	0.1969	0.2054	0.1962	0.2224	0.2188
	Avg.	0.1591	0.2324	0.1387	0.1664	0.1469	0.1514	0.1380	0.1706	0.1629
AutoTimes	1D	0.0967	0.1369	0.0747	0.0770	0.0768	0.0784	0.0660	0.0963	0.0878
	3D	0.1453	0.2057	0.1196	0.1296	0.1164	0.1203	0.1099	0.1409	0.1360
	5D	0.1733	0.2664	0.1499	0.1751	0.1498	0.1534	0.1411	0.1739	0.1729
	7D	0.1902	0.3029	0.1814	0.2113	0.1780	0.1823	0.1691	0.2032	0.2023
	Avg.	0.1514	0.2280	0.1314	0.1482	0.1302	0.1336	0.1215	0.1536	0.1497

Table 9: Benchmark(MSE) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.1472	0.1254	0.0376	0.0610	0.0379	0.0275	0.0248	0.0523	0.0642
	3D	0.2661	0.2014	0.0621	0.1232	0.0681	0.0536	0.0465	0.0913	0.1140
	5D	0.3569	0.2609	0.0875	0.1714	0.0999	0.0772	0.0670	0.1226	0.1554
	7D	0.2963	0.3124	0.1097	0.2215	0.1199	0.0975	0.0871	0.1498	0.1743
	Avg.	0.2666	0.2250	0.0742	0.1443	0.0814	0.0640	0.0564	0.1040	0.1270
MLP	1D	0.1649	0.1283	0.0423	0.0696	0.0380	0.0312	0.0661	0.0633	0.0755
	3D	0.2915	0.2214	0.0679	0.1255	0.0694	0.0575	0.0862	0.1000	0.1274
	5D	0.1769	0.2903	0.0936	0.1673	0.0981	0.0812	0.1010	0.1325	0.1626
	7D	0.2817	0.3469	0.1190	0.2048	0.1248	0.1031	0.1384	0.1644	0.1854
	Avg.	0.2688	0.2467	0.0807	0.1418	<u>0.0826</u>	0.0682	0.0979	0.1150	0.1377
MLP	1D	0.1468	0.1294	0.0449	0.0678	0.0405	0.0308	0.0291	0.0580	0.0684
	3D	0.2830	0.2223	0.0699	0.1257	0.0720	0.0575	0.0510	0.0974	0.1223
	5D	0.4103	0.2915	0.0950	0.1689	0.1013	0.0812	0.0722	0.1300	0.1688
	7D	0.3001	0.3481	0.1204	0.2080	0.1291	0.1029	0.0924	0.1617	0.1828
	Avg.	0.2851	0.2478	0.0825	<u>0.1426</u>	0.0857	<u>0.0681</u>	0.0612	0.1118	0.1356
TCN	1D	0.3257	0.1416	0.1462	0.8044	0.1152	0.0344	0.0567	0.1260	0.2188
	3D	0.2475	0.2295	0.1342	0.9118	0.1694	0.0597	0.0731	0.1423	0.2459
	5D	0.2906	0.3102	0.1367	0.6579	0.1763	0.1011	0.1119	0.1884	0.2466
	7D	0.3160	0.3587	0.1805	1.3613	0.2556	0.1105	0.1087	0.2119	0.3629
	Avg.	0.2950	0.2600	0.1494	0.9338	0.1791	0.0764	0.0876	0.1671	0.2686
CNN	1D	0.1391	0.1432	0.0407	0.0778	0.0392	0.0279	0.0255	0.0520	0.0682
	3D	0.2746	0.2966	0.1185	0.1814	0.0799	0.0575	0.0505	0.0921	0.1439
	5D	0.3575	0.3305	0.1278	0.2059	0.1303	0.0850	0.0723	0.1257	0.1794
	7D	0.3067	0.4557	0.1821	0.2844	0.1657	0.1096	0.0950	0.1579	0.2196
	Avg.	0.2695	0.3065	0.1173	0.1874	0.1038	0.0700	0.0608	0.1069	0.1528
CNN	1D	0.1770	0.1473	0.0429	0.0742	0.0546	0.0449	0.0394	0.0761	0.0821
	3D	0.3474	0.2545	0.0758	0.1626	0.1049	0.0829	0.0737	0.1365	0.1548
	5D	0.3436	0.3207	0.1069	0.2093	0.1475	0.1193	0.0939	0.1907	0.1915
	7D	0.3053	0.4134	0.1426	0.2522	0.2020	0.1648	0.1283	0.2267	0.2294
	Avg.	0.2933	0.2840	0.0921	0.1746	0.1273	0.1030	0.0838	0.1575	0.1644
Transformer	1D	0.1277	0.1337	0.0357	0.0674	0.0382	0.0314	0.0295	0.0619	0.0657
	3D	0.2760	0.2269	0.0688	0.1343	0.0755	0.0627	0.0529	0.1057	0.1254
	5D	0.3581	0.3195	0.1004	0.1846	0.1081	0.0855	0.0769	0.1483	0.1727
	7D	0.2999	0.3668	0.1288	0.2222	0.1384	0.1129	0.1003	0.1810	0.1938
	Avg.	0.2654	0.2617	0.0834	0.1521	0.0901	0.0731	0.0649	0.1242	0.1394
Transformer	1D	0.1315	0.1368	0.0370	0.0608	0.0385	0.0325	0.0321	0.0593	0.0661
	3D	0.3008	0.2298	0.0845	0.1276	0.0713	0.0645	0.0535	0.1041	0.1295
	5D	0.3504	0.2951	0.1222	0.1736	0.1042	0.0946	0.0777	0.1368	0.1693
	7D	0.3145	0.3605	0.1365	0.2170	0.1412	0.1198	0.0968	0.1708	0.1946
	Avg.	0.2743	0.2555	0.0951	0.1448	0.0888	0.0778	0.0650	0.1177	0.1399

Table 10: Benchmark(MSE) on part 1(Orlando area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.2408	0.2451	0.0505	0.2788	0.0655	0.0496	0.0435	0.0846	0.1323
	3D	0.2765	0.3219	0.0784	0.3653	0.1097	0.0768	0.0837	0.1162	0.1786
	5D	0.3053	0.3853	0.1085	0.4750	0.1569	0.1011	0.0953	0.1525	0.2225
	7D	0.3058	0.4057	0.1427	0.5133	0.1855	0.1226	0.1103	0.1818	0.2459
	Avg.	0.2821	0.3395	0.0950	0.4081	0.1294	0.0875	0.0832	0.1338	0.1948
RNN DeepAR	1D	0.2406	0.2367	0.0480	0.2190	0.0615	0.0435	0.0386	0.0782	0.1208
	3D	0.2720	0.3088	0.0793	0.4084	0.1117	0.0783	0.0652	0.1212	0.1806
	5D	0.3033	0.3441	0.1072	0.5899	0.1537	0.1003	0.0883	0.1564	0.2304
	7D	0.3046	0.4056	0.1356	0.5250	0.1639	0.1221	0.1070	0.1800	0.2430
	Avg.	0.2801	0.3238	0.0925	0.4356	0.1227	0.0860	0.0748	0.1340	0.1937
DilatedRNN	1D	0.1818	0.1370	0.0372	0.3297	0.0496	0.0356	0.0304	0.0860	0.1109
	3D	0.2517	0.2674	0.0689	0.3124	0.0887	0.0645	0.0722	0.1150	0.1551
	5D	0.2822	0.3308	0.1004	0.6438	0.1338	0.0915	0.0916	0.1423	0.2270
	7D	0.3125	0.3989	0.1242	0.5068	0.1702	0.1159	0.1257	0.0714	0.2282
	Avg.	0.2571	0.2835	0.0827	0.4482	0.1106	0.0769	0.0800	0.1037	0.1803
GCN	1D	0.1465	0.5157	0.0443	1.1053	0.0519	0.0353	0.0275	0.0518	0.2473
	3D	0.2526	1.6031	0.0656	0.7137	0.0944	0.0654	0.0494	0.0891	0.3667
	5D	0.3055	1.9044	0.0867	1.1121	0.1266	0.0895	0.0730	0.1184	0.4770
	7D	0.2687	2.2856	0.1071	1.4113	0.1651	0.1087	0.0928	0.1471	0.5733
	Avg.	0.2433	1.5772	0.0759	1.0856	0.1095	0.0747	0.0607	0.1016	0.4161
GNN FourierGNN	1D	0.1619	0.1324	0.0412	0.0901	0.0442	0.0326	0.0275	0.0575	0.0734
	3D	0.2714	0.2058	0.0649	0.2944	0.0746	0.0560	0.0489	0.0941	0.1388
	5D	0.3066	0.2721	0.0891	0.4409	0.1106	0.0806	0.0705	0.1238	0.1868
	7D	0.2766	0.3139	0.1154	0.4880	0.1381	0.1065	0.0919	0.1528	0.2104
	Avg.	0.2541	0.2310	0.0777	0.3283	0.0919	0.0689	0.0597	0.1071	0.1523
StemGNN	1D	0.2290	0.1703	0.0403	1.1688	0.0652	0.0410	0.0325	0.0723	0.2274
	3D	0.2703	0.4534	0.0903	1.2569	0.1323	0.0896	0.0743	0.1263	0.3117
	5D	0.3017	0.7711	0.1305	3.3143	0.1663	0.1663	0.1417	0.2080	0.6500
	7D	0.3225	0.7901	0.1768	3.4285	0.1979	0.2323	0.1489	0.1864	0.6854
	Avg.	0.2809	0.5462	0.1095	2.2921	0.1404	0.1323	0.0994	0.1483	0.4686
LLM GPT4TS	1D	0.1984	0.1399	0.0456	0.0882	0.0542	0.0431	0.0387	0.0935	0.0877
	3D	0.3883	0.2385	0.0811	0.1648	0.0986	0.0788	0.0690	0.1343	0.1567
	5D	0.3668	0.3102	0.0997	0.2037	0.1304	0.1091	0.0870	0.1876	0.1868
	7D	0.2928	0.3700	0.1378	0.2503	0.1622	0.1494	0.1319	0.2233	0.2147
	Avg.	0.3116	0.2646	0.0911	0.1768	0.1114	0.0951	0.0816	0.1597	0.1615
AutoTimes	1D	0.1531	0.1507	0.0401	0.0671	0.0390	0.0310	0.0287	0.0623	0.0715
	3D	0.3072	0.2333	0.0683	0.1278	0.0686	0.0572	0.0508	0.0978	0.1264
	5D	0.3527	0.2975	0.0916	0.1700	0.0992	0.0811	0.0715	0.1319	0.1619
	7D	0.2913	0.3428	0.1181	0.2061	0.1265	0.1031	0.0916	0.1612	0.1801
	Avg.	0.2761	0.2561	0.0795	0.1427	0.0833	0.0681	0.0606	0.1133	0.1350

Table 11: Benchmark(SEDI(10%)) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.7864	0.7812	0.8759	0.8936	0.5616	0.7207	0.8520	0.6714	0.7678
	3D	0.6292	0.5742	0.7217	0.8116	0.4534	0.5071	0.8848	0.4950	0.6346
	5D	0.3454	0.5374	0.5817	0.5394	0.2466	0.6411	0.7525	0.4886	0.5166
	7D	0.4239	0.4298	0.5850	0.4289	0.3422	0.4863	0.6125	0.3879	0.4621
	Avg.	0.5462	0.5806	0.6911	0.6684	0.4010	0.5888	0.7755	0.5107	0.5953
MLP	1D	0.7048	0.8216	0.8532	0.8846	0.5473	0.7329	0.8553	0.7092	0.7636
	3D	0.5612	0.7146	0.7743	0.8119	0.4318	0.6636	0.7766	0.6406	0.6718
	5D	0.4763	0.6439	0.7178	0.7482	0.3704	0.6106	0.7299	0.5839	0.6101
	7D	0.4169	0.5895	0.6672	0.6981	0.3318	0.5673	0.6902	0.5322	0.5616
	Avg.	0.5398	0.6924	0.7531	0.7857	0.4203	0.6436	0.7630	0.6165	0.6518
NLinear	1D	0.7194	0.8371	0.8693	0.8895	0.5597	0.7424	0.9270	0.7199	0.7830
	3D	0.5637	0.7241	0.7856	0.8139	0.4313	0.6657	0.8537	0.6444	0.6853
	5D	0.4787	0.6501	0.7253	0.7514	0.3672	0.6099	0.7961	0.5884	0.6209
	7D	0.4221	0.5970	0.6711	0.7013	0.3247	0.5648	0.7467	0.5361	0.5705
	Avg.	0.5460	0.7021	0.7628	0.7890	0.4207	0.6457	0.8309	0.6222	0.6649
TCN	1D	0.2114	0.2592	0.1635	0.5951	0.1360	0.2353	0.0159	0.1838	0.2250
	3D	0.2580	0.6474	0.1253	0.6241	0.1660	0.1922	0.0016	0.2078	0.2778
	5D	0.2739	0.6196	0.1078	0.6335	0.0822	0.1795	0.0042	0.1559	0.2571
	7D	0.1665	0.5082	0.0866	0.6118	0.1509	0.1327	0.0015	0.1184	0.2221
	Avg.	0.2275	0.5086	0.1208	0.6161	0.1338	0.1849	0.0058	0.1665	0.2455
CNN	1D	0.8035	0.8603	0.8865	0.9105	0.6118	0.7813	0.9242	0.7484	0.8158
	3D	0.6507	0.7626	0.8142	0.8435	0.5140	0.6944	0.8481	0.6713	0.7248
	5D	0.5411	0.6858	0.7491	0.7815	0.4423	0.6347	0.7851	0.6129	0.6541
	7D	0.4649	0.6315	0.6967	0.7253	0.4070	0.5847	0.7424	0.5611	0.6017
	Avg.	0.6150	0.7351	0.7866	0.8152	0.4938	0.6738	0.8249	0.6484	0.6991
TimesNet	1D	0.7318	0.8206	0.8768	0.8816	0.5354	0.7174	0.9262	0.7095	0.7749
	3D	0.5932	0.7320	0.8222	0.8192	0.4502	0.6755	0.8460	0.6211	0.6949
	5D	0.5099	0.6536	0.7561	0.7491	0.3574	0.6132	0.7838	0.5683	0.6239
	7D	0.4334	0.5895	0.6931	0.6956	0.3655	0.5801	0.7424	0.5018	0.5752
	Avg.	0.5671	0.6989	0.7870	0.7864	0.4271	0.6466	0.8246	0.6002	0.6672
Transformer	1D	0.7801	0.8399	0.8911	0.9096	0.5962	0.7881	0.9297	0.7340	0.8086
	3D	0.6106	0.7327	0.8207	0.8230	0.4748	0.6938	0.8714	0.6657	0.7116
	5D	0.5000	0.6422	0.7584	0.7629	0.4100	0.6334	0.8106	0.6079	0.6407
	7D	0.4416	0.5900	0.7224	0.7117	0.3721	0.5893	0.7646	0.5631	0.5944
	Avg.	0.5831	0.7012	0.7982	0.8018	0.4633	0.6761	0.8441	0.6427	0.6888
PatchTST	1D	0.7807	0.8541	0.8903	0.9017	0.5822	0.7739	0.9368	0.7388	0.8073
	3D	0.6322	0.7404	0.8227	0.8299	0.4753	0.6916	0.8753	0.6693	0.7171
	5D	0.5390	0.6651	0.7676	0.7680	0.4122	0.6324	0.8157	0.6075	0.6509
	7D	0.4604	0.6027	0.7225	0.7115	0.3739	0.5853	0.7706	0.5585	0.5982
	Avg.	0.6031	0.7156	0.8008	0.8028	0.4609	0.6708	0.8496	0.6435	0.6934

Table 12: Benchmark(SEDI(10%)) on part 1(Orlando area) stations(RNN, GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.5609	0.7803	0.7524	0.7912	0.4446	0.6691	0.8597	0.5862	0.6805
	3D	0.4845	0.4138	0.6461	0.5190	0.2755	0.6331	0.6811	0.4614	0.5143
	5D	0.3516	0.2382	0.5651	0.4990	0.2518	0.4117	0.6005	0.3917	0.4137
	7D	0.3015	0.2847	0.3757	0.4407	0.1476	0.4200	0.6180	0.4238	0.3765
	Avg.	0.4246	0.4292	0.5848	0.5625	0.2799	0.5335	0.6898	0.4658	0.4963
RNN DeepAR	1D	0.6365	0.7378	0.7444	0.6196	0.3993	0.6907	0.9190	0.6512	0.6748
	3D	0.4103	0.5953	0.6350	0.6830	0.2931	0.5966	0.6518	0.4432	0.5385
	5D	0.4231	0.3288	0.4959	0.3929	0.1312	0.5495	0.6462	0.3630	0.4163
	7D	0.2563	0.2015	0.3758	0.2432	0.1920	0.3033	0.5191	0.1891	0.2850
	Avg.	0.4315	0.4658	0.5628	0.4847	0.2539	0.5350	0.6840	0.4116	0.4787
DilatedRNN	1D	0.7387	0.8252	0.8307	0.8204	0.5269	0.7872	0.8584	0.5999	0.7484
	3D	0.5365	0.5276	0.8109	0.7547	0.4164	0.7141	0.8946	0.5329	0.6485
	5D	0.5228	0.5105	0.7001	0.6505	0.3770	0.5818	0.7581	0.3955	0.5620
	7D	0.3678	0.4182	0.5574	0.5107	0.2812	0.4989	0.6595	0.4477	0.4677
	Avg.	0.5414	0.5704	0.7248	0.6841	0.4004	0.6455	0.7927	0.4940	0.6067
GCN	1D	0.4046	0.6415	0.5966	0.7780	0.1645	0.4610	0.6537	0.3519	0.5065
	3D	0.3726	0.5581	0.6180	0.7132	0.0493	0.3374	0.6494	0.1105	0.4261
	5D	0.1951	0.5385	0.5924	0.7068	0.0753	0.2661	0.6071	0.0779	0.3824
	7D	0.3213	0.4485	0.5717	0.5943	0.0581	0.2512	0.5159	0.1533	0.3643
	Avg.	0.3234	0.5467	0.5947	0.6981	0.0868	0.3289	0.6065	0.1734	0.4198
GNN FourierGNN	1D	0.6658	0.8088	0.8470	0.7987	0.4921	0.5838	0.9344	0.6798	0.7263
	3D	0.5485	0.6963	0.8076	0.7115	0.3680	0.6447	0.8552	0.5561	0.6485
	5D	0.4276	0.5196	0.7487	0.6496	0.2064	0.5572	0.7322	0.4515	0.5366
	7D	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Avg.	0.4105	0.5062	0.6008	0.5399	0.2666	0.4464	0.6305	0.4219	0.4778
StemGNN	1D	0.5792	0.6391	0.8858	0.2337	0.5054	0.6999	0.8500	0.6941	0.6359
	3D	0.4938	0.3418	0.6541	0.1064	0.3230	0.5675	0.6905	0.4331	0.4513
	5D	0.3437	0.1567	0.4210	0.0723	0.2016	0.4045	0.6286	0.3792	0.3259
	7D	0.3112	0.2233	0.4052	0.0754	0.1406	0.3908	0.4314	0.2839	0.2827
	Avg.	0.4320	0.3402	0.5915	0.1220	0.2927	0.5157	0.6501	0.4476	0.4240
LLM GPT4TS	1D	0.7516	0.8292	0.8806	0.9015	0.5472	0.7470	0.9197	0.7224	0.7874
	3D	0.6089	0.7259	0.8109	0.8253	0.4655	0.6666	0.8548	0.6335	0.6989
	5D	0.5089	0.6604	0.7543	0.7710	0.4040	0.5947	0.7901	0.5795	0.6329
	7D	0.4288	0.6194	0.7052	0.6977	0.3788	0.5422	0.7208	0.4999	0.5741
	Avg.	0.5745	0.7087	0.7878	0.7989	0.4488	0.6376	0.8214	0.6088	0.6733
AutoTimes	1D	0.7309	0.8282	0.8781	0.8932	0.5675	0.7435	0.9318	0.7280	0.7876
	3D	0.5771	0.7171	0.8003	0.8198	0.4692	0.6811	0.8533	0.6595	0.6972
	5D	0.4934	0.6467	0.7364	0.7547	0.3728	0.6204	0.8093	0.6073	0.6301
	7D	0.4271	0.5959	0.6974	0.7030	0.3523	0.5638	0.7742	0.5483	0.5827
	Avg.	0.5571	0.6970	0.7780	0.7927	0.4405	0.6522	0.8421	0.6358	0.6744

Table 13: Benchmark(SEDI(5%)) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.4689	0.7533	0.8301	0.8850	0.3539	0.6561	0.7502	0.5351	0.6541
	3D	0.3180	0.5296	0.4839	0.7465	0.2869	0.3768	0.8140	0.2908	0.4808
	5D	0.0805	0.4723	0.3498	0.4559	0.1390	0.5192	0.6084	0.2483	0.3592
	7D	0.1822	0.3669	0.3667	0.3342	0.1868	0.3331	0.3853	0.1790	0.2918
	Avg.	0.2624	0.5306	0.5076	0.6054	0.2416	0.4713	0.6395	0.3133	0.4465
MLP	1D	0.3895	0.8083	0.7305	0.8724	0.3649	0.6597	0.7756	0.5515	0.6441
	3D	0.2869	0.6851	0.6306	0.7616	0.2921	0.5692	0.6816	0.4193	0.5408
	5D	0.2272	0.6029	0.5627	0.6654	0.2399	0.4985	0.6239	0.3319	0.4690
	7D	0.1810	0.5436	0.5061	0.5977	0.2074	0.4439	0.5472	0.2696	0.4121
	Avg.	0.2711	0.6600	0.6075	0.7243	0.2761	0.5428	0.6571	0.3931	0.5165
NLinear	1D	0.4063	0.8239	0.7535	0.8781	0.3729	0.6715	0.8501	0.5726	0.6661
	3D	0.2888	0.6956	0.6440	0.7626	0.2941	0.5698	0.7364	0.4278	0.5524
	5D	0.2269	0.6092	0.5723	0.6661	0.2402	0.4984	0.6635	0.3386	0.4769
	7D	0.1823	0.5510	0.5088	0.5971	0.2081	0.4426	0.5951	0.2740	0.4199
	Avg.	0.2761	0.6699	0.6197	0.7260	0.2788	0.5456	0.7113	0.4033	0.5288
TCN	1D	0.0904	0.2365	0.0355	0.6726	0.0601	0.1777	0.0000	0.1539	0.1783
	3D	0.0644	0.6592	0.0357	0.7267	0.0692	0.1249	0.0000	0.1242	0.2255
	5D	0.0781	0.6191	0.0220	0.7086	0.0472	0.1066	0.0000	0.1456	0.2159
	7D	0.0692	0.4961	0.0162	0.6670	0.0598	0.0681	0.0000	0.0902	0.1833
	Avg.	0.0755	0.5027	0.0274	0.6937	0.0591	0.1193	0.0000	0.1285	0.2008
CNN	1D	0.5813	0.8499	0.8135	0.9149	0.4038	0.7116	0.8792	0.6556	0.7262
	3D	0.4037	0.7353	0.7020	0.8165	0.3552	0.6041	0.7509	0.5147	0.6103
	5D	0.3168	0.6380	0.6203	0.7250	0.3213	0.5264	0.6747	0.4165	0.5299
	7D	0.2622	0.5682	0.5523	0.6502	0.2858	0.4714	0.6070	0.3486	0.4682
	Avg.	0.3910	0.6979	0.6720	0.7766	0.3415	0.5784	0.7279	0.4838	0.5837
TimesNet	1D	0.4963	0.8057	0.7888	0.8751	0.3616	0.6363	0.8313	0.5658	0.6701
	3D	0.3499	0.7130	0.7169	0.7632	0.3035	0.5742	0.7399	0.4469	0.5759
	5D	0.2842	0.6150	0.6111	0.6662	0.2352	0.4649	0.6938	0.3600	0.4913
	7D	0.2136	0.5427	0.5451	0.6053	0.2383	0.4457	0.6072	0.2963	0.4368
	Avg.	0.3360	0.6691	0.6655	0.7274	0.2846	0.5303	0.7181	0.4173	0.5435
Transformer	1D	0.5339	0.8220	0.8005	0.9039	0.3823	0.7226	0.8579	0.6208	0.7055
	3D	0.3627	0.6993	0.7409	0.7699	0.3055	0.5886	0.7792	0.5172	0.5954
	5D	0.2737	0.5880	0.6522	0.6846	0.2615	0.5178	0.6879	0.4260	0.5114
	7D	0.2218	0.5256	0.6113	0.6180	0.2310	0.4643	0.6272	0.3555	0.4569
	Avg.	<u>0.3480</u>	0.6587	0.7012	0.7441	0.2951	<u>0.5733</u>	<u>0.7380</u>	<u>0.4799</u>	0.5673
PatchTST	1D	0.5188	0.8360	0.8061	0.8954	0.3815	0.6915	0.8645	0.6200	0.7017
	3D	0.3580	0.7099	0.7235	0.7964	0.3198	0.5939	0.7783	0.5003	0.5975
	5D	0.2903	0.6186	0.6554	0.7030	0.2698	0.5185	0.6945	0.3977	0.5185
	7D	0.2205	0.5470	0.5951	0.6183	0.2401	0.4617	0.6365	0.3319	0.4564
	Avg.	0.3469	<u>0.6779</u>	<u>0.6950</u>	<u>0.7533</u>	<u>0.3028</u>	0.5664	0.7435	0.4625	<u>0.5685</u>

Table 14: Benchmark(SEDI(5%)) on part 1(Orlando area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.1865	0.7157	0.6846	0.7911	0.3022	0.6181	0.7183	0.4020	0.5523
	3D	0.1732	0.3087	0.5155	0.4562	0.1632	0.5043	0.3897	0.2527	0.3454
	5D	0.0804	0.1507	0.3768	0.3735	0.1444	0.2692	0.3643	0.1433	0.2378
	7D	0.0824	0.1802	0.1650	0.3368	0.0946	0.2447	0.3626	0.1930	0.2074
	Avg.	0.1306	0.3388	0.4355	0.4894	0.1761	0.4091	0.4587	0.2477	0.3357
RNN DeepAR	1D	0.2195	0.6513	0.7199	0.6546	0.2093	0.5973	0.8096	0.4660	0.5409
	3D	0.0994	0.4842	0.5131	0.6412	0.1952	0.4451	0.3791	0.2111	0.3711
	5D	0.1516	0.2169	0.3219	0.2775	0.0788	0.3864	0.4042	0.1503	0.2484
	7D	0.0521	0.1250	0.1591	0.1357	0.1175	0.1689	0.2765	0.0582	0.1366
	Avg.	0.1307	0.3694	0.4285	0.4273	0.1502	0.3994	0.4674	0.2214	0.3243
DilatedRNN	1D	0.4293	0.7865	0.7306	0.8093	0.3334	0.7120	0.7360	0.4368	0.6217
	3D	0.2092	0.4563	0.7102	0.6971	0.2610	0.6008	0.7979	0.3049	0.5047
	5D	0.2419	0.4152	0.5496	0.5746	0.2462	0.4473	0.5624	0.1760	0.4017
	7D	0.1639	0.3365	0.4385	0.4249	0.1768	0.3593	0.4711	0.2315	0.3253
	Avg.	0.2611	0.4986	0.6072	0.6265	0.2543	0.5298	0.6419	0.2873	0.4633
GCN	1D	0.2245	0.6332	0.4674	0.8024	0.0513	0.3851	0.5477	0.2553	0.4209
	3D	0.1671	0.5823	0.5272	0.7293	0.0245	0.2475	0.4494	0.0417	0.3461
	5D	0.0577	0.5161	0.5331	0.7090	0.0208	0.1781	0.4557	0.0147	0.3106
	7D	0.1061	0.4156	0.4653	0.6042	0.0143	0.1622	0.3817	0.0264	0.2720
	Avg.	0.1388	0.5368	0.4982	0.7112	0.0277	0.2432	0.4586	0.0845	0.3374
GNN FourierGNN	1D	0.3952	0.7906	0.7008	0.7820	0.3138	0.4377	0.8682	0.5139	0.6003
	3D	0.2862	0.6614	0.7273	0.6802	0.2185	0.5534	0.7375	0.3481	0.5266
	5D	0.1981	0.4685	0.6289	0.5985	0.1152	0.4331	0.5852	0.2422	0.4087
	7D	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Avg.	0.2199	0.4801	0.5142	0.5152	0.1619	0.3561	0.5477	0.2760	0.3839
StemGNN	1D	0.2809	0.5362	0.7391	0.1403	0.3245	0.6182	0.7072	0.5281	0.4843
	3D	0.2238	0.2364	0.4764	0.0558	0.2164	0.4179	0.4960	0.2882	0.3014
	5D	0.1175	0.0823	0.2467	0.0490	0.1018	0.3428	0.6017	0.1714	0.2142
	7D	0.1019	0.1424	0.2231	0.0433	0.0722	0.3268	0.2812	0.1275	0.1648
	Avg.	0.1810	0.2493	0.4213	0.0721	0.1787	0.4264	0.5215	0.2788	0.2912
LLM GPT4TS	1D	0.4741	0.8136	0.7639	0.8954	0.3627	0.6754	0.8547	0.6139	0.6817
	3D	0.3590	0.7011	0.6964	0.7845	0.2968	0.5789	0.7585	0.4680	0.5804
	5D	0.2704	0.6083	0.6133	0.6943	0.2622	0.4925	0.6912	0.3753	0.5009
	7D	0.1934	0.5552	0.5596	0.6083	0.2345	0.4420	0.5751	0.3155	0.4354
	Avg.	0.3242	0.6696	0.6583	0.7456	0.2890	0.5472	0.7199	0.4432	0.5496
AutoTimes	1D	0.4118	0.8162	0.7727	0.8817	0.3661	0.6693	0.8491	0.5905	0.6697
	3D	0.2907	0.6894	0.6914	0.7738	0.3074	0.5801	0.7521	0.4588	0.5680
	5D	0.2436	0.6064	0.5992	0.6760	0.2393	0.5027	0.6838	0.3734	0.4905
	7D	0.1916	0.5429	0.5674	0.6062	0.2155	0.4391	0.6313	0.2945	0.4360
	Avg.	0.2844	0.6637	0.6577	0.7344	0.2821	0.5478	0.7291	0.4293	0.5411

Table 15: Benchmark(SEDI(1%)) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.1779	0.5839	0.4092	0.7549	0.1845	0.0960	0.1547	0.1705	0.3165
	3D	0.0992	0.2754	0.1454	0.4582	0.0937	0.0457	0.3397	0.0142	0.1839
	5D	0.0379	0.2232	0.0934	0.1592	0.0319	0.0439	0.2012	0.0224	0.1017
	7D	0.0414	0.1542	0.0770	0.0976	0.0557	0.0440	0.0543	0.0174	0.0677
	Avg.	0.0891	0.3092	0.1813	0.3675	0.0915	0.0574	0.1875	0.0561	0.1674
MLP	1D	0.1797	0.7203	0.3411	0.7288	0.1974	0.1141	0.2192	0.0923	0.3241
	3D	0.1123	0.5652	0.2414	0.5123	0.1239	0.0869	0.1741	0.0547	0.2339
	5D	0.0710	0.4629	0.1773	0.3898	0.0954	0.0660	0.1499	0.0355	0.1810
	7D	0.0457	0.3786	0.1352	0.3123	0.0787	0.0529	0.1111	0.0259	0.1425
	Avg.	0.1022	0.5318	0.2238	0.4858	0.1238	0.0800	0.1636	0.0521	0.2204
NLinear	1D	0.1896	0.7415	0.3636	0.7393	0.2129	0.1214	0.3349	0.1212	0.3531
	3D	0.1167	0.5773	0.2543	0.5140	0.1262	0.0901	0.2602	0.0657	0.2506
	5D	0.0732	0.4684	0.1861	0.3879	0.0958	0.0675	0.2168	0.0426	0.1923
	7D	0.0468	0.3854	0.1376	0.3039	0.0755	0.0528	0.1754	0.0301	0.1509
	Avg.	0.1066	0.5432	0.2354	0.4863	0.1276	0.0830	0.2468	0.0649	0.2367
TCN	1D	0.0397	0.1212	0.0051	0.7399	0.0447	0.0000	0.0000	0.0054	0.1195
	3D	0.0177	0.6617	0.0000	0.7624	0.0562	0.0000	0.0000	0.0022	0.1875
	5D	0.0141	0.6104	0.0003	0.7485	0.0169	0.0038	0.0000	0.0048	0.1748
	7D	0.0110	0.4625	0.0000	0.6858	0.0528	0.0000	0.0000	0.0011	0.1517
	Avg.	0.0206	0.4639	0.0014	0.7342	0.0427	0.0010	0.0000	0.0034	0.1584
CNN	1D	0.2454	0.7908	0.4015	0.8634	0.2252	0.1487	0.3321	0.2147	0.4027
	3D	0.1422	0.6460	0.3200	0.6801	0.1425	0.1042	0.2606	0.1291	0.3031
	5D	0.0876	0.5359	0.2394	0.5430	0.1099	0.0788	0.2303	0.0968	0.2402
	7D	0.0584	0.4666	0.1858	0.4412	0.0883	0.0659	0.2029	0.0855	0.1993
	Avg.	0.1334	0.6098	0.2866	0.6319	0.1415	0.0994	0.2565	0.1315	0.2863
TimesNet	1D	0.1929	0.6805	0.4112	0.7304	0.2111	0.1093	0.3149	0.1015	0.3440
	3D	0.1385	0.5621	0.3526	0.5907	0.1684	0.0843	0.2438	0.1168	0.2822
	5D	0.0866	0.4737	0.2962	0.4546	0.1363	0.0592	0.2430	0.1009	0.2313
	7D	0.0478	0.3389	0.2187	0.3775	0.1147	0.0450	0.2044	0.0759	0.1779
	Avg.	0.1165	0.5138	0.3197	0.5383	0.1576	0.0744	0.2515	0.0988	0.2588
Transformer	1D	0.2115	0.7672	0.3883	0.8135	0.2255	0.1465	0.3263	0.1734	0.3815
	3D	0.1407	0.6055	0.3318	0.5916	0.1422	0.0995	0.2523	0.1404	0.2880
	5D	0.0900	0.4964	0.2614	0.4640	0.1084	0.0713	0.2279	0.0948	0.2268
	7D	0.0598	0.3922	0.2230	0.3780	0.0913	0.0644	0.1827	0.0733	0.1831
	Avg.	0.1255	0.5653	0.3011	0.5618	0.1418	0.0954	0.2473	0.1205	0.2698
PatchTST	1D	0.2100	0.7859	0.3868	0.7935	0.2204	0.1244	0.3310	0.1787	0.3788
	3D	0.1306	0.6186	0.3123	0.6206	0.1494	0.0971	0.2570	0.1091	0.2868
	5D	0.0852	0.5129	0.2530	0.4784	0.1061	0.0748	0.2222	0.0804	0.2266
	7D	0.0558	0.4093	0.2074	0.3418	0.0878	0.0574	0.1940	0.0625	0.1770
	Avg.	0.1204	0.5817	0.2899	0.5586	0.1409	0.0884	0.2510	0.1077	0.2673

Table 16: Benchmark(SEDI(1%)) on part 1(Orlando area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.0453	0.3104	0.2330	0.3873	0.1873	0.1146	0.0987	0.0689	0.1807
	3D	0.0554	0.0603	0.1232	0.1141	0.1423	0.1027	0.0269	0.0340	0.0823
	5D	0.0106	0.0295	0.0621	0.0487	0.0979	0.0315	0.0320	0.0069	0.0399
	7D	0.0080	0.0418	0.0189	0.0710	0.0580	0.0026	0.0302	0.0354	0.0332
	Avg.	0.0298	0.1105	0.1093	0.1553	0.1214	0.0628	0.0469	0.0363	0.0840
RNN DeepAR	1D	0.0566	0.3096	0.1705	0.2465	0.1637	0.1377	0.1660	0.0815	0.1665
	3D	0.0274	0.1400	0.1083	0.2401	0.0774	0.0840	0.0316	0.0045	0.0892
	5D	0.0261	0.0597	0.0569	0.0369	0.0386	0.0706	0.0328	0.0083	0.0412
	7D	0.0041	0.0228	0.0199	0.0122	0.0574	0.0027	0.0165	0.0004	0.0170
	Avg.	0.0286	0.1330	0.0889	0.1339	0.0843	0.0737	0.0617	0.0237	0.0785
DilatedRNN	1D	0.1559	0.4928	0.3281	0.4393	0.2232	0.1702	0.1036	0.1106	0.2530
	3D	0.0638	0.1765	0.2754	0.3851	0.1390	0.1160	0.2855	0.0669	0.1885
	5D	0.0547	0.1639	0.1659	0.2360	0.1243	0.0570	0.1354	0.0216	0.1198
	7D	0.0403	0.1313	0.1366	0.1638	0.0905	0.0284	0.1025	0.0681	0.0952
	Avg.	0.0787	0.2411	0.2265	0.3060	0.1442	0.0929	0.1567	0.0668	0.1641
GCN	1D	0.0884	0.5829	0.1324	0.7469	0.0199	0.0966	0.0929	0.0599	0.2275
	3D	0.0417	0.5323	0.1576	0.6933	0.0113	0.0010	0.1498	0.0130	0.2000
	5D	0.0196	0.3964	0.1654	0.6615	0.0030	0.0006	0.0740	0.0034	0.1655
	7D	0.0167	0.2674	0.1065	0.6168	0.0029	0.0000	0.0741	0.0043	0.1361
	Avg.	0.0416	0.4448	0.1405	0.6796	0.0093	0.0246	0.0977	0.0201	0.1823
GNN FourierGNN	1D	0.1456	0.6623	0.2734	0.5818	0.1506	0.0832	0.3269	0.1528	0.2971
	3D	0.0915	0.4698	0.2704	0.4247	0.0601	0.0586	0.2105	0.0649	0.2063
	5D	0.0560	0.2342	0.1852	0.3349	0.0231	0.0258	0.1322	0.0383	0.1287
	7D	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Avg.	0.0733	0.3416	0.1823	0.3353	0.0584	0.0419	0.1674	0.0640	0.1580
StemGNN	1D	0.1073	0.1712	0.2926	0.0254	0.1481	0.1117	0.0857	0.1783	0.1400
	3D	0.0683	0.0233	0.1672	0.0099	0.0966	0.0626	0.0417	0.0869	0.0695
	5D	0.0257	0.0177	0.0445	0.0088	0.0513	0.0275	0.1592	0.0370	0.0465
	7D	0.0163	0.0200	0.0784	0.0047	0.0312	0.0147	0.0371	0.0203	0.0278
	Avg.	0.0544	0.0581	0.1457	0.0122	0.0818	0.0542	0.0809	0.0806	0.0710
LLM GPT4TS	1D	0.1951	0.7144	0.3922	0.7640	0.2087	0.1158	0.2950	0.1505	0.3545
	3D	0.1160	0.5546	0.3427	0.5552	0.1764	0.0802	0.2384	0.0979	0.2702
	5D	0.0732	0.4710	0.2864	0.4675	0.1690	0.0739	0.2255	0.0925	0.2324
	7D	0.0485	0.3988	0.2381	0.3505	0.1101	0.0880	0.1745	0.0952	0.1880
	Avg.	0.1082	0.5347	0.3148	0.5343	0.1661	0.0895	0.2334	0.1090	0.2612
AutoTimes	1D	0.1860	0.7398	0.3775	0.7625	0.2213	0.1164	0.3134	0.1337	0.3563
	3D	0.1117	0.5977	0.2956	0.5645	0.1442	0.0913	0.2333	0.0828	0.2652
	5D	0.0700	0.4938	0.2123	0.3913	0.0975	0.0724	0.1973	0.0624	0.1996
	7D	0.0417	0.4171	0.1844	0.3174	0.0776	0.0502	0.1755	0.0373	0.1627
	Avg.	0.1024	0.5621	0.2675	0.5089	0.1352	0.0826	0.2299	0.0791	0.2459

Table 17: Benchmark(MAE) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.0739	0.0969	0.1538	0.1234	0.1172	0.1069	0.1168	0.1548	0.1180	
	3D	0.1588	0.1621	0.2502	0.2129	0.1952	0.1638	0.1855	0.2459	0.1968	
	5D	0.2147	0.2079	0.3080	0.2637	0.2568	0.1984	0.2268	0.3070	0.2479	
	7D	0.2650	0.2436	0.3517	0.3046	0.2989	0.2236	0.2595	0.3454	0.2865	
	Avg.	0.1781	0.1776	0.2659	0.2262	0.2170	0.1732	0.1971	0.2633	0.2123	
MLP	1D	0.0932	0.1147	0.1777	0.1381	0.1322	0.1152	0.1279	0.2124	0.1389	
	3D	0.1637	0.1770	0.2667	0.2123	0.2056	0.1680	0.1926	0.2856	0.2089	
	5D	0.2179	0.2209	0.3239	0.2646	0.2615	0.2014	0.2349	0.3376	0.2578	
	7D	0.2621	0.2558	0.3663	0.3055	0.3081	0.2258	0.2663	0.3779	0.2960	
	Avg.	0.1842	0.1921	0.2836	0.2301	0.2269	0.1776	0.2054	0.3034	0.2254	
NLinear	1D	0.0857	0.1055	0.1672	0.1274	0.1207	0.1085	0.1191	0.1484	0.1228	
	3D	0.1603	0.1709	0.2606	0.2059	0.1988	0.1642	0.1882	0.2414	0.1988	
	5D	0.2153	0.2167	0.3190	0.2608	0.2560	0.1984	0.2311	0.3010	0.2498	
	7D	0.2603	0.2526	0.3618	0.3022	0.3031	0.2237	0.2635	0.3457	0.2891	
	Avg.	0.1804	0.1864	0.2772	0.2241	0.2196	0.1737	0.2005	0.2591	0.2151	
TCN	1D	0.1747	0.1498	0.1937	0.1790	0.1882	0.1957	0.1439	0.7995	0.2531	
	3D	0.2303	0.1938	0.2971	0.2766	0.2811	0.2347	0.2049	0.9442	0.3328	
	5D	0.3770	0.2350	0.3711	0.3214	0.3427	0.2733	0.2633	0.8945	0.3848	
	7D	0.3407	0.2960	0.3925	0.3659	0.3887	0.3089	0.2950	1.0090	0.4246	
	Avg.	0.2807	0.2186	0.3136	0.2857	0.3002	0.2531	0.2268	0.9118	0.3488	
CNN	1D	0.0777	0.1049	0.1764	0.1385	0.1134	0.1034	0.1154	0.1565	0.1233	
	3D	0.1611	0.1813	0.2938	0.2362	0.1982	0.1669	0.1906	0.2810	0.2136	
	5D	0.2289	0.2449	0.3621	0.3098	0.2576	0.2038	0.2351	0.3383	0.2726	
	7D	0.2809	0.2903	0.4201	0.3565	0.3054	0.2314	0.2702	0.3758	0.3163	
	Avg.	0.1871	0.2054	0.3131	0.2603	0.2186	0.1764	0.2028	0.2879	0.2315	
TimesNet	1D	0.0962	0.1269	0.1984	0.1537	0.1505	0.1309	0.1520	0.1927	0.1502	
	3D	0.1684	0.1943	0.2871	0.2323	0.2249	0.1854	0.2122	0.2767	0.2227	
	5D	0.2212	0.2409	0.3516	0.2849	0.2861	0.2200	0.2587	0.3347	0.2748	
	7D	0.2722	0.2761	0.3937	0.3329	0.3326	0.2409	0.2915	0.3775	0.3147	
	Avg.	0.1895	0.2095	0.3077	0.2509	0.2485	0.1943	0.2286	0.2954	0.2406	
Transformer	iTransformer	1D	0.0731	0.0925	0.1477	0.1178	0.1129	0.1024	0.1104	0.1409	0.1122
		3D	0.1446	0.1603	0.2490	0.1975	0.1901	0.1593	0.1807	0.2334	0.1894
		5D	0.2057	0.2052	0.3095	0.2518	0.2498	0.1938	0.2248	0.3007	0.2427
		7D	0.2513	0.2460	0.3529	0.2933	0.2968	0.2193	0.2555	0.3480	0.2829
		Avg.	0.1687	0.1760	0.2648	0.2151	0.2124	0.1687	0.1928	0.2558	0.2068
	PatchTST	1D	0.0691	0.0926	0.1472	0.1158	0.1114	0.1012	0.1104	0.1394	0.1109
		3D	0.1419	0.1610	0.2449	0.1975	0.1927	0.1590	0.1828	0.2376	0.1897
		5D	0.2002	0.2094	0.3042	0.2524	0.2504	0.1943	0.2251	0.2999	0.2420
		7D	0.2463	0.2461	0.3499	0.2947	0.3005	0.2200	0.2574	0.3444	0.2824
		Avg.	0.1644	0.1773	0.2615	0.2151	0.2138	0.1686	0.1939	0.2553	0.2062

Table 18: Benchmark(MAE) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.1150	0.1373	0.2165	0.1678	0.1691	0.1378	0.1531	0.2537	0.1688
	3D	0.2287	0.2081	0.3074	0.2574	0.2650	0.2002	0.2272	0.3706	0.2581
	5D	0.2646	0.2556	0.3832	0.3241	0.3190	0.2378	0.2694	0.4339	0.3109
	7D	0.3326	0.2989	0.4425	0.3728	0.3855	0.2579	0.2909	0.4743	0.3569
	Avg.	0.2352	0.2250	0.3374	0.2805	0.2847	0.2084	0.2352	0.3831	0.2737
RNN DeepAR	1D	0.1196	0.1356	0.2067	0.1678	0.1659	0.1410	0.1546	0.2557	0.1684
	3D	0.2086	0.2169	0.3113	0.2615	0.2693	0.2047	0.2301	0.3624	0.2581
	5D	0.2839	0.2586	0.3710	0.3123	0.3166	0.2299	0.2790	0.4224	0.3092
	7D	0.3404	0.2992	0.4478	0.3695	0.3657	0.2598	0.3007	0.4670	0.3562
	Avg.	0.2381	0.2276	0.3342	0.2778	0.2794	0.2088	0.2411	0.3769	0.2730
DilatedRNN	1D	0.0781	0.1134	0.1706	0.1359	0.1309	0.1155	0.1217	0.2111	0.1347
	3D	0.1724	0.1856	0.2843	0.2268	0.2393	0.1769	0.2149	0.3164	0.2271
	5D	0.2463	0.2351	0.3483	0.2967	0.3012	0.2133	0.2548	0.3866	0.2853
	7D	0.3312	0.2732	0.4052	0.3472	0.3469	0.2460	0.2917	0.4338	0.3344
	Avg.	0.2070	0.2018	0.3021	0.2516	0.2546	0.1879	0.2208	0.3370	0.2454
GCN	1D	0.0910	0.1254	0.1880	0.1761	0.1626	0.1374	0.1528	0.9648	0.2498
	3D	0.1718	0.1904	0.2843	0.2485	0.2410	0.1928	0.2173	0.5350	0.2601
	5D	0.2460	0.2340	0.3390	0.3061	0.3085	0.2319	0.2571	0.7279	0.3313
	7D	0.2998	0.2700	0.3911	0.3452	0.3508	0.2609	0.2914	0.6410	0.3563
	Avg.	0.2022	0.2050	0.3006	0.2690	0.2657	0.2057	0.2297	0.7172	0.2994
GNN FourierGNN	1D	0.0825	0.1072	0.1604	0.1365	0.1226	0.1165	0.1234	0.2444	0.1367
	3D	0.1593	0.1722	0.2527	0.2059	0.2024	0.1734	0.1925	0.3399	0.2123
	5D	0.2109	0.2189	0.3114	0.2658	0.2568	0.2123	0.2320	0.4031	0.2639
	7D	0.2896	0.2506	0.3590	0.3121	0.3067	0.2357	0.2624	0.4222	0.3048
	Avg.	0.1856	0.1872	0.2709	0.2301	0.2221	0.1845	0.2026	0.3524	0.2294
StemGNN	1D	0.0859	0.1404	0.1762	0.1382	0.1444	0.1204	0.1252	0.3656	0.1620
	3D	0.2047	0.2609	0.2958	0.2616	0.2763	0.2432	0.2227	0.4821	0.2809
	5D	0.2874	0.3173	0.3743	0.3535	0.4289	0.2839	0.2765	0.5247	0.3558
	7D	0.3603	0.3402	0.4218	0.4172	0.4253	0.3211	0.3283	0.6507	0.4081
	Avg.	0.2346	0.2647	0.3170	0.2926	0.3187	0.2422	0.2382	0.5058	0.3017
LLM GPT4TS	1D	0.0931	0.1264	0.1860	0.1431	0.1392	0.1304	0.1510	0.1974	0.1458
	3D	0.1639	0.1959	0.2833	0.2283	0.2281	0.1895	0.2159	0.2917	0.2246
	5D	0.2256	0.2429	0.3469	0.2834	0.2897	0.2195	0.2535	0.3389	0.2751
	7D	0.2728	0.2842	0.3941	0.3275	0.3342	0.2466	0.2878	0.3960	0.3179
	Avg.	0.1888	0.2124	0.3026	0.2456	0.2478	0.1965	0.2271	0.3060	0.2408
AutoTimes	1D	0.0863	0.1029	0.1665	0.1318	0.1270	0.1140	0.1312	0.1587	0.1273
	3D	0.1530	0.1663	0.2545	0.2069	0.2008	0.1668	0.1926	0.2450	0.1982
	5D	0.2065	0.2109	0.3079	0.2619	0.2556	0.2017	0.2342	0.3065	0.2481
	7D	0.2552	0.2469	0.3515	0.3009	0.3029	0.2267	0.2654	0.3482	0.2872
	Avg.	0.1752	0.1817	0.2701	0.2254	0.2216	0.1773	0.2058	0.2646	0.2152

Table 19: Benchmark(MSE) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.0315	0.0629	0.1405	0.0774	0.0602	0.0525	0.0787	0.1295	0.0792
	3D	0.0808	0.1311	0.2777	0.1506	0.1289	0.0935	0.1471	0.2762	0.1607
	5D	0.1300	0.1819	0.3613	0.2040	0.1912	0.1197	0.1905	0.3950	0.2217
	7D	0.1783	0.2232	0.4256	0.2480	0.2456	0.1399	0.2230	0.4617	0.2682
	Avg.	0.1052	0.1498	0.3012	0.1700	0.1565	0.1014	0.1599	0.3156	0.1824
MLP	1D	0.0397	0.0747	0.1695	0.0877	0.0697	0.0578	0.0868	0.2450	0.1039
	3D	0.0932	0.1486	0.3241	0.1611	0.1419	0.0985	0.1605	0.3833	0.1889
	5D	0.1460	0.2061	0.4276	0.2192	0.2111	0.1263	0.2114	0.4777	0.2532
	7D	0.1968	0.2537	0.5054	0.2679	0.2786	0.1475	0.2509	0.5544	0.3069
	Avg.	0.1189	0.1708	0.3566	0.1840	0.1753	0.1075	0.1774	0.4151	0.2132
NLinear	1D	0.0374	0.0687	0.1605	0.0830	0.0639	0.0554	0.0816	0.1269	0.0847
	3D	0.0924	0.1442	0.3186	0.1581	0.1376	0.0974	0.1568	0.2825	0.1734
	5D	0.1450	0.2026	0.4224	0.2174	0.2065	0.1253	0.2083	0.3881	0.2395
	7D	0.1962	0.2511	0.5021	0.2666	0.2743	0.1471	0.2484	0.4696	0.2944
	Avg.	0.1177	0.1666	0.3509	0.1813	0.1706	0.1063	0.1738	<u>0.3168</u>	0.1980
TCN	1D	0.0976	0.0963	0.1756	0.1090	0.1377	0.1694	0.0984	5.8002	0.8355
	3D	0.1318	0.1480	0.3271	0.2042	0.2212	0.2113	0.1527	7.0577	1.0567
	5D	0.3490	0.2009	0.4521	0.2532	0.3033	0.2986	0.2079	5.1319	0.8996
	7D	0.2479	0.2662	0.4783	0.2929	0.3587	0.2911	0.2391	6.7494	1.1155
	Avg.	0.2066	0.1779	0.3583	0.2148	0.2552	0.2426	0.1745	6.1848	0.9768
CNN	1D	0.0395	0.0838	0.2180	0.1137	0.0630	0.0539	0.0818	0.3165	0.1213
	3D	0.1168	0.1894	0.4307	0.2323	0.1464	0.1036	0.1678	2.9814	0.5460
	5D	0.1869	0.3223	0.5611	0.3355	0.2198	0.1352	0.2221	2.7558	0.5923
	7D	0.2499	0.4475	0.7525	0.3950	0.2877	0.1613	0.2656	1.4768	0.5045
	Avg.	0.1483	0.2608	0.4906	0.2691	0.1792	0.1135	0.1843	1.8826	0.4411
TimesNet	1D	0.0453	0.0899	0.1964	0.1033	0.0884	0.0700	0.1090	0.1729	0.1094
	3D	0.1100	0.1839	0.3598	0.1995	0.1694	0.1154	0.1857	0.3311	0.2069
	5D	0.1666	0.2634	0.4952	0.2586	0.2486	0.1484	0.2477	0.4500	0.2848
	7D	0.2283	0.3207	0.5949	0.3303	0.3296	0.1674	0.2892	0.5382	0.3498
	Avg.	0.1376	0.2145	0.4116	0.2230	0.2090	0.1253	0.2079	0.3731	0.2377
Transformer	1D	0.0337	0.0645	0.1413	0.0798	0.0614	0.0536	0.0766	0.1210	0.0790
	3D	0.0880	0.1424	0.3086	0.1614	0.1361	0.0964	0.1512	0.2755	0.1700
	5D	0.1486	0.2002	0.4253	0.2206	0.2052	0.1266	0.2082	0.3973	0.2415
	7D	0.2032	0.2658	0.4903	0.2697	0.2727	0.1490	0.2408	0.5022	0.2992
	Avg.	0.1184	0.1682	0.3414	0.1828	0.1688	0.1064	0.1692	0.3240	0.1974
	1D	0.0326	0.0667	0.1392	0.0789	0.0615	0.0525	0.0781	0.1213	0.0789
	3D	0.0872	0.1459	0.2950	0.1614	0.1389	0.0963	0.1586	0.2825	0.1707
	5D	0.1402	0.2096	0.3952	0.2208	0.2068	0.1249	0.2112	0.3960	0.2381
	7D	0.1899	0.2624	0.4758	0.2686	0.2763	0.1473	0.2494	0.4766	0.2933
	Avg.	<u>0.1125</u>	0.1711	0.3263	0.1824	0.1709	<u>0.1053</u>	0.1743	0.3191	<u>0.1952</u>

Table 20: Benchmark(MSE) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	Metric	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
LSTM	1D	0.0575	0.1123	0.2405	0.1158	0.1092	0.0775	0.1224	1.1111	0.2433	
	3D	0.1531	0.1869	0.3854	0.2070	0.2091	0.1206	0.1944	1.4368	0.3617	
	5D	0.1983	0.2390	0.4905	0.2732	0.2894	0.1479	0.2368	1.3811	0.4070	
	7D	0.2734	0.2901	0.5826	0.3426	0.3981	0.1677	0.2615	1.8380	0.5192	
	Avg.	0.1706	0.2071	0.4248	0.2346	0.2515	0.1284	0.2038	1.4417	0.3828	
RNN	DeepAR	1D	0.0611	0.1108	0.2390	0.1248	0.1152	0.0790	0.1230	0.9006	0.2192
	3D	0.1276	0.1894	0.3864	0.2041	0.2173	0.1226	0.1953	1.3494	0.3490	
	5D	0.1973	0.2385	0.4877	0.2623	0.2918	0.1463	0.2344	1.4248	0.4104	
	7D	0.2820	0.3010	0.5995	0.3402	0.3564	0.1667	0.2657	1.4515	0.4704	
	Avg.	0.1670	0.2099	0.4282	0.2328	0.2452	0.1286	0.2046	1.2816	0.3622	
DilatedRNN	1D	0.0366	0.0808	0.1810	0.0931	0.0823	0.0635	0.0944	0.8270	0.1823	
	3D	0.1137	0.1710	0.3481	0.1894	0.2205	0.1151	0.1746	1.0864	0.3023	
	5D	0.1970	0.2275	0.4660	0.2738	0.3055	0.1469	0.2277	1.3925	0.4046	
	7D	0.2899	0.2742	0.5486	0.3313	0.3679	0.1781	0.2746	1.3743	0.4549	
	Avg.	0.1593	0.1884	0.3859	0.2219	0.2441	0.1259	0.1928	1.1701	0.3360	
GCN	1D	0.0333	0.0964	0.1750	0.1817	0.1209	0.0909	0.0948	170.0996	21.3616	
	3D	0.0829	0.1800	0.3166	0.2245	0.1894	0.1255	0.1616	32.4988	4.2224	
	5D	0.1407	0.2346	0.4015	0.2996	0.2709	0.1500	0.2053	44.7123	5.8019	
	7D	0.2021	0.2599	0.4775	0.3266	0.3373	0.1828	0.2379	17.1457	2.3962	
	Avg.	0.1147	0.1927	0.3426	0.2581	0.2296	0.1373	0.1749	66.1141	8.4455	
GNN	FourierGNN	1D	0.0339	0.0703	0.1442	0.0825	0.0631	0.0563	0.0809	0.6083	0.1425
	3D	0.0851	0.1359	0.2816	0.1522	0.1320	0.0988	0.1493	0.9670	0.2502	
	5D	0.1331	0.1885	0.3656	0.2117	0.1916	0.1290	0.1920	1.6289	0.3800	
	7D	0.2053	0.2240	0.4330	0.2525	0.2511	0.1483	0.2250	1.0897	0.3536	
	Avg.	0.1144	0.1547	0.3061	0.1747	0.1594	0.1081	0.1618	1.0735	0.2816	
StemGNN	1D	0.0391	0.1055	0.1724	0.0912	0.0873	0.0665	0.0898	1.2556	0.2384	
	3D	0.1283	0.2207	0.3498	0.2164	0.2680	0.1644	0.1796	1.5368	0.3830	
	5D	0.2199	0.2945	0.4663	0.3239	0.5381	0.2094	0.2414	1.6949	0.4985	
	7D	0.3149	0.3582	0.5362	0.3986	0.4688	0.2426	0.3031	2.0180	0.5801	
	Avg.	0.1755	0.2447	0.3812	0.2575	0.3406	0.1707	0.2035	1.6263	0.4250	
LLM	GPT4TS	1D	0.0473	0.0949	0.1828	0.0977	0.0819	0.0735	0.1124	0.1921	0.1103
	3D	0.1163	0.1995	0.3605	0.2009	0.1762	0.1251	0.1953	0.3751	0.2186	
	5D	0.1908	0.2827	0.4853	0.2619	0.2654	0.1534	0.2487	0.4691	0.2947	
	7D	0.2394	0.3739	0.5955	0.3207	0.3350	0.1826	0.2946	0.5944	0.3670	
	Avg.	0.1484	0.2377	0.4060	0.2203	0.2146	0.1337	0.2127	0.4077	0.2477	
AutoTimes	1D	0.0375	0.0718	0.1582	0.0879	0.0687	0.0581	0.0900	0.1373	0.0887	
	3D	0.0889	0.1438	0.3015	0.1598	0.1410	0.0984	0.1615	0.2986	0.1742	
	5D	0.1406	0.2011	0.3952	0.2175	0.2057	0.1265	0.2115	0.4012	0.2374	
	7D	0.1944	0.2496	0.4659	0.2653	0.2754	0.1485	0.2510	0.4759	0.2908	
	Avg.	0.1154	0.1666	0.3302	0.1826	0.1727	0.1079	0.1785	0.3283	0.1978	

Table 21: Benchmark(SEDI10) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.8818	0.7184	0.7414	0.7247	0.8200	0.3737	0.3398	0.5508	0.6438
	3D	0.7665	0.5503	0.5961	0.5659	0.6510	0.1472	0.2172	0.4868	0.4976
	5D	0.5602	0.5282	0.4028	0.4240	0.5721	0.1895	0.1261	0.2954	0.3873
	7D	0.5497	0.3427	0.3481	0.3862	0.4955	0.1744	0.1154	0.2842	0.3370
	Avg.	0.6895	0.5349	0.5221	0.5252	0.6347	0.2212	0.1996	0.4043	0.4664
MLP	1D	0.8058	0.7257	0.7262	0.6834	0.8167	0.3643	0.3446	0.6240	0.6363
	3D	0.7185	0.6182	0.6290	0.5804	0.7417	0.2687	0.2462	0.4971	0.5375
	5D	0.6519	0.5507	0.5684	0.5174	0.6849	0.2191	0.1954	0.4089	0.4746
	7D	0.5975	0.5004	0.5246	0.4770	0.6386	0.1878	0.1653	0.3525	0.4305
	Avg.	0.6934	0.5988	0.6120	0.5645	0.7205	0.2600	0.2379	0.4706	0.5197
NLinear	1D	0.8288	0.7478	0.7501	0.7158	0.8417	0.3967	0.3814	0.6633	0.6657
	3D	0.7342	0.6294	0.6440	0.5981	0.7567	0.2817	0.2669	0.5051	0.5520
	5D	0.6635	0.5558	0.5801	0.5287	0.6958	0.2278	0.2130	0.4102	0.4844
	7D	0.6061	0.5082	0.5352	0.4845	0.6486	0.1949	0.1812	0.3478	0.4383
	Avg.	0.7081	0.6103	0.6274	0.5818	0.7357	0.2753	0.2606	0.4816	0.5351
TCN	1D	0.0431	0.6078	0.3406	0.0637	0.2487	0.0808	0.0715	0.3472	0.2254
	3D	0.0567	0.5585	0.2838	0.0416	0.1401	0.0277	0.0530	0.2676	0.1786
	5D	0.0184	0.5116	0.2562	0.0373	0.1397	0.0250	0.0363	0.2671	0.1614
	7D	0.0186	0.4692	0.2239	0.0317	0.0847	0.0265	0.0212	0.2158	0.1365
	Avg.	0.0342	0.5368	0.2761	0.0436	0.1533	0.0400	0.0455	0.2744	0.1755
CNN	1D	0.8640	0.7711	0.7702	0.7476	0.8616	0.4340	0.4051	0.6863	0.6925
	3D	0.7784	0.6467	0.6583	0.6233	0.7737	0.3144	0.2890	0.5221	0.5757
	5D	0.7120	0.5774	0.5855	0.5506	0.7132	0.2545	0.2342	0.4435	0.5089
	7D	0.6548	0.5222	0.5345	0.5048	0.6640	0.2148	0.1971	0.3859	0.4598
	Avg.	0.7523	0.6294	0.6371	0.6066	0.7531	0.3044	0.2814	0.5094	0.5592
TimesNet	1D	0.8337	0.7222	0.7209	0.7025	0.8136	0.3596	0.3287	0.6174	0.6373
	3D	0.7562	0.6169	0.6290	0.5857	0.7337	0.2810	0.2499	0.4779	0.5413
	5D	0.6648	0.5550	0.5633	0.5488	0.6977	0.2211	0.2120	0.3950	0.4822
	7D	0.6015	0.5281	0.5277	0.5205	0.6550	0.1957	0.1803	0.3386	0.4434
	Avg.	0.7141	0.6056	0.6102	0.5894	0.7250	0.2643	0.2427	0.4572	0.5261
Transformer	1D	0.8659	0.7631	0.7879	0.7573	0.8621	0.4461	0.4166	0.6868	0.6982
	3D	0.7734	0.6422	0.6638	0.6423	0.7706	0.3221	0.2848	0.5011	0.5750
	5D	0.6801	0.5696	0.5953	0.5755	0.7140	0.2659	0.2376	0.4227	0.5076
	7D	0.6419	0.5160	0.5492	0.5275	0.6594	0.2241	0.1957	0.3605	0.4593
	Avg.	0.7403	0.6227	0.6490	0.6256	0.7515	0.3145	0.2837	0.4928	0.5600
	1D	0.8590	0.7637	0.7698	0.7506	0.8585	0.4215	0.3905	0.6882	0.6877
	3D	0.7758	0.6388	0.6537	0.6250	0.7703	0.2996	0.2812	0.5160	0.5701
	5D	0.7032	0.5647	0.5871	0.5635	0.7103	0.2452	0.2350	0.4122	0.5026
	7D	0.6466	0.5090	0.5363	0.5138	0.6570	0.2072	0.1959	0.3525	0.4523
	Avg.	0.7462	0.6190	0.6367	0.6132	0.7490	0.2934	0.2756	0.4922	0.5532

Table 22: Benchmark(SED110) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.8083	0.6454	0.6079	0.5562	0.7041	0.2763	0.2681	0.4926	0.5449
	3D	0.5921	0.4417	0.4968	0.4188	0.5596	0.1513	0.1253	0.2533	0.3799
	5D	0.5695	0.3029	0.3213	0.2700	0.4679	0.0608	0.0898	0.1559	0.2798
	7D	0.4313	0.2660	0.2389	0.2622	0.3556	0.0713	0.0706	0.1512	0.2309
	Avg.	0.6003	0.4140	0.4162	0.3768	0.5218	0.1399	0.1385	0.2632	0.3588
RNN DeepAR	1D	0.7362	0.6359	0.6459	0.6021	0.7733	0.2473	0.2189	0.4341	0.5367
	3D	0.5426	0.4889	0.4593	0.3482	0.5423	0.1834	0.1369	0.2752	0.3721
	5D	0.4483	0.3053	0.3537	0.2949	0.4634	0.1024	0.0758	0.1629	0.2758
	7D	0.4221	0.2386	0.2580	0.2948	0.3905	0.0559	0.0651	0.1325	0.2322
	Avg.	0.5373	0.4172	0.4292	0.3850	0.5424	0.1472	0.1242	0.2512	0.3542
DilatedRNN	1D	0.8506	0.7323	0.7212	0.6417	0.8066	0.3305	0.3625	0.6252	0.6338
	3D	0.6793	0.5691	0.5292	0.5888	0.6736	0.2884	0.2448	0.4514	0.5031
	5D	0.6265	0.4423	0.4842	0.4554	0.5612	0.1994	0.1708	0.3447	0.4106
	7D	0.5085	0.3899	0.4256	0.3499	0.5194	0.1390	0.1381	0.3058	0.3470
	Avg.	0.6662	0.5334	0.5401	0.5090	0.6402	0.2393	0.2290	0.4318	0.4736
GCN	1D	0.6518	0.5570	0.4700	0.4942	0.6120	0.1855	0.1511	0.3621	0.4354
	3D	0.5192	0.4007	0.3853	0.3893	0.5453	0.1505	0.0687	0.2266	0.3357
	5D	0.4771	0.3253	0.3348	0.3428	0.4210	0.0771	0.0502	0.2179	0.2808
	7D	0.3201	0.3215	0.2548	0.2575	0.3693	0.0644	0.0448	0.1812	0.2267
	Avg.	0.4920	0.4011	0.3612	0.3709	0.4869	0.1194	0.0787	0.2469	0.3197
GNN FourierGNN	1D	0.8202	0.6697	0.7106	0.5979	0.8159	0.3118	0.3331	0.5605	0.6025
	3D	0.7025	0.4850	0.5521	0.5582	0.6706	0.2101	0.1943	0.3597	0.4665
	5D	0.6813	0.4581	0.4406	0.4683	0.5734	0.1673	0.1316	0.3081	0.4036
	7D	0.3345	0.2409	0.0220	0.2379	0.2917	0.0836	0.0000	0.1621	0.1716
	Avg.	0.6346	0.4634	0.4313	0.4656	0.5879	0.1932	0.1648	0.3476	0.4111
StemGNN	1D	0.8138	0.6649	0.7160	0.7050	0.7574	0.3779	0.3439	0.4236	0.6003
	3D	0.6362	0.3999	0.5483	0.6300	0.3625	0.1748	0.2228	0.3599	0.4168
	5D	0.5210	0.2598	0.4265	0.4668	0.2569	0.1445	0.1705	0.2618	0.3135
	7D	0.5100	0.2602	0.3649	0.5142	0.2265	0.1454	0.1180	0.1969	0.2920
	Avg.	0.6203	0.3962	0.5139	0.5790	0.4008	0.2106	0.2138	0.3105	0.4056
LLM GPT4TS	1D	0.8319	0.7426	0.7364	0.7378	0.8351	0.3639	0.3322	0.6284	0.6511
	3D	0.7633	0.6308	0.6306	0.6308	0.7516	0.2830	0.2576	0.4905	0.5548
	5D	0.6801	0.5679	0.5677	0.5609	0.6889	0.2466	0.2190	0.4091	0.4925
	7D	0.6033	0.5121	0.5246	0.5080	0.6128	0.2084	0.2058	0.3591	0.4418
	Avg.	0.7196	0.6134	0.6148	0.6094	0.7221	0.2755	0.2537	0.4718	0.5350
AutoTimes	1D	0.8531	0.7521	0.7430	0.7179	0.8305	0.3906	0.3425	0.6501	0.6600
	3D	0.7662	0.6229	0.6394	0.6149	0.7558	0.2781	0.2571	0.4953	0.5537
	5D	0.7053	0.5431	0.5788	0.5370	0.6996	0.2228	0.2107	0.3837	0.4851
	7D	0.6394	0.4979	0.5350	0.4873	0.6540	0.1912	0.1722	0.3345	0.4390
	Avg.	0.7410	0.6040	0.6240	0.5893	0.7350	0.2707	0.2456	0.4659	0.5344

Table 23: Benchmark(SED15) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.8342	0.6509	0.7189	0.5818	0.8104	0.1846	0.2289	0.4477	0.5572
	3D	0.6624	0.4428	0.5498	0.4211	0.5509	0.0714	0.1161	0.3521	0.3958
	5D	0.4035	0.4136	0.3370	0.2744	0.4775	0.0775	0.0608	0.2023	0.2808
	7D	0.3444	0.2324	0.2942	0.2387	0.3680	0.0719	0.0487	0.1784	0.2221
	Avg.	0.5611	0.4349	0.4750	0.3790	0.5517	0.1014	0.1136	0.2951	0.3640
MLP	1D	0.7236	0.6638	0.7095	0.5405	0.7799	0.1768	0.2334	0.4945	0.5403
	3D	0.5999	0.5424	0.6030	0.4380	0.6965	0.1204	0.1453	0.3640	0.4387
	5D	0.5054	0.4671	0.5328	0.3770	0.6327	0.0929	0.1048	0.2848	0.3747
	7D	0.4306	0.4096	0.4805	0.3363	0.5808	0.0777	0.0841	0.2368	0.3296
	Avg.	0.5649	0.5207	0.5815	0.4230	0.6725	0.1170	0.1419	0.3450	0.4208
NLinear	1D	0.7534	0.6911	0.7338	0.5774	0.8137	0.2061	0.2718	0.5285	0.5720
	3D	0.6184	0.5555	0.6175	0.4565	0.7167	0.1345	0.1656	0.3748	0.4550
	5D	0.5186	0.4731	0.5441	0.3893	0.6472	0.1032	0.1217	0.2885	0.3857
	7D	0.4410	0.4185	0.4908	0.3459	0.5935	0.0863	0.0983	0.2338	0.3385
	Avg.	0.5829	0.5345	0.5965	0.4423	0.6928	0.1325	0.1644	0.3564	0.4378
TCN	1D	0.0118	0.5797	0.3884	0.0267	0.1376	0.0484	0.0211	0.2944	0.1885
	3D	0.0040	0.5137	0.3141	0.0246	0.0547	0.0226	0.0160	0.2058	0.1444
	5D	0.0006	0.4627	0.2702	0.0261	0.0496	0.0176	0.0127	0.1736	0.1266
	7D	0.0017	0.4034	0.2274	0.0168	0.0263	0.0212	0.0049	0.1324	0.1043
	Avg.	0.0045	0.4899	0.3000	0.0235	0.0670	0.0274	0.0137	0.2016	0.1410
CNN	1D	0.8093	0.7314	0.7508	0.6299	0.8348	0.2384	0.2924	0.5532	0.6050
	3D	0.6845	0.5954	0.6251	0.4936	0.7366	0.1714	0.1852	0.3895	0.4852
	5D	0.5898	0.5139	0.5470	0.4167	0.6707	0.1357	0.1424	0.3192	0.4169
	7D	0.5054	0.4468	0.4917	0.3746	0.6193	0.1136	0.1166	0.2659	0.3667
	Avg.	0.6472	0.5719	0.6037	0.4787	0.7154	0.1648	0.1841	0.3820	0.4685
TimesNet	1D	0.7676	0.6709	0.6935	0.5627	0.7861	0.2032	0.2121	0.4738	0.5462
	3D	0.6634	0.5501	0.5961	0.4573	0.6974	0.1583	0.1456	0.3630	0.4539
	5D	0.5322	0.4809	0.5175	0.4121	0.6554	0.1097	0.1187	0.2822	0.3886
	7D	0.4505	0.4466	0.4755	0.3817	0.6156	0.0908	0.0969	0.2365	0.3493
	Avg.	0.6034	0.5371	0.5706	0.4535	0.6886	0.1405	0.1433	0.3389	0.4345
Transformer	1D	0.8159	0.7126	0.7751	0.6343	0.8312	0.2433	0.3077	0.5579	0.6098
	3D	0.6860	0.5796	0.6393	0.5059	0.7310	0.1663	0.1820	0.3744	0.4831
	5D	0.5466	0.4946	0.5556	0.4381	0.6620	0.1345	0.1418	0.3024	0.4094
	7D	0.5014	0.4312	0.5017	0.3931	0.6026	0.1117	0.1071	0.2477	0.3621
	Avg.	0.6375	0.5545	0.6179	0.4928	0.7067	0.1640	0.1847	0.3706	0.4661
PatchTST	1D	0.7949	0.7187	0.7563	0.6163	0.8293	0.2320	0.2837	0.5581	0.5987
	3D	0.6819	0.5758	0.6293	0.4861	0.7305	0.1552	0.1803	0.3896	0.4786
	5D	0.5748	0.4898	0.5514	0.4254	0.6642	0.1178	0.1409	0.2952	0.4074
	7D	0.4961	0.4226	0.4913	0.3773	0.6033	0.0979	0.1098	0.2402	0.3548
	Avg.	0.6369	0.5517	0.6071	0.4763	0.7068	0.1507	0.1787	0.3708	0.4599

Table 24: Benchmark(SED15) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.7332	0.5713	0.5663	0.3973	0.6438	0.1152	0.1544	0.4112	0.4491
	3D	0.4665	0.3101	0.4458	0.2647	0.4646	0.0569	0.0435	0.1711	0.2779
	5D	0.4118	0.1630	0.2707	0.1366	0.3549	0.0219	0.0237	0.0855	0.1835
	7D	0.2363	0.1584	0.1859	0.1436	0.2318	0.0164	0.0116	0.0808	0.1331
	Avg.	0.4619	0.3007	0.3672	0.2356	0.4238	0.0526	0.0583	0.1872	0.2609
RNN DeepAR	1D	0.6083	0.5414	0.6110	0.4122	0.7179	0.1127	0.1176	0.3538	0.4344
	3D	0.3862	0.3916	0.4057	0.1641	0.4000	0.0645	0.0464	0.1911	0.2562
	5D	0.2697	0.1970	0.2997	0.1120	0.3417	0.0330	0.0206	0.0904	0.1705
	7D	0.2144	0.1297	0.2115	0.1583	0.2708	0.0149	0.0153	0.0675	0.1353
	Avg.	0.3697	0.3150	0.3820	0.2116	0.4326	0.0563	0.0500	0.1757	0.2491
DilatedRNN	1D	0.7899	0.6688	0.7067	0.5014	0.7638	0.1727	0.2554	0.5057	0.5455
	3D	0.5246	0.4989	0.4769	0.4444	0.5866	0.1338	0.1463	0.3257	0.3921
	5D	0.5002	0.3712	0.4380	0.3043	0.4636	0.0808	0.0983	0.2348	0.3114
	7D	0.3440	0.3058	0.3608	0.2246	0.4235	0.0556	0.0727	0.1851	0.2465
	Avg.	0.5397	0.4612	0.4956	0.3687	0.5594	0.1107	0.1432	0.3128	0.3739
GNN GCN	1D	0.5082	0.4420	0.4160	0.3446	0.5456	0.0585	0.0730	0.2261	0.3267
	3D	0.3708	0.2814	0.3344	0.2126	0.4741	0.0322	0.0285	0.1289	0.2329
	5D	0.3660	0.2239	0.2841	0.1705	0.3472	0.0283	0.0190	0.1080	0.1934
	7D	0.2101	0.2369	0.1928	0.1174	0.3013	0.0214	0.0139	0.0751	0.1461
	Avg.	0.3638	0.2960	0.3068	0.2113	0.4171	0.0351	0.0336	0.1345	0.2248
GNN FourierGNN	1D	0.7366	0.5968	0.6861	0.4622	0.7762	0.1503	0.2150	0.4446	0.5085
	3D	0.5907	0.3768	0.5108	0.4087	0.6033	0.0794	0.1057	0.2671	0.3678
	5D	0.5702	0.3447	0.3859	0.3252	0.4975	0.0653	0.0574	0.2065	0.3066
	7D	0.1979	0.1722	0.0218	0.1190	0.1703	0.0245	0.0000	0.1025	0.1010
	Avg.	0.5239	0.3726	0.4012	0.3288	0.5118	0.0799	0.0945	0.2552	0.3210
GNN StemGNN	1D	0.7373	0.5495	0.6823	0.5692	0.6799	0.1761	0.2236	0.3510	0.4961
	3D	0.4293	0.2531	0.4840	0.4764	0.2654	0.0843	0.1171	0.2404	0.2938
	5D	0.3508	0.1486	0.3510	0.2997	0.1361	0.0596	0.0929	0.1794	0.2023
	7D	0.3049	0.1494	0.2887	0.3710	0.1265	0.0532	0.0529	0.1182	0.1831
	Avg.	0.4556	0.2752	0.4515	0.4291	0.3020	0.0933	0.1216	0.2223	0.2938
LLM GPT4TS	1D	0.7690	0.6909	0.7162	0.5834	0.8045	0.1787	0.2197	0.4808	0.5554
	3D	0.6664	0.5792	0.5939	0.4798	0.7217	0.1361	0.1662	0.3546	0.4622
	5D	0.5390	0.5007	0.5301	0.4063	0.6407	0.1114	0.1208	0.2802	0.3912
	7D	0.4387	0.4361	0.4769	0.3686	0.5478	0.0865	0.1206	0.2392	0.3393
	Avg.	0.6033	0.5517	0.5793	0.4595	0.6787	0.1282	0.1568	0.3387	0.4370
LLM AutoTimes	1D	0.7971	0.7077	0.7325	0.5873	0.7981	0.2034	0.2337	0.5131	0.5716
	3D	0.6631	0.5604	0.6141	0.4775	0.7168	0.1289	0.1570	0.3674	0.4606
	5D	0.5817	0.4669	0.5437	0.3959	0.6487	0.1001	0.1188	0.2663	0.3903
	7D	0.4821	0.4121	0.4920	0.3482	0.5998	0.0813	0.0891	0.2221	0.3408
	Avg.	0.6310	0.5368	0.5956	0.4522	0.6908	0.1284	0.1497	0.3422	0.4408

Table 25: Benchmark(SED1) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.6573	0.5100	0.6316	0.3043	0.7196	0.0689	0.0676	0.2726	0.4040	
	3D	0.3895	0.2703	0.3182	0.1519	0.1471	0.0237	0.0293	0.1641	0.1868	
	5D	0.1531	0.2351	0.1278	0.0776	0.1391	0.0239	0.0150	0.0801	0.1065	
	7D	0.1288	0.1139	0.0992	0.0620	0.0954	0.0233	0.0115	0.0621	0.0745	
	Avg.	0.3322	0.2823	0.2942	0.1489	0.2753	0.0349	0.0308	0.1447	0.1929	
MLP	1D	0.5207	0.5562	0.6165	0.2440	0.6082	0.0591	0.0721	0.2813	0.3698	
	3D	0.3807	0.3950	0.4371	0.1491	0.4736	0.0313	0.0367	0.1506	0.2567	
	5D	0.2848	0.2999	0.3244	0.1064	0.3780	0.0208	0.0217	0.1009	0.1921	
	7D	0.2143	0.2278	0.2535	0.0832	0.3032	0.0154	0.0159	0.0762	0.1487	
	Avg.	0.3501	0.3697	0.4079	0.1457	0.4408	0.0316	0.0366	0.1522	0.2418	
NLinear	1D	0.5611	0.5933	0.6453	0.2940	0.6585	0.0746	0.1005	0.3234	0.4063	
	3D	0.4026	0.4130	0.4504	0.1715	0.4983	0.0376	0.0481	0.1656	0.2734	
	5D	0.2988	0.3081	0.3325	0.1193	0.3940	0.0249	0.0293	0.1075	0.2018	
	7D	0.2243	0.2382	0.2600	0.0924	0.3169	0.0187	0.0214	0.0785	0.1563	
	Avg.	0.3717	0.3882	0.4220	0.1693	0.4669	0.0390	0.0498	0.1688	0.2595	
TCN	1D	0.0066	0.5271	0.4001	0.0007	0.0213	0.0162	0.0032	0.1545	0.1412	
	3D	0.0015	0.4500	0.2571	0.0007	0.0091	0.0013	0.0009	0.0784	0.0999	
	5D	0.0012	0.3930	0.2123	0.0052	0.0086	0.0016	0.0007	0.0595	0.0853	
	7D	0.0008	0.3182	0.1512	0.0014	0.0040	0.0011	0.0002	0.0381	0.0644	
	Avg.	0.0025	0.4221	0.2552	0.0020	0.0107	0.0050	0.0013	0.0826	0.0977	
CNN	1D	0.6485	0.6556	0.6666	0.3565	0.7208	0.1142	0.1183	0.3829	0.4579	
	3D	0.5067	0.4810	0.4757	0.2192	0.5761	0.0708	0.0628	0.2076	0.3250	
	5D	0.4065	0.3772	0.3571	0.1551	0.4734	0.0500	0.0432	0.1545	0.2521	
	7D	0.3154	0.2962	0.2908	0.1239	0.3883	0.0352	0.0348	0.1236	0.2010	
	Avg.	0.4693	0.4525	0.4475	0.2137	0.5396	0.0676	0.0648	0.2171	0.3090	
TimesNet	1D	0.5695	0.5793	0.5418	0.2885	0.6165	0.0793	0.0665	0.3141	0.3820	
	3D	0.4430	0.4012	0.3915	0.1719	0.4484	0.0588	0.0338	0.1733	0.2652	
	5D	0.3449	0.3556	0.3061	0.1343	0.3016	0.0396	0.0268	0.1137	0.2028	
	7D	0.2569	0.3039	0.2662	0.1502	0.2378	0.0329	0.0171	0.0836	0.1686	
	Avg.	0.4036	0.4100	0.3764	0.1862	0.4011	0.0527	0.0360	0.1712	0.2546	
Transformer	iTransformer	1D	0.6414	0.6492	0.7335	0.3589	0.7157	0.1346	0.1328	0.3828	0.4686
		3D	0.4722	0.4587	0.5177	0.2267	0.5546	0.0652	0.0535	0.1698	0.3148
		5D	0.3254	0.3369	0.3727	0.1694	0.4024	0.0424	0.0432	0.1264	0.2273
		7D	0.2713	0.2676	0.2885	0.1307	0.3039	0.0351	0.0263	0.1016	0.1781
		Avg.	0.4276	0.4281	0.4781	0.2214	0.4941	0.0693	0.0639	0.1952	0.2972
	PatchTST	1D	0.6252	0.6566	0.6956	0.3576	0.7156	0.1057	0.1119	0.3680	0.4545
		3D	0.4853	0.4471	0.4602	0.2096	0.5551	0.0559	0.0550	0.1808	0.3061
		5D	0.3605	0.3380	0.3390	0.1545	0.4579	0.0366	0.0385	0.1168	0.2302
		7D	0.2783	0.2549	0.2570	0.1169	0.3595	0.0270	0.0272	0.0890	0.1762
		Avg.	0.4373	0.4241	0.4380	0.2096	0.5220	0.0563	0.0582	0.1887	0.2918

Table 26: Benchmark(SED11) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.4291	0.4087	0.3496	0.1019	0.2429	0.0293	0.0425	0.2103	0.2268
	3D	0.2253	0.1166	0.1295	0.0430	0.0740	0.0102	0.0015	0.0263	0.0783
	5D	0.1955	0.0355	0.0244	0.0035	0.0392	0.0068	0.0001	0.0118	0.0396
	7D	0.0624	0.0424	0.0114	0.0123	0.0152	0.0096	0.0000	0.0172	0.0213
	Avg.	0.2281	0.1508	0.1287	0.0402	0.0928	0.0140	0.0110	0.0664	0.0915
RNN DeepAR	1D	0.2812	0.3768	0.3740	0.1118	0.3160	0.0380	0.0321	0.1645	0.2118
	3D	0.1413	0.1842	0.0789	0.0012	0.0224	0.0113	0.0020	0.0412	0.0603
	5D	0.0638	0.0405	0.0379	0.0001	0.0261	0.0098	0.0000	0.0097	0.0235
	7D	0.0509	0.0226	0.0147	0.0099	0.0078	0.0038	0.0000	0.0086	0.0148
	Avg.	0.1343	0.1560	0.1263	0.0307	0.0931	0.0157	0.0085	0.0560	0.0776
DilatedRNN	1D	0.5056	0.5262	0.6002	0.2310	0.4296	0.0608	0.0933	0.2888	0.3419
	3D	0.2602	0.3661	0.2122	0.2188	0.2341	0.0468	0.0419	0.1524	0.1916
	5D	0.2760	0.2381	0.2124	0.1143	0.1930	0.0294	0.0254	0.0943	0.1479
	7D	0.1493	0.1823	0.1490	0.0520	0.1680	0.0163	0.0192	0.0660	0.1003
	Avg.	0.2978	0.3282	0.2934	0.1540	0.2562	0.0383	0.0450	0.1504	0.1954
GCN	1D	0.3231	0.2377	0.2717	0.0811	0.3542	0.0041	0.0109	0.1084	0.1739
	3D	0.2107	0.1584	0.1549	0.0344	0.1992	0.0013	0.0017	0.0220	0.0978
	5D	0.2340	0.1057	0.1296	0.0405	0.1187	0.0016	0.0006	0.0149	0.0807
	7D	0.1392	0.1227	0.0528	0.0238	0.1049	0.0005	0.0007	0.0087	0.0567
	Avg.	0.2267	0.1561	0.1523	0.0449	0.1942	0.0019	0.0035	0.0385	0.1023
GNN FourierGNN	1D	0.4777	0.4612	0.5958	0.1421	0.5984	0.0562	0.0684	0.2740	0.3342
	3D	0.3302	0.2082	0.2986	0.1587	0.2871	0.0221	0.0272	0.1077	0.1800
	5D	0.3435	0.1698	0.1615	0.1190	0.1834	0.0178	0.0135	0.0668	0.1344
	7D	0.0674	0.1128	0.0205	0.0193	0.0121	0.0065	0.0000	0.0385	0.0346
	Avg.	0.3047	0.2380	0.2691	0.1098	0.2702	0.0256	0.0273	0.1218	0.1708
StemGNN	1D	0.3696	0.2974	0.4643	0.3041	0.2775	0.0648	0.0807	0.2241	0.2603
	3D	0.1062	0.0767	0.1976	0.2546	0.1082	0.0213	0.0238	0.0741	0.1078
	5D	0.0501	0.0430	0.1616	0.1482	0.0252	0.0109	0.0207	0.0624	0.0653
	7D	0.0836	0.0342	0.0738	0.2426	0.0306	0.0089	0.0087	0.0357	0.0647
	Avg.	0.1524	0.1128	0.2243	0.2374	0.1104	0.0265	0.0335	0.0991	0.1245
LLM GPT4TS	1D	0.5969	0.6126	0.6000	0.3151	0.6995	0.0589	0.0816	0.3198	0.4106
	3D	0.4723	0.4811	0.4031	0.1917	0.5157	0.0371	0.0536	0.1539	0.2886
	5D	0.3476	0.3804	0.3272	0.1408	0.3431	0.0286	0.0259	0.0843	0.2098
	7D	0.2432	0.3079	0.2592	0.1095	0.2058	0.0258	0.0320	0.0852	0.1586
	Avg.	0.4150	0.4455	0.3974	0.1893	0.4410	0.0376	0.0483	0.1608	0.2669
AutoTimes	1D	0.6320	0.6440	0.6541	0.3166	0.6710	0.0912	0.0855	0.3150	0.4262
	3D	0.4754	0.4335	0.4585	0.1965	0.5345	0.0417	0.0434	0.1679	0.2939
	5D	0.3847	0.3112	0.3382	0.1197	0.4343	0.0285	0.0265	0.0990	0.2178
	7D	0.2695	0.2412	0.2606	0.0943	0.3648	0.0192	0.0180	0.0764	0.1680
	Avg.	0.4404	0.4075	0.4278	0.1818	0.5012	0.0451	0.0433	0.1646	0.2765

Table 27: Benchmark(MAE) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.0251	0.0246	0.0207	0.0463	0.0302	0.0628	0.0505	0.0380	0.0373	
	3D	0.0526	0.0607	0.0523	0.0777	0.0603	0.1228	0.0977	0.0757	0.0750	
	5D	0.0839	0.0940	0.0814	0.1102	0.0958	0.1610	0.1305	0.1105	0.1084	
	7D	0.1005	0.1202	0.1013	0.1327	0.1143	0.1891	0.1558	0.1281	0.1303	
	Avg.	0.0655	0.0749	0.0639	0.0917	0.0752	0.1339	0.1086	0.0881	0.0877	
MLP	1D	0.0247	0.0327	0.0265	0.0477	0.0353	0.0684	0.0548	0.0461	0.0420	
	3D	0.0503	0.0683	0.0560	0.0821	0.0665	0.1226	0.0987	0.0811	0.0782	
	5D	0.0734	0.0989	0.0827	0.1093	0.0899	0.1596	0.1307	0.1108	0.1069	
	7D	0.0946	0.1254	0.1081	0.1330	0.1100	0.1873	0.1561	0.1368	0.1314	
	Avg.	0.0608	0.0813	0.0683	0.0930	0.0754	0.1345	0.1101	0.0937	0.0896	
NLinear	1D	0.0184	0.0243	0.0200	0.0427	0.0307	0.0615	0.0480	0.0377	0.0354	
	3D	0.0463	0.0613	0.0516	0.0798	0.0636	0.1196	0.0947	0.0752	0.0740	
	5D	0.0705	0.0931	0.0796	0.1089	0.0880	0.1577	0.1279	0.1055	0.1039	
	7D	0.0933	0.1207	0.1053	0.1334	0.1078	0.1861	0.1545	0.1322	0.1292	
	Avg.	0.0571	0.0749	0.0641	0.0912	0.0725	0.1312	0.1063	0.0877	0.0856	
TCN	1D	0.0577	0.0505	0.0318	0.0970	0.1129	0.0920	0.2015	0.1758	0.1024	
	3D	0.0968	0.1382	0.0689	0.1048	0.1373	0.1937	0.1485	0.2512	0.1424	
	5D	0.1406	0.1002	0.0931	0.1558	0.1816	0.2363	0.1885	0.2557	0.1690	
	7D	0.2256	0.1780	0.1241	0.1633	0.2116	0.2240	0.2846	0.2676	0.2098	
	Avg.	0.1302	0.1167	0.0795	0.1302	0.1608	0.1865	0.2058	0.2376	0.1559	
CNN	1D	0.0221	0.0265	0.0205	0.0443	0.0278	0.0654	0.0491	0.0357	0.0364	
	3D	0.0567	0.0702	0.0510	0.0861	0.0603	0.1332	0.1010	0.0719	0.0788	
	5D	0.0871	0.1048	0.0834	0.1215	0.0883	0.1774	0.1384	0.1015	0.1128	
	7D	0.1202	0.1362	0.1056	0.1499	0.1094	0.2057	0.1686	0.1280	0.1405	
	Avg.	0.0715	0.0845	0.0651	0.1005	0.0714	0.1454	0.1143	0.0843	0.0921	
TimesNet	1D	0.0199	0.0288	0.0242	0.0513	0.0445	0.0809	0.0651	0.0528	0.0459	
	3D	0.0467	0.0651	0.0558	0.0875	0.0724	0.1426	0.1164	0.0866	0.0841	
	5D	0.0703	0.0970	0.0838	0.1148	0.0988	0.1852	0.1484	0.1127	0.1139	
	7D	0.0923	0.1284	0.1121	0.1438	0.1158	0.2145	0.1759	0.1417	0.1406	
	Avg.	0.0573	0.0798	0.0690	0.0993	0.0829	0.1558	0.1264	0.0984	0.0961	
Transformer	iTransformer	1D	0.0190	0.0253	0.0192	0.0383	0.0278	0.0641	0.0482	0.0345	0.0346
		3D	0.0469	0.0617	0.0487	0.0726	0.0576	0.1244	0.0959	0.0706	0.0723
		5D	0.0709	0.0948	0.0766	0.1023	0.0834	0.1633	0.1331	0.0977	0.1028
		7D	0.0914	0.1226	0.1041	0.1275	0.1048	0.1889	0.1596	0.1214	0.1275
		Avg.	0.0571	0.0761	0.0621	0.0852	0.0684	0.1352	0.1092	0.0811	0.0843
	PatchTST	1D	0.0183	0.0241	0.0177	0.0381	0.0288	0.0637	0.0481	0.0349	0.0342
		3D	0.0464	0.0654	0.0545	0.0765	0.0588	0.1242	0.1007	0.0714	0.0747
		5D	0.0762	0.0951	0.0761	0.1057	0.0835	0.1651	0.1354	0.1014	0.1048
		7D	0.0984	0.1266	0.1032	0.1319	0.1053	0.1955	0.1608	0.1257	0.1309
		Avg.	0.0598	0.0778	0.0629	0.0881	0.0691	0.1371	0.1112	0.0833	0.0862

Table 28: Benchmark(MAE) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.0486	0.0447	0.0393	0.0640	0.0465	0.1120	0.0839	0.0606	0.0625
	3D	0.0895	0.0878	0.0788	0.1048	0.0828	0.1775	0.1299	0.1091	0.1075
	5D	0.1246	0.1247	0.1167	0.1344	0.1172	0.2057	0.1590	0.1281	0.1388
	7D	0.1621	0.1584	0.1496	0.1725	0.1389	0.2203	0.1823	0.1567	0.1676
	Avg.	0.1062	0.1039	0.0961	0.1189	0.0964	0.1789	0.1388	0.1136	0.1191
RNN DeepAR	1D	0.0569	0.0461	0.0381	0.0632	0.0486	0.1023	0.0763	0.0694	0.0626
	3D	0.0859	0.0950	0.0851	0.1180	0.0949	0.1640	0.1403	0.0993	0.1103
	5D	0.1086	0.1219	0.1137	0.1424	0.1388	0.2005	0.1742	0.1466	0.1433
	7D	0.1496	0.1542	0.1549	0.1745	0.1467	0.2247	0.2012	0.1485	0.1693
	Avg.	0.1003	0.1043	0.0980	0.1245	0.1073	0.1729	0.1480	0.1159	0.1214
DilatedRNN	1D	0.0394	0.0268	0.0277	0.0441	0.0472	0.0691	0.0596	0.0372	0.0439
	3D	0.0726	0.0714	0.0679	0.0978	0.0648	0.1390	0.1186	0.0837	0.0895
	5D	0.1100	0.1028	0.0887	0.1257	0.0966	0.2000	0.1578	0.1128	0.1243
	7D	0.1312	0.1500	0.1175	0.1499	0.1234	0.2363	0.1870	0.1478	0.1554
	Avg.	0.0883	0.0877	0.0754	0.1043	0.0830	0.1611	0.1308	0.0954	0.1033
GNN GCN	1D	0.0543	0.0596	0.0264	0.3354	0.1422	0.0789	0.0946	0.1151	0.1133
	3D	0.0858	0.0942	0.0598	0.3643	0.1756	0.1443	0.1374	0.1593	0.1526
	5D	0.1006	0.1192	0.0851	0.3859	0.2002	0.1767	0.1713	0.1952	0.1793
	7D	0.1196	0.1408	0.1141	0.4079	0.2197	0.2033	0.2068	0.2315	0.2055
	Avg.	0.0901	0.1034	0.0713	0.3734	0.1845	0.1508	0.1525	0.1753	0.1627
GNN FourierGNN	1D	10.0224	0.0295	0.0262	0.0462	0.0370	0.0732	0.0543	0.0436	0.0415
	3D	0.0548	0.0673	0.0538	0.0829	0.0784	0.1336	0.1087	0.0841	0.0830
	5D	0.0830	0.0978	0.0814	0.1153	0.0967	0.1668	0.1511	0.1109	0.1129
	7D	0.1064	0.1236	0.1114	0.1452	0.1335	0.2015	0.1823	0.1354	0.1424
	Avg.	0.0666	0.0796	0.0682	0.0974	0.0864	0.1438	0.1241	0.0935	0.0949
GNN StemGNN	1D	0.0636	0.0445	0.0335	0.0635	0.0586	0.1001	0.0599	0.0489	0.0591
	3D	0.1114	0.1001	0.0788	0.1273	0.1028	0.1524	0.1421	0.1408	0.1195
	5D	0.1690	0.1688	0.1022	0.1698	0.1352	0.1980	0.2366	0.2161	0.1745
	7D	0.2094	0.2090	0.1719	0.2016	0.1625	0.2325	0.2514	0.2730	0.2139
	Avg.	0.1384	0.1306	0.0966	0.1406	0.1148	0.1707	0.1725	0.1697	0.1417
LLM GPT4TS	1D	0.0209	0.0287	0.0261	0.0482	0.0407	0.0740	0.0596	0.7614	0.1324
	3D	0.0467	0.0638	0.0553	0.0845	0.0730	0.1431	0.1105	0.0869	0.0830
	5D	0.0702	0.0946	0.0832	0.1141	0.0980	0.1823	0.1455	0.1120	0.1125
	7D	0.0909	0.1231	0.1105	0.1392	0.1160	0.2087	0.1715	0.1397	0.1375
	Avg.	0.0572	0.0776	0.0688	0.0965	0.0819	0.1520	0.1218	0.2750	0.1163
LLM AutoTimes	1D	0.0228	0.0282	0.0199	0.0429	0.0321	0.0680	0.0542	0.0407	0.0386
	3D	0.0475	0.0615	0.0486	0.0760	0.0638	0.1240	0.0993	0.0751	0.0745
	5D	0.0692	0.0965	0.0737	0.1075	0.0892	0.1632	0.1315	0.1056	0.1045
	7D	0.0921	0.1218	0.1038	0.1304	0.1083	0.1900	0.1573	0.1285	0.1290
	Avg.	0.0579	0.0770	0.0615	0.0892	0.0734	0.1363	0.1106	0.0875	0.0867

Table 29: Benchmark(MSE) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.0036	0.0046	0.0052	0.0165	0.0137	0.0368	0.0139	0.0095	0.0130
	3D	0.0156	0.0159	0.0199	0.0359	0.0318	0.0888	0.0374	0.0252	0.0338
	5D	0.0295	0.0300	0.0345	0.0538	0.0486	0.1253	0.0552	0.0426	0.0524
	7D	0.0461	0.0442	0.0494	0.0729	0.0635	0.1537	0.0701	0.0562	0.0695
	Avg.	0.0237	0.0237	0.0272	0.0448	0.0394	0.1011	0.0441	0.0334	0.0422
MLP	1D	0.0047	0.0056	0.0070	0.0182	0.0163	0.0420	0.0159	0.0116	0.0152
	3D	0.0176	0.0174	0.0217	0.0381	0.0353	0.0983	0.0403	0.0280	0.0371
	5D	0.0348	0.0315	0.0372	0.0561	0.0522	0.1410	0.0605	0.0452	0.0573
	7D	0.0548	0.0465	0.0531	0.0740	0.0679	0.1735	0.0778	0.0624	0.0763
	Avg.	0.0280	0.0253	0.0297	0.0466	0.0429	0.1137	0.0486	0.0368	0.0465
MLP	1D	0.0031	0.0044	0.0050	0.0174	0.0158	0.0392	0.0138	0.0098	0.0136
	3D	0.0153	0.0159	0.0202	0.0378	0.0349	0.0970	0.0383	0.0263	0.0357
	5D	0.0321	0.0298	0.0362	0.0561	0.0518	0.1392	0.0581	0.0433	0.0558
	7D	0.0520	0.0449	0.0526	0.0742	0.0677	0.1724	0.0754	0.0607	0.0750
	Avg.	0.0256	0.0238	0.0285	<u>0.0464</u>	0.0426	0.1120	<u>0.0464</u>	0.0350	0.0450
TCN	1D	0.0925	0.0196	0.0078	0.0383	0.0743	0.0452	0.0900	0.1310	0.0623
	3D	0.1029	0.1063	0.0244	0.0440	0.0804	0.1231	0.0514	0.1998	0.0916
	5D	0.1186	0.0304	0.0379	0.0681	0.1249	0.1691	0.0767	0.2183	0.1055
	7D	0.1905	0.1158	0.0575	0.0808	0.1354	0.1655	0.1613	0.1934	0.1375
	Avg.	0.1261	0.0680	0.0319	0.0578	0.1038	0.1257	0.0949	0.1856	0.0992
CNN	1D	0.0048	0.0050	0.0045	0.0208	0.0142	0.0451	0.0148	0.0100	0.0149
	3D	0.0219	0.0210	0.0187	0.0451	0.0345	0.1297	0.0455	0.0276	0.0430
	5D	0.0466	0.0406	0.0363	0.0757	0.0548	0.1949	0.0722	0.0451	0.0708
	7D	0.0710	0.0626	0.0527	0.1014	0.0705	0.2231	0.0981	0.0619	0.0927
	Avg.	0.0361	0.0323	0.0280	0.0607	0.0435	0.1482	0.0576	0.0362	0.0553
CNN	1D	0.0035	0.0052	0.0066	0.0215	0.0262	0.0586	0.0206	0.0163	0.0198
	3D	0.0159	0.0181	0.0214	0.0461	0.0445	0.1318	0.0553	0.0331	0.0458
	5D	0.0315	0.0351	0.0369	0.0654	0.0684	0.2087	0.0811	0.0509	0.0723
	7D	0.0514	0.0543	0.0537	0.0919	0.0871	0.2343	0.1012	0.0706	0.0931
	Avg.	0.0256	0.0282	0.0297	0.0562	0.0565	0.1584	0.0645	0.0427	0.0577
Transformer	1D	0.0035	0.0047	0.0050	0.0177	0.0157	0.0436	0.0140	0.0094	0.0142
	3D	0.0171	0.0171	0.0209	0.0403	0.0352	0.1103	0.0410	0.0255	0.0384
	5D	0.0343	0.0335	0.0356	0.0623	0.0527	0.1582	0.0674	0.0423	0.0608
	7D	0.0518	0.0522	0.0539	0.0807	0.0714	0.1897	0.0847	0.0590	0.0804
	Avg.	0.0267	0.0269	0.0288	0.0503	0.0437	0.1254	0.0518	<u>0.0341</u>	0.0485
Transformer	1D	0.0032	0.0048	0.0044	0.0179	0.0154	0.0444	0.0144	0.0096	0.0143
	3D	0.0161	0.0205	0.0212	0.0420	0.0353	0.1100	0.0432	0.0266	0.0393
	5D	0.0385	0.0364	0.0374	0.0608	0.0531	0.1632	0.0668	0.0438	0.0625
	7D	0.0564	0.0578	0.0514	0.0832	0.0700	0.2020	0.0825	0.0598	0.0829
	Avg.	0.0285	0.0299	0.0286	0.0510	0.0435	0.1299	0.0517	0.0350	0.0498

Table 30: Benchmark(MSE) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.0183	0.0087	0.0109	0.0287	0.0217	0.0748	0.0274	0.0181	0.0261
	3D	0.0420	0.0271	0.0312	0.0588	0.0444	0.1455	0.0540	0.0381	0.0551
	5D	0.0624	0.0427	0.0506	0.0774	0.0630	0.1761	0.0717	0.0552	0.0749
	7D	0.0949	0.0621	0.0773	0.0985	0.0797	0.2013	0.0880	0.0714	0.0967
	Avg.	0.0544	0.0352	0.0425	0.0659	0.0522	0.1494	0.0603	0.0457	0.0632
RNN DeepAR	1D	0.0408	0.0090	0.0108	0.0298	0.0224	0.0750	0.0266	0.0173	0.0290
	3D	0.0634	0.0251	0.0316	0.0568	0.0463	0.1353	0.0580	0.0370	0.0567
	5D	0.0682	0.0414	0.0506	0.0743	0.0642	0.1775	0.0754	0.0564	0.0760
	7D	0.0910	0.0569	0.0728	0.0969	0.0799	0.1947	0.0899	0.0703	0.0941
	Avg.	0.0659	0.0331	0.0414	0.0644	0.0532	0.1457	0.0625	0.0453	0.0639
DilatedRNN	1D	0.0172	0.0048	0.0056	0.0194	0.0162	0.0408	0.0175	0.0104	0.0165
	3D	0.0395	0.0193	0.0225	0.0442	0.0356	0.1128	0.0489	0.0300	0.0441
	5D	0.0583	0.0347	0.0401	0.0675	0.0560	0.1787	0.0786	0.0509	0.0706
	7D	0.0808	0.0583	0.0578	0.0958	0.0747	0.2167	0.0990	0.0714	0.0943
	Avg.	0.0489	0.0293	0.0315	0.0567	0.0457	0.1372	0.0610	0.0407	0.0564
GCN	1D	0.0108	0.0094	0.0060	0.3682	0.0982	0.0392	0.0227	0.0923	0.0809
	3D	0.0267	0.0246	0.0196	0.3854	0.1184	0.0918	0.0472	0.1217	0.1044
	5D	0.0434	0.0359	0.0339	0.4006	0.1337	0.1285	0.0674	0.1440	0.1234
	7D	0.0569	0.0499	0.0486	0.4154	0.1489	0.1565	0.0868	0.1794	0.1428
	Avg.	0.0344	0.0300	0.0270	0.3924	0.1248	0.1040	0.0560	0.1343	0.1129
GNN FourierGNN	1D	0.0037	0.0049	0.0063	0.0178	0.0153	0.0430	0.0145	0.0108	0.0145
	3D	0.0154	0.0166	0.0207	0.0379	0.0364	0.0938	0.0399	0.0268	0.0359
	5D	0.0298	0.0302	0.0359	0.0555	0.0496	0.1312	0.0609	0.0432	0.0545
	7D	0.0477	0.0434	0.0542	0.0769	0.0691	0.1594	0.0787	0.0591	0.0736
	Avg.	0.0241	0.0237	0.0293	0.0471	0.0426	0.1069	0.0485	0.0350	0.0446
StemGNN	1D	0.0390	0.0081	0.0070	0.0221	0.0281	0.0535	0.0164	0.0145	0.0236
	3D	0.0816	0.0405	0.0270	0.0609	0.0599	0.1179	0.0722	0.0861	0.0683
	5D	0.1100	0.0850	0.0433	0.1015	0.0809	0.1587	0.1540	0.1312	0.1081
	7D	0.1554	0.1173	0.1096	0.1232	0.1079	0.1982	0.1746	0.2072	0.1492
	Avg.	0.0965	0.0628	0.0467	0.0769	0.0692	0.1321	0.1043	0.1098	0.0873
LLM GPT4TS	1D	0.0038	0.0050	0.0060	0.0202	0.0223	0.0542	0.0183	0.6054	0.0919
	3D	0.0151	0.0174	0.0205	0.0442	0.0474	0.1530	0.0496	0.0339	0.0476
	5D	0.0321	0.0319	0.0371	0.0648	0.0716	0.2071	0.0829	0.0526	0.0725
	7D	0.0494	0.0489	0.0528	0.0872	0.0841	0.2397	0.0996	0.0734	0.0919
	Avg.	0.0251	0.0258	0.0291	0.0541	0.0564	0.1635	0.0626	0.1913	0.0760
AutoTimes	1D	0.0037	0.0052	0.0048	0.0183	0.0159	0.0447	0.0161	0.0106	0.0149
	3D	0.0159	0.0167	0.0189	0.0396	0.0342	0.1026	0.0409	0.0263	0.0369
	5D	0.0321	0.0314	0.0337	0.0564	0.0519	0.1485	0.0614	0.0429	0.0573
	7D	0.0522	0.0466	0.0508	0.0745	0.0661	0.1807	0.0781	0.0586	0.0760
	Avg.	0.0260	0.0250	0.0271	0.0472	0.0420	0.1191	0.0491	0.0346	0.0463

Table 31: Benchmark(SEDI10) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.6277	0.9549	0.7386	0.8116	0.6054	0.5258	0.4129	0.7049	0.6727	
	3D	0.5476	0.8745	0.6759	0.7109	0.6014	0.2654	0.2066	0.5505	0.5541	
	5D	0.6912	0.7397	0.5334	0.4254	0.3311	0.1757	0.2171	0.7550	0.4836	
	7D	0.4139	0.7251	0.5557	0.4168	0.2230	0.1996	0.2779	0.4330	0.4056	
	Avg.	0.5701	0.8236	0.6259	0.5912	0.4402	0.2916	0.2787	0.6108	0.5290	
MLP	1D	0.6638	0.9234	0.7067	0.7650	0.5962	0.6661	0.5693	0.7943	0.7106	
	3D	0.6002	0.8344	0.6524	0.6845	0.5402	0.5045	0.4339	0.7120	0.6203	
	5D	0.5675	0.7951	0.6033	0.6396	0.5106	0.4231	0.3434	0.6583	0.5676	
	7D	0.5468	0.7584	0.5638	0.6033	0.4908	0.3665	0.2743	0.6150	0.5274	
	Avg.	0.5946	0.8278	0.6316	0.6731	0.5344	0.4901	0.4052	0.6949	0.6065	
NLinear	1D	0.6873	0.9462	0.7181	0.7919	0.6840	0.7250	0.6069	0.8282	0.7484	
	3D	0.6075	0.8573	0.6627	0.6996	0.5720	0.5455	0.4497	0.7289	0.6404	
	5D	0.5611	0.8142	0.6102	0.6446	0.5887	0.4524	0.3461	0.6703	0.5859	
	7D	0.5297	0.7778	0.5694	0.6049	0.5667	0.3843	0.2728	0.6247	0.5413	
	Avg.	0.5964	0.8489	0.6401	0.6852	0.6029	0.5268	0.4189	0.7130	0.6290	
TCN	1D	0.0000	0.5706	0.0827	0.6326	0.0352	0.2239	0.0323	0.1379	0.2144	
	3D	0.0000	0.5064	0.1207	0.4291	0.0457	0.1531	0.0371	0.0961	0.1735	
	5D	0.0000	0.6164	0.2791	0.3773	0.0210	0.1591	0.0051	0.0720	0.1912	
	7D	0.0000	0.4764	0.2020	0.3618	0.0504	0.1032	0.0058	0.0897	0.1612	
	Avg.	0.0000	0.5424	0.1712	0.4502	0.0381	0.1598	0.0201	0.0989	0.1851	
CNN	1D	0.6698	0.9523	0.7217	0.8100	0.6071	0.6887	0.6051	0.8452	0.7375	
	3D	0.5945	0.8768	0.6748	0.7005	0.5454	0.5162	0.4809	0.7553	0.6430	
	5D	0.5511	0.8238	0.6236	0.6393	0.5137	0.4306	0.3992	0.6931	0.5843	
	7D	0.4826	0.7900	0.5874	0.5979	0.5182	0.3631	0.3035	0.6538	0.5371	
	Avg.	0.5745	0.8607	0.6519	0.6869	0.5461	0.4996	0.4472	0.7369	0.6255	
TimesNet	1D	0.6552	0.9272	0.7150	0.7544	0.6641	0.6907	0.5531	0.7868	0.7183	
	3D	0.5754	0.8511	0.6457	0.6942	0.6398	0.5445	0.4613	0.7510	0.6454	
	5D	0.5569	0.7771	0.6128	0.6588	0.5925	0.4988	0.3881	0.7040	0.5986	
	7D	0.5420	0.7563	0.5579	0.5900	0.5868	0.4348	0.3305	0.6759	0.5593	
	Avg.	0.5824	0.8279	0.6329	0.6744	0.6208	0.5422	0.4332	0.7294	0.6304	
Transformer	iTransformer	1D	0.7003	0.9512	0.7272	0.8163	0.6347	0.7277	0.6053	0.8355	0.7498
		3D	0.6180	0.8700	0.6747	0.7233	0.5542	0.5795	0.4833	0.7460	0.6561
		5D	0.5504	0.8174	0.6210	0.6422	0.5322	0.4709	0.4083	0.7281	0.5963
		7D	0.5340	0.7791	0.5854	0.6087	0.5170	0.4027	0.3178	0.6779	0.5528
		Avg.	0.6007	0.8545	0.6521	0.6976	0.5595	0.5452	0.4537	0.7469	0.6388
	PatchTST	1D	0.6792	0.9499	0.7239	0.8126	0.6161	0.7059	0.5947	0.8288	0.7389
		3D	0.6237	0.8522	0.6693	0.7078	0.5701	0.5418	0.4613	0.7455	0.6465
		5D	0.5874	0.8176	0.6237	0.6493	0.5349	0.4162	0.3692	0.6891	0.5859
		7D	0.5685	0.7710	0.5751	0.6068	0.5180	0.3580	0.3234	0.6535	0.5468
		Avg.	0.6147	0.8477	0.6480	0.6941	0.5598	0.5055	0.4372	0.7292	0.6295

Table 32: Benchmark(SED110) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.5986	0.9119	0.6797	0.7175	0.4453	0.3662	0.3615	0.8138	0.6118
	3D	0.4081	0.7852	0.6322	0.5370	0.5730	0.2313	0.0634	0.7151	0.4932
	5D	0.3945	0.7141	0.5173	0.5120	0.4875	0.1908	0.1536	0.5050	0.4343
	7D	0.3128	0.6101	0.3420	0.3272	0.3638	0.1886	0.1527	0.5237	0.3526
	Avg.	0.4285	0.7553	0.5428	0.5234	0.4674	0.2442	0.1828	0.6394	0.4730
RNN DeepAR	1D	0.7169	0.8493	0.6995	0.6674	0.5813	0.4386	0.2829	0.6361	0.6090
	3D	0.6225	0.8410	0.5679	0.3894	0.4384	0.3261	0.1375	0.4309	0.4692
	5D	0.5846	0.7292	0.4498	0.4753	0.2992	0.1870	0.0540	0.3966	0.3970
	7D	0.3725	0.5653	0.4430	0.3004	0.2407	0.1125	0.0338	0.4785	0.3183
	Avg.	0.5741	0.7462	0.5400	0.4581	0.3899	0.2660	0.1270	0.4855	0.4484
DilatedRNN	1D	0.7312	0.9593	0.6869	0.7442	0.6992	0.5015	0.5966	0.8414	0.7200
	3D	0.5753	0.8802	0.6644	0.7700	0.6316	0.4108	0.2628	0.6683	0.6079
	5D	0.4241	0.8156	0.6258	0.6074	0.4600	0.3191	0.3749	0.6353	0.5328
	7D	0.4604	0.8182	0.5153	0.5517	0.3548	0.2705	0.2351	0.4838	0.4612
	Avg.	0.5478	0.8683	0.6231	0.6683	0.5364	0.3755	0.3673	0.6572	0.5805
GCN	1D	0.4854	0.9003	0.6324	0.1708	0.2040	0.2328	0.2320	0.6176	0.4344
	3D	0.6238	0.8072	0.5102	0.2442	0.2452	0.1719	0.0390	0.5450	0.3983
	5D	0.3944	0.6931	0.5014	0.1669	0.1642	0.1136	0.0259	0.4580	0.3147
	7D	0.3085	0.6024	0.4438	0.1910	0.1941	0.1250	0.0132	0.4392	0.2897
	Avg.	0.4530	0.7508	0.5220	0.1932	0.2019	0.1608	0.0776	0.5149	0.3593
GNN FourierGNN	1D	0.6691	0.9421	0.6985	0.7937	0.5549	0.5872	0.4493	0.7657	0.6826
	3D	0.5334	0.7993	0.6534	0.6598	0.4105	0.3459	0.1875	0.5586	0.5186
	5D	0.5633	0.7879	0.5943	0.6006	0.4668	0.2598	0.1198	0.5730	0.4957
	7D	0.0554	0.7839	0.2819	0.0166	0.3390	0.4959	0.3091	0.4102	0.3365
	Avg.	0.4553	0.8283	0.5570	0.5177	0.4428	0.4222	0.2664	0.5769	0.5083
StemGNN	1D	0.5132	0.9430	0.6959	0.7223	0.6158	0.5858	0.5300	0.8021	0.6760
	3D	0.4147	0.8389	0.5985	0.6249	0.4637	0.3827	0.3055	0.2724	0.4877
	5D	0.3527	0.7613	0.5997	0.4161	0.3047	0.3077	0.1978	0.1839	0.3905
	7D	0.3412	0.5044	0.5294	0.4240	0.4111	0.2068	0.1485	0.2098	0.3469
	Avg.	0.4054	0.7619	0.6059	0.5468	0.4489	0.3708	0.2954	0.3671	0.4753
LLM GPT4TS	1D	0.6787	0.9443	0.7168	0.7884	0.6702	0.6890	0.6025	0.8041	0.7367
	3D	0.6011	0.8545	0.6597	0.7083	0.6279	0.5487	0.4981	0.7420	0.6550
	5D	0.5491	0.8160	0.6111	0.6604	0.6258	0.4968	0.3814	0.7025	0.6054
	7D	0.5232	0.7625	0.5718	0.6086	0.6078	0.4304	0.3385	0.6751	0.5647
	Avg.	0.5880	0.8443	0.6399	0.6914	0.6329	0.5412	0.4551	0.7309	0.6405
AutoTimes	1D	0.6448	0.9341	0.7184	0.7720	0.6195	0.6573	0.5476	0.7961	0.7112
	3D	0.6021	0.8554	0.6698	0.6963	0.5620	0.5466	0.4127	0.7164	0.6326
	5D	0.5356	0.8042	0.6136	0.6414	0.5401	0.3865	0.3336	0.6615	0.5646
	7D	0.5157	0.7672	0.5732	0.5856	0.4905	0.3349	0.2588	0.6415	0.5209
	Avg.	0.5745	0.8402	0.6438	0.6738	0.5530	0.4813	0.3882	0.7038	0.6073

Table 33: Benchmark(SED15) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.4833	0.8342	0.7370	0.6299	0.4489	0.3851	0.2482	0.4830	0.5312
	3D	0.4492	0.7331	0.6381	0.5034	0.4117	0.1543	0.0802	0.3220	0.4115
	5D	0.4828	0.5492	0.3042	0.2790	0.1729	0.0880	0.0122	0.5843	0.3091
	7D	0.3626	0.5619	0.4138	0.2012	0.0665	0.1109	0.0606	0.2154	0.2491
	Avg.	0.4445	0.6696	0.5233	0.4034	0.2750	0.1846	0.1003	0.4012	0.3752
MLP	1D	0.4868	0.7569	0.7065	0.5871	0.4078	0.5087	0.3327	0.6228	0.5512
	3D	0.4710	0.6937	0.6470	0.5058	0.3758	0.3225	0.1684	0.5400	0.4655
	5D	0.4580	0.6670	0.5897	0.4482	0.3555	0.2428	0.0945	0.4839	0.4175
	7D	0.4494	0.6453	0.5390	0.4123	0.3390	0.1858	0.0560	0.4394	0.3833
	Avg.	0.4663	0.6907	0.6205	0.4883	0.3695	0.3149	0.1629	0.5215	0.4544
NLinear	1D	0.4930	0.8307	0.7161	0.6046	0.5018	0.5945	0.3810	0.6592	0.5976
	3D	0.4742	0.7212	0.6579	0.5146	0.3980	0.3657	0.2050	0.5564	0.4866
	5D	0.4587	0.6854	0.5962	0.4499	0.4339	0.2739	0.1190	0.4926	0.4387
	7D	0.4464	0.6589	0.5434	0.4055	0.4184	0.2062	0.0775	0.4460	0.4003
	Avg.	0.4681	0.7240	0.6284	0.4937	0.4380	0.3601	0.1956	0.5385	0.4808
TCN	1D	0.0000	0.3567	0.0136	0.4318	0.0107	0.1672	0.0000	0.1150	0.1369
	3D	0.0000	0.1261	0.0040	0.2543	0.0116	0.1255	0.0043	0.0396	0.0707
	5D	0.0000	0.3681	0.0067	0.1475	0.0071	0.1120	0.0011	0.0407	0.0854
	7D	0.0000	0.3236	0.0071	0.1737	0.0090	0.0733	0.0029	0.0409	0.0788
	Avg.	0.0000	0.2936	0.0079	0.2518	0.0096	0.1195	0.0021	0.0590	0.0929
CNN	1D	0.4957	0.8587	0.7286	0.6188	0.4264	0.5775	0.3903	0.6648	0.5951
	3D	0.4817	0.7316	0.6798	0.5397	0.3828	0.3614	0.2360	0.5689	0.4977
	5D	0.4690	0.7103	0.6015	0.4761	0.3429	0.2800	0.1671	0.5037	0.4438
	7D	0.4586	0.6798	0.5713	0.4426	0.3498	0.2129	0.1045	0.4676	0.4109
	Avg.	0.4763	0.7451	0.6453	0.5193	0.3755	0.3580	0.2245	0.5513	0.4869
TimesNet	1D	0.4965	0.8383	0.7227	0.5684	0.4971	0.5364	0.3178	0.6377	0.5769
	3D	0.4830	0.7219	0.6260	0.4994	0.4545	0.3832	0.2066	0.5937	0.4960
	5D	0.4702	0.6861	0.5996	0.4549	0.4377	0.3555	0.1495	0.5097	0.4579
	7D	0.4631	0.6700	0.5338	0.4061	0.3978	0.2709	0.0735	0.4908	0.4133
	Avg.	0.4782	0.7291	0.6205	0.4822	0.4468	0.3865	0.1868	0.5580	0.4860
Transformer	1D	0.4980	0.8183	0.7279	0.6254	0.4587	0.5977	0.3443	0.6536	0.5905
	3D	0.4872	0.7191	0.6889	0.5409	0.3748	0.4086	0.2513	0.5751	0.5057
	5D	0.4677	0.6923	0.6284	0.4678	0.3331	0.2985	0.1772	0.5567	0.4527
	7D	0.4621	0.6499	0.5764	0.4284	0.3209	0.2356	0.0949	0.5118	0.4100
	Avg.	0.4788	0.7199	0.6554	0.5156	0.3719	0.3851	0.2169	0.5743	0.4897
PatchTST	1D	0.4960	0.8721	0.7236	0.6265	0.4401	0.5827	0.3574	0.6710	0.5962
	3D	0.4818	0.7280	0.6681	0.5293	0.4099	0.3688	0.1969	0.5886	0.4964
	5D	0.4706	0.6798	0.6242	0.4671	0.3653	0.2530	0.1281	0.5333	0.4402
	7D	0.4632	0.6601	0.5685	0.4300	0.3357	0.2045	0.0910	0.4903	0.4054
	Avg.	0.4779	0.7350	0.6461	0.5132	0.3877	0.3523	0.1934	0.5708	0.4845

Table 34: Benchmark(SED15) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
LSTM	1D	0.4641	0.7095	0.6824	0.5011	0.2974	0.2451	0.2112	0.6269	0.4672	
	3D	0.3891	0.6514	0.4065	0.2849	0.4403	0.0936	0.0088	0.5336	0.3510	
	5D	0.2231	0.5646	0.3061	0.2470	0.3562	0.0632	0.0009	0.2817	0.2553	
	7D	0.2076	0.4678	0.2152	0.1831	0.2072	0.0674	0.0005	0.3638	0.2141	
	Avg.	0.3210	0.5983	0.4026	0.3040	0.3253	0.1173	0.0553	0.4515	0.3219	
RNN	DeepAR	1D	0.4971	0.6304	0.7049	0.4552	0.4365	0.2756	0.0297	0.4078	0.4296
	3D	0.4781	0.6997	0.2977	0.1989	0.2794	0.1441	0.0243	0.2278	0.2938	
	5D	0.4216	0.5716	0.2539	0.1953	0.1395	0.0621	0.0067	0.1784	0.2286	
	7D	0.2083	0.3185	0.2788	0.1505	0.1244	0.0288	0.0014	0.3022	0.1766	
	Avg.	0.4013	0.5551	0.3838	0.2500	0.2449	0.1276	0.0155	0.2790	0.2822	
DilatedRNN	1D	0.4995	0.8446	0.6623	0.5381	0.5445	0.3585	0.3783	0.6746	0.5626	
	3D	0.3358	0.7483	0.5946	0.5782	0.4993	0.2727	0.2162	0.4331	0.4598	
	5D	0.3087	0.6095	0.4686	0.3822	0.3043	0.1886	0.1554	0.4542	0.3589	
	7D	0.2428	0.6708	0.3055	0.3207	0.2195	0.1360	0.0831	0.2931	0.2839	
	Avg.	0.3467	0.7183	0.5078	0.4548	0.3919	0.2390	0.2082	0.4638	0.4163	
GCN	1D	0.2212	0.6122	0.5943	0.0559	0.1395	0.0918	0.0419	0.4499	0.2758	
	3D	0.3721	0.4720	0.3438	0.0940	0.1325	0.0795	0.0066	0.3889	0.2362	
	5D	0.2164	0.4499	0.2933	0.0681	0.0992	0.0281	0.0156	0.2755	0.1808	
	7D	0.1905	0.3415	0.2091	0.0704	0.1321	0.0426	0.0031	0.2659	0.1569	
	Avg.	0.2500	0.4689	0.3601	0.0721	0.1258	0.0605	0.0168	0.3450	0.2124	
GNN	FourierGNN	1D	0.4919	0.7914	0.6990	0.6032	0.3669	0.4685	0.2353	0.5503	0.5258
	3D	0.4326	0.6570	0.6282	0.4592	0.2182	0.2079	0.0710	0.3348	0.3761	
	5D	0.3897	0.6169	0.5050	0.3899	0.3039	0.1432	0.0397	0.3653	0.3442	
	7D	0.0438	0.7500	0.2826	0.0114	0.2771	0.4944	0.2234	0.4051	0.3110	
	Avg.	0.3395	0.7038	0.5287	0.3659	0.2915	0.3285	0.1423	0.4139	0.3893	
StemGNN	1D	0.4364	0.7987	0.6266	0.5491	0.4181	0.4067	0.3369	0.6388	0.5264	
	3D	0.3888	0.4209	0.4605	0.4211	0.2311	0.2059	0.1343	0.0727	0.2919	
	5D	0.3336	0.3623	0.3915	0.2047	0.1320	0.1726	0.1182	0.1248	0.2300	
	7D	0.2888	0.1800	0.4382	0.2561	0.2205	0.0900	0.0297	0.1272	0.2038	
	Avg.	0.3619	0.4405	0.4792	0.3578	0.2504	0.2188	0.1548	0.2409	0.3130	
LLM	GPT4TS	1D	0.4935	0.7486	0.7258	0.6062	0.4816	0.5495	0.3292	0.6464	0.5726
	3D	0.4864	0.7065	0.6642	0.5368	0.4544	0.3838	0.2071	0.5906	0.5037	
	5D	0.4749	0.6675	0.5934	0.4731	0.4366	0.3370	0.1492	0.5413	0.4591	
	7D	0.4733	0.6325	0.5443	0.4185	0.4246	0.2941	0.1148	0.5117	0.4267	
	Avg.	0.4820	0.6887	0.6319	0.5087	0.4493	0.3911	0.2001	0.5725	0.4905	
AutoTimes	1D	0.4978	0.7505	0.7208	0.5839	0.4112	0.4900	0.2914	0.6368	0.5478	
	3D	0.4891	0.6864	0.6746	0.5202	0.3868	0.3387	0.1463	0.5478	0.4737	
	5D	0.4757	0.6591	0.6115	0.4611	0.3379	0.2262	0.0936	0.4891	0.4193	
	7D	0.4676	0.6384	0.5580	0.4129	0.3100	0.1690	0.0401	0.4792	0.3844	
	Avg.	0.4825	0.6836	0.6412	0.4945	0.3615	0.3060	0.1429	0.5382	0.4563	

Table 35: Benchmark(SED11) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.3658	0.4400	0.6106	0.2791	0.2523	0.1913	0.0017	0.1085	0.2811	
	3D	0.1993	0.5306	0.2093	0.1940	0.2057	0.0560	0.0006	0.0238	0.1774	
	5D	0.4240	0.0000	0.0009	0.0696	0.0153	0.0313	0.0003	0.3653	0.1133	
	7D	0.0304	0.1521	0.0885	0.0771	0.0010	0.0364	0.0002	0.0318	0.0522	
	Avg.	0.2549	0.2807	0.2273	0.1549	0.1186	0.0787	0.0007	0.1323	0.1560	
MLP	1D	0.4225	0.5053	0.5287	0.2443	0.2607	0.2196	0.0000	0.2934	0.3093	
	3D	0.3675	0.4036	0.3464	0.1685	0.2435	0.0814	0.0000	0.1774	0.2235	
	5D	0.3138	0.3536	0.2750	0.1473	0.2184	0.0495	0.0000	0.1288	0.1858	
	7D	0.2753	0.3030	0.2604	0.1273	0.1921	0.0361	0.0000	0.0967	0.1614	
	Avg.	0.3448	0.3914	0.3526	0.1719	0.2287	0.0966	0.0000	0.1741	0.2200	
NLinear	1D	0.4559	0.6520	0.5873	0.2726	0.3445	0.3222	0.0017	0.3720	0.3760	
	3D	0.3803	0.5081	0.3681	0.1790	0.2541	0.1208	0.0006	0.2236	0.2543	
	5D	0.3199	0.4094	0.2875	0.1484	0.3054	0.0743	0.0003	0.1590	0.2130	
	7D	0.2696	0.3856	0.2603	0.1257	0.2874	0.0523	0.0002	0.1216	0.1878	
	Avg.	0.3564	0.4888	0.3758	0.1814	0.2979	0.1424	0.0007	0.2190	0.2578	
TCN	1D	0.0000	0.2500	0.0000	0.1225	0.0000	0.0350	0.0000	0.0295	0.0546	
	3D	0.0000	0.0000	0.0000	0.0375	0.0102	0.0479	0.0000	0.0244	0.0150	
	5D	0.0000	0.2500	0.0000	0.0495	0.0000	0.0293	0.0000	0.0231	0.0440	
	7D	0.0000	0.2500	0.0000	0.0397	0.0000	0.0220	0.0000	0.0241	0.0420	
	Avg.	0.0000	0.1875	0.0000	0.0623	0.0026	0.0335	0.0000	0.0253	0.0389	
CNN	1D	0.4633	0.6128	0.6185	0.2791	0.2458	0.3428	0.0017	0.3970	0.3701	
	3D	0.4056	0.4835	0.4551	0.1877	0.2257	0.1355	0.0006	0.2692	0.2704	
	5D	0.3481	0.2539	0.3577	0.1425	0.1908	0.0806	0.0003	0.2015	0.1969	
	7D	0.3072	0.2509	0.3038	0.1435	0.1752	0.0579	0.0002	0.1575	0.1745	
	Avg.	0.3811	0.4003	0.4338	0.1882	0.2094	0.1542	0.0007	0.2563	0.2530	
TimesNet	1D	0.4500	0.4217	0.6072	0.2355	0.3251	0.2351	0.0000	0.3089	0.3230	
	3D	0.3962	0.3883	0.3045	0.1760	0.2791	0.0921	0.0000	0.2841	0.2400	
	5D	0.3561	0.2643	0.2874	0.1676	0.2668	0.0771	0.0000	0.2030	0.2028	
	7D	0.3043	0.3146	0.3029	0.1534	0.2012	0.0484	0.0000	0.2037	0.1911	
	Avg.	0.3767	0.3472	0.3755	0.1831	0.2680	0.1132	0.0000	0.2499	0.2392	
Transformer	iTransformer	1D	0.4618	0.4321	0.6016	0.2978	0.2928	0.2689	0.0017	0.3874	0.3430
		3D	0.4367	0.3200	0.4293	0.2319	0.2021	0.1303	0.0006	0.2640	0.2519
		5D	0.3434	0.2868	0.3233	0.2029	0.1550	0.0719	0.0000	0.2448	0.2035
		7D	0.3251	0.2145	0.3189	0.1756	0.1470	0.0723	0.0002	0.1811	0.1794
		Avg.	0.3917	0.3134	0.4183	0.2271	0.1992	0.1358	0.0006	0.2693	0.2444
	PatchTST	1D	0.4679	0.6537	0.6137	0.3004	0.2673	0.2930	0.0017	0.3825	0.3725
		3D	0.4271	0.3566	0.4240	0.2139	0.2474	0.1196	0.0006	0.2748	0.2580
		5D	0.3933	0.2809	0.3304	0.1799	0.1971	0.0623	0.0003	0.2138	0.2072
		7D	0.3624	0.2612	0.2706	0.1685	0.1637	0.0470	0.0002	0.1633	0.1796
		Avg.	0.4127	0.3881	0.4097	0.2157	0.2189	0.1305	0.0007	0.2586	0.2543

Table 36: Benchmark(SED1) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.0001	0.2733	0.0378	0.2081	0.1774	0.0409	0.0000	0.3703	0.1385
	3D	0.0000	0.1555	0.0000	0.1002	0.2794	0.0039	0.0000	0.2858	0.1031
	5D	0.0000	0.0737	0.0000	0.1202	0.1893	0.0008	0.0000	0.0477	0.0540
	7D	0.0000	0.0957	0.0000	0.0289	0.0794	0.0055	0.0000	0.1361	0.0432
	Avg.	0.0000	0.1495	0.0094	0.1143	0.1814	0.0128	0.0000	0.2100	0.0847
RNN	DeepAR	1D	0.2069	0.0000	0.1219	0.1144	0.2667	0.0738	0.0000	0.1061
		3D	0.0367	0.0022	0.0000	0.0315	0.1257	0.0019	0.0000	0.0250
		5D	0.0024	0.0290	0.0000	0.0132	0.0656	0.0026	0.0000	0.0141
		7D	0.0000	0.0067	0.0015	0.0362	0.0675	0.0024	0.0000	0.0263
		Avg.	0.0615	0.0095	0.0308	0.0488	0.1314	0.0202	0.0000	0.0429
DilatedRNN	1D	0.2500	0.7031	0.1696	0.2419	0.3680	0.1416	0.0017	0.4663	0.2928
	3D	0.0393	0.5124	0.0613	0.3114	0.3394	0.0586	0.0006	0.1103	0.1792
	5D	0.0608	0.1556	0.0784	0.1157	0.1599	0.0458	0.0000	0.2260	0.1053
	7D	0.0093	0.1339	0.0520	0.1387	0.1010	0.0266	0.0000	0.0268	0.0611
	Avg.	0.0898	0.3763	0.0903	0.2019	0.2421	0.0681	0.0006	0.2074	0.1596
GCN	1D	0.0000	0.2501	0.0863	0.0000	0.0447	0.0696	0.0000	0.2556	0.0883
	3D	0.0018	0.2434	0.1158	0.0591	0.0594	0.0400	0.0000	0.1857	0.0881
	5D	0.0001	0.2293	0.1161	0.0636	0.0495	0.0156	0.0000	0.1070	0.0727
	7D	0.0000	0.2132	0.0493	0.0394	0.0866	0.0129	0.0000	0.1207	0.0653
	Avg.	0.0005	0.2340	0.0919	0.0405	0.0601	0.0345	0.0000	0.1672	0.0786
GNN	FourierGNN	1D	0.3857	0.2666	0.4037	0.2520	0.1738	0.2141	0.0017	0.2611
		3D	0.1599	0.0238	0.2421	0.1815	0.0594	0.0633	0.0000	0.0808
		5D	0.0866	0.1047	0.1099	0.1791	0.1155	0.0366	0.0000	0.1541
		7D	0.0512	0.5000	0.2843	0.0077	0.2030	0.3503	0.0253	0.2878
		Avg.	0.1708	0.2238	0.2600	0.1550	0.1379	0.1661	0.0068	0.1959
StemGNN	1D	0.1644	0.3046	0.0190	0.2558	0.2438	0.1763	0.0017	0.3469	0.1891
	3D	0.1045	0.0746	0.0213	0.1809	0.1293	0.0529	0.0000	0.0152	0.0723
	5D	0.1407	0.0966	0.0252	0.1099	0.0238	0.0233	0.0003	0.0483	0.0585
	7D	0.2351	0.0170	0.1665	0.1207	0.0341	0.0050	0.0002	0.0657	0.0805
	Avg.	0.1612	0.1232	0.0580	0.1668	0.1077	0.0644	0.0006	0.1190	0.1001
LLM	GPT4TS	1D	0.4499	0.3964	0.5837	0.2308	0.3138	0.2181	0.0000	0.2992
		3D	0.4427	0.3339	0.3435	0.1856	0.2937	0.1022	0.0000	0.2416
		5D	0.3916	0.3441	0.2802	0.1616	0.2576	0.0685	0.0000	0.2240
		7D	0.3466	0.3160	0.2302	0.1480	0.2485	0.0620	0.0000	0.2211
		Avg.	0.4077	0.3476	0.3594	0.1815	0.2784	0.1127	0.0000	0.2417
	AutoTimes	1D	0.4461	0.3905	0.5992	0.2552	0.2755	0.2245	0.0000	0.3387
		3D	0.4272	0.2062	0.4152	0.2065	0.2348	0.1022	0.0000	0.1955
		5D	0.4054	0.1724	0.2943	0.1710	0.1900	0.0538	0.0000	0.1463
		7D	0.3764	0.1710	0.2486	0.1452	0.1564	0.0356	0.0000	0.1701
		Avg.	0.4138	0.2350	0.3893	0.1945	0.2142	0.1040	0.0000	0.2126

Table 37: Benchmark(MAE) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.1324	0.1424	0.1441	0.1043	0.1098	0.1192	0.1423	0.1310	0.1282
	3D	0.2158	0.2352	0.2471	0.1838	0.1875	0.1903	0.2516	0.2082	0.2150
	5D	0.2810	0.2902	0.3178	0.2277	0.2437	0.2354	0.3228	0.2565	0.2719
	7D	0.3329	0.3374	0.3666	0.2661	0.2837	0.2719	0.3793	0.2915	0.3162
	Avg.	0.2405	0.2513	0.2689	0.1955	0.2062	0.2042	0.2740	0.2218	0.2328
TCN	1D	0.3079	0.2613	0.2634	0.2414	0.2079	0.1885	0.2957	0.2740	0.2550
	3D	0.4332	0.3215	0.3490	0.3061	0.2900	0.2554	0.3944	0.3465	0.3370
	5D	0.4324	0.3788	0.4040	0.3563	0.3457	0.3070	0.4547	0.3992	0.3848
	7D	0.4842	0.3951	0.4429	0.3915	0.3803	0.3397	0.4979	0.4242	0.4195
	Avg.	0.4144	0.3392	0.3648	0.3238	0.3060	0.2727	0.4107	0.3610	0.3491
LSTM	1D	0.1867	0.2613	0.2736	0.1595	0.2496	0.1697	0.3087	0.1836	0.2241
	3D	0.2862	0.3516	0.3751	0.2391	0.3291	0.2446	0.4267	0.2704	0.3153
	5D	0.3475	0.4012	0.4475	0.2892	0.3753	0.2944	0.4867	0.3172	0.3699
	7D	0.4174	0.4312	0.5125	0.3261	0.4193	0.3261	0.5173	0.3429	0.4116
	Avg.	0.3094	0.3613	0.4022	<u>0.2535</u>	0.3433	0.2587	0.4348	0.2785	0.3302
GCN	1D	0.1850	0.2317	0.2365	0.2061	0.1953	0.1681	0.1944	0.2152	0.2040
	3D	0.2668	0.3177	0.3426	0.2735	0.2782	0.2385	0.3119	0.2914	0.2901
	5D	0.3329	0.3673	0.4137	0.3184	0.3256	0.2796	0.3977	0.3336	0.3461
	7D	0.3863	0.3860	0.4624	0.3484	0.3646	0.3102	0.4265	0.3620	0.3808
	Avg.	<u>0.2928</u>	<u>0.3257</u>	<u>0.3638</u>	0.2866	<u>0.2909</u>	<u>0.2491</u>	<u>0.3326</u>	<u>0.3005</u>	<u>0.3053</u>

Table 38: Benchmark(MSE) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.1144	0.1360	0.1284	0.0684	0.1124	0.0747	0.1674	0.1256	0.1159
	3D	0.2226	0.3080	0.2875	0.1478	0.3256	0.1424	0.5486	0.2405	0.2779
	5D	0.3117	0.4687	0.4318	0.2081	0.6883	0.1928	1.2099	0.3317	0.4804
	7D	0.3808	0.6967	0.6094	0.2648	1.0917	0.2369	1.8233	0.3899	0.6867
	Avg.	0.2574	0.4024	0.3643	0.1723	0.5545	0.1617	0.9373	0.2719	0.3902
TCN	1D	0.3964	0.5800	0.4714	0.2575	0.3783	0.2180	1.3093	0.5552	0.5208
	3D	0.6306	0.6849	0.6180	0.3337	0.5102	0.2900	1.5333	0.6517	0.6566
	5D	0.5709	0.7997	0.7216	0.3930	0.6323	0.3534	1.6885	0.7298	0.7362
	7D	0.6880	0.8404	0.7966	0.4639	0.7027	0.3941	1.8442	0.7434	0.8092
	Avg.	0.5715	0.7263	<u>0.6519</u>	0.3620	<u>0.5559</u>	0.3139	1.5938	0.6700	<u>0.6807</u>
LSTM	1D	0.2550	1.4732	1.0367	0.1620	5.2575	0.1497	6.8509	0.4084	1.9492
	3D	0.3871	1.3765	1.0735	0.2642	5.7927	0.2476	8.5420	0.5052	2.2736
	5D	0.4911	1.5590	1.4324	0.3373	5.8335	0.3032	8.3886	0.5652	2.3638
	7D	0.6198	1.6812	1.8663	0.3982	6.0814	0.3494	8.5648	0.6176	2.5224
	Avg.	0.4383	1.5225	1.3522	<u>0.2904</u>	5.7413	0.2625	8.0866	<u>0.5241</u>	2.2772
GCN	1D	0.1994	0.2886	0.7728	0.3590	0.2534	0.1165	0.2171	6.2444	1.0564
	3D	0.3102	0.4645	1.0667	0.4061	0.5848	0.1883	0.9433	3.7576	0.9652
	5D	0.4273	0.5908	1.3546	0.4490	0.8592	0.2408	2.4139	2.7351	1.1338
	7D	0.5166	0.5968	1.5264	0.4728	1.3609	0.2752	1.8411	2.3212	1.1139
	Avg.	<u>0.3634</u>	<u>0.4852</u>	1.1801	0.4217	0.7646	<u>0.2052</u>	<u>1.3539</u>	3.7646	1.0673

Table 39: Benchmark(SED10) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.7211	0.6967	0.7119	0.7136	0.6164	0.6352	0.5654	0.5655	0.6532
	3D	0.4915	0.5725	0.5866	0.6120	0.4218	0.3578	0.3615	0.4789	0.4853
	5D	0.4848	0.5067	0.3817	0.4218	0.4414	0.3132	0.2858	0.2961	0.3914
	7D	0.3894	0.3587	0.3566	0.3627	0.3969	0.2975	0.3130	0.2304	0.3382
	Avg.	0.5217	0.5336	0.5092	0.5275	0.4691	0.4010	0.3814	0.3927	0.4670
TCN	1D	0.0784	0.2495	0.2068	0.2437	0.1814	0.1960	0.1446	0.3070	0.2009
	3D	0.0532	0.2022	0.1792	0.2623	0.1446	0.1581	0.1133	0.2063	0.1649
	5D	0.0396	0.3425	0.1477	0.2496	0.1148	0.1110	0.0962	0.1845	0.1608
	7D	0.0296	0.2988	0.1870	0.2038	0.1204	0.0959	0.0846	0.1729	0.1491
	Avg.	0.0502	0.2732	0.1802	0.2399	0.1403	0.1403	0.1097	0.2177	0.1689
LSTM	1D	0.5963	0.5745	0.5905	0.6192	0.4984	0.4388	0.4556	0.4579	0.5289
	3D	0.3988	0.3716	0.4186	0.4427	0.3756	0.2807	0.2706	0.2667	0.3532
	5D	0.3112	0.3202	0.3298	0.3156	0.3155	0.1589	0.1675	0.1616	0.2600
	7D	0.2194	0.2752	0.2372	0.2119	0.2776	0.1530	0.1409	0.1215	0.2046
	Avg.	0.3814	0.3854	0.3940	0.3973	0.3668	0.2579	0.2586	0.2519	0.3367
GCN	1D	0.6376	0.6056	0.6599	0.6106	0.5677	0.4922	0.4984	0.5006	0.5716
	3D	0.4726	0.4649	0.5055	0.5054	0.4442	0.3484	0.3468	0.2876	0.4219
	5D	0.3874	0.4223	0.4138	0.4359	0.3351	0.2257	0.3323	0.2870	0.3549
	7D	0.3222	0.3495	0.3559	0.3843	0.3540	0.2325	0.2193	0.2286	0.3058
	Avg.	<u>0.4550</u>	<u>0.4606</u>	<u>0.4838</u>	<u>0.4841</u>	<u>0.4253</u>	<u>0.3247</u>	<u>0.3492</u>	<u>0.3260</u>	<u>0.4136</u>

Table 40: Benchmark(SED15) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.6332	0.6362	0.6144	0.6326	0.5096	0.5134	0.4330	0.4133	0.5482
	3D	0.3649	0.4971	0.4727	0.5064	0.3081	0.2306	0.2453	0.3332	0.3698
	5D	0.3839	0.4192	0.2439	0.2904	0.3384	0.1945	0.1835	0.1870	0.2801
	7D	0.2661	0.2370	0.2318	0.2391	0.2680	0.1891	0.2082	0.1318	0.2214
	Avg.	0.4120	0.4474	0.3907	0.4171	0.3560	0.2819	0.2675	0.2663	0.3549
TCN	1D	0.0430	0.1756	0.1576	0.2303	0.0816	0.1463	0.0996	0.2425	0.1471
	3D	0.0297	0.1503	0.1236	0.2540	0.0569	0.1100	0.0737	0.1527	0.1188
	5D	0.0184	0.2776	0.1009	0.2379	0.0433	0.0725	0.0580	0.1319	0.1176
	7D	0.0144	0.2439	0.1245	0.1805	0.0438	0.0607	0.0515	0.1198	0.1049
	Avg.	0.0264	0.2118	0.1266	0.2257	0.0564	0.0974	0.0707	0.1617	0.1221
LSTM	1D	0.4770	0.4954	0.4591	0.5050	0.3977	0.2952	0.3280	0.3177	0.4094
	3D	0.2547	0.2300	0.2644	0.2704	0.2521	0.1542	0.1588	0.1499	0.2168
	5D	0.1636	0.1575	0.1770	0.1499	0.1813	0.0687	0.0753	0.0678	0.1302
	7D	0.0944	0.1259	0.1061	0.0900	0.1473	0.0655	0.0635	0.0448	0.0922
	Avg.	0.2474	0.2522	0.2516	0.2538	0.2446	0.1459	0.1564	0.1451	0.2121
GCN	1D	0.5252	0.5452	0.5692	0.5273	0.4744	0.3749	0.3826	0.3792	0.4722
	3D	0.3509	0.3754	0.3954	0.3991	0.3506	0.2435	0.2377	0.1980	0.3188
	5D	0.2737	0.3348	0.2919	0.3214	0.2387	0.1391	0.2282	0.1910	0.2523
	7D	0.2059	0.2485	0.2320	0.2662	0.2589	0.1385	0.1329	0.1483	0.2039
	Avg.	<u>0.3389</u>	<u>0.3760</u>	<u>0.3721</u>	<u>0.3785</u>	<u>0.3307</u>	<u>0.2240</u>	<u>0.2453</u>	<u>0.2291</u>	<u>0.3118</u>

Table 41: Benchmark(SEDI1) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.4740	0.4959	0.4062	0.4271	0.2444	0.2396	0.2085	0.1731	0.3336
	3D	0.1546	0.3007	0.2568	0.2787	0.0906	0.0572	0.0829	0.1196	0.1676
	5D	0.2301	0.2313	0.0694	0.0856	0.1254	0.0529	0.0529	0.0553	0.1129
	7D	0.1192	0.0830	0.0748	0.0743	0.0600	0.0622	0.0772	0.0339	0.0731
	Avg.	0.2445	0.2777	0.2018	0.2164	0.1301	0.1030	0.1054	0.0955	0.1718
TCN	1D	0.0189	0.0567	0.0819	0.1791	0.0102	0.0441	0.0177	0.1081	0.0646
	3D	0.0118	0.0500	0.0595	0.2010	0.0112	0.0321	0.0156	0.0523	0.0542
	5D	0.0069	0.1495	0.0454	0.1835	0.0099	0.0237	0.0102	0.0432	0.0590
	7D	0.0050	0.1320	0.0650	0.1251	0.0093	0.0155	0.0112	0.0421	0.0506
	Avg.	0.0106	0.0970	0.0629	0.1722	0.0101	0.0288	0.0137	0.0614	0.0571
LSTM	1D	0.2279	0.2413	0.2024	0.1892	0.1417	0.0773	0.1187	0.0833	0.1602
	3D	0.0538	0.0479	0.0459	0.0548	0.0255	0.0255	0.0213	0.0149	0.0362
	5D	0.0247	0.0309	0.0125	0.0212	0.0107	0.0055	0.0052	0.0041	0.0144
	7D	0.0081	0.0172	0.0055	0.0100	0.0077	0.0036	0.0043	0.0024	0.0073
	Avg.	0.0786	0.0844	0.0666	0.0688	0.0464	0.0280	0.0374	0.0262	0.0545
GCN	1D	0.3746	0.4027	0.3847	0.2958	0.2711	0.1353	0.1841	0.1716	0.2775
	3D	0.1810	0.1967	0.1936	0.1876	0.1355	0.0689	0.0853	0.0646	0.1392
	5D	0.1242	0.1687	0.1118	0.1442	0.0571	0.0347	0.0949	0.0610	0.0996
	7D	0.0821	0.1073	0.0759	0.1182	0.0863	0.0343	0.0453	0.0467	0.0745
	Avg.	0.1905	0.2188	0.1915	0.1865	0.1375	0.0683	0.1024	0.0860	0.1477

Table 42: MAE results of input factors ablation study on S6. All, G, R, and C represent all factors: groundwater, rainfall, and human control(pump and gate).

Method	T	w/ All	w/o G	w/o R	w/o C	w/o GR	w/o RC	w/o GC	w/o GRC
iTransformer	1D	0.0738	0.0735	0.0729	0.0733	0.0739	0.0737	0.0739	0.0742
	3D	0.1275	0.1279	0.1278	0.1275	0.1290	0.1280	0.1265	0.1276
	5D	0.1660	0.1662	0.1664	0.1662	0.1660	0.1660	0.1658	0.1671
	7D	0.1953	0.1953	0.1953	0.1948	0.1954	0.1961	0.1947	0.1955
	Avg.	0.1406	0.1407	0.1406	0.1405	0.1411	0.1410	0.1402	0.1411
PatchTST	1D	0.0726	0.0731	0.0713	0.0719	0.0732	0.0722	0.0727	0.0772
	3D	0.1252	0.1262	0.1252	0.1255	0.1273	0.1273	0.1254	0.1295
	5D	0.1615	0.1639	0.1624	0.1618	0.1649	0.1649	0.1634	0.1667
	7D	0.1910	0.1933	0.1923	0.1916	0.1938	0.1941	0.1927	0.1951
	Avg.	0.1376	0.1391	0.1378	0.1377	0.1398	0.1396	0.1385	0.1421
TSMixer	1D	0.1004	0.0781	0.1599	0.1443	0.0778	0.1789	0.0782	0.0782
	3D	0.1471	0.1296	0.2611	0.2216	0.1297	0.2506	0.1299	0.1299
	5D	0.1802	0.1658	0.3104	0.2350	0.1657	0.3018	0.1664	0.1663
	7D	0.2107	0.1941	0.2776	0.2435	0.1939	0.3914	0.1947	0.1944
	Avg.	0.1596	0.1419	0.2523	0.2111	0.1418	0.2807	0.1423	0.1422
NLinear	1D	0.0937	0.0977	0.0854	0.0933	0.0850	0.0804	0.0983	0.0780
	3D	0.1429	0.1461	0.1363	0.1422	0.1366	0.1328	0.1462	0.1314
	5D	0.1769	0.1796	0.1713	0.1764	0.1717	0.1689	0.1797	0.1678
	7D	0.2050	0.2074	0.2003	0.2043	0.2008	0.1978	0.2072	0.1970
	Avg.	0.1546	0.1577	0.1483	0.1540	0.1485	0.1450	0.1579	0.1435
TimesNet	1D	0.0986	0.0936	0.0986	0.0986	0.0928	0.0970	0.0924	0.0930
	3D	0.1513	0.1446	0.1511	0.1525	0.1465	0.1529	0.1448	0.1468
	5D	0.1883	0.1838	0.1889	0.1890	0.1843	0.1908	0.1830	0.1814
	7D	0.2187	0.2177	0.2262	0.2200	0.2131	0.2219	0.2111	0.2109
	Avg.	0.1642	0.1599	0.1662	0.1650	0.1592	0.1656	0.1578	0.1580

Table 43: MSE results of input factors ablation study on S6. All, G, R, and C represent all factors: groundwater, rainfall, and human control(pump and gate).

Method	T	w/ All	w/o G	w/o R	w/o C	w/o GR	w/o RC	w/o GC	w/o GRC
iTransformer	1D	0.0400	0.0407	0.0398	0.0402	0.0405	0.0400	0.0405	0.0409
	3D	0.0817	0.0828	0.0821	0.0818	0.0827	0.0825	0.0818	0.0832
	5D	0.1175	0.1172	0.1170	0.1167	0.1166	0.1169	0.1164	0.1184
	7D	0.1419	0.1430	0.1407	0.1430	0.1429	0.1431	0.1407	0.1424
	Avg.	<u>0.0953</u>	0.0960	0.0949	0.0954	0.0957	0.0956	0.0949	0.0962
PatchTST	1D	0.0395	0.0393	0.0392	0.0393	0.0405	0.0399	0.0389	0.0432
	3D	0.0798	0.0807	0.0806	0.0797	0.0831	0.0819	0.0797	0.0850
	5D	0.1107	0.1128	0.1122	0.1105	0.1154	0.1140	0.1123	0.1173
	7D	0.1368	0.1400	0.1388	0.1367	0.1412	0.1393	0.1385	0.1428
	Avg.	<u>0.0917</u>	0.0932	0.0927	0.0916	0.0950	0.0938	0.0923	0.0971
TSMixer	1D	0.0562	0.0415	0.1891	0.2590	0.0414	0.4396	0.0419	0.0416
	3D	0.0957	0.0827	0.4979	0.2580	0.0830	0.4833	0.0825	0.0826
	5D	0.1243	0.1145	0.5722	0.3232	0.1145	0.4855	0.1139	0.1141
	7D	0.1557	0.1396	0.3653	0.2389	0.1394	1.2701	0.1389	0.1382
	Avg.	0.1080	0.0946	0.4061	0.2698	0.0946	0.6697	<u>0.0943</u>	0.0941
NLinear	1D	0.0477	0.0497	0.0450	0.0471	0.0459	0.0434	0.0494	0.0426
	3D	0.0878	0.0898	0.0848	0.0871	0.0856	0.0837	0.0895	0.0829
	5D	0.1178	0.1196	0.1154	0.1171	0.1160	0.1145	0.1191	0.1139
	7D	0.1434	0.1451	0.1411	0.1425	0.1417	0.1400	0.1444	0.1395
	Avg.	0.0992	0.1011	0.0966	0.0984	0.0973	<u>0.0954</u>	0.1006	<u>0.0947</u>
TimesNet	1D	0.0564	0.0518	0.0580	0.0573	0.0521	0.0550	0.0515	0.0517
	3D	0.1049	0.0970	0.1043	0.1049	0.0985	0.1086	0.0966	0.1015
	5D	0.1409	0.1357	0.1419	0.1416	0.1397	0.1485	0.1338	0.1338
	7D	0.1729	0.1770	0.1863	0.1770	0.1677	0.1826	0.1624	0.1624
	Avg.	0.1188	0.1154	0.1226	0.1202	0.1145	0.1237	0.1111	<u>0.1123</u>

Table 44: SEDI(10%) results of input factors ablation study on S6. All, G, R, and C represent all factors: groundwater, rainfall, and human control(pump and gate).

Method	T	w/ All	w/o G	w/o R	w/o C	w/o GR	w/o RC	w/o GC	w/o GRC
iTransformer	1D	0.6508	0.6479	0.6510	0.6551	0.6525	0.6548	0.6512	0.6520
	3D	0.5526	0.5429	0.5485	0.5510	0.5393	0.5485	0.5513	0.5533
	5D	0.4856	0.4793	0.4831	0.4877	0.4786	0.4864	0.4824	0.4779
	7D	0.4384	0.4323	0.4408	0.4399	0.4301	0.4380	0.4320	0.4326
	Avg.	0.5318	0.5256	0.5308	0.5334	0.5251	<u>0.5319</u>	0.5292	0.5289
PatchTST	1D	0.6632	0.6629	0.6665	0.6644	0.6686	0.6618	0.6621	0.6561
	3D	0.5696	0.5614	0.5704	0.5717	0.5616	0.5586	0.5658	0.5455
	5D	0.5028	0.5013	0.4997	0.5043	0.4926	0.4945	0.4993	0.4857
	7D	0.4562	0.4518	0.4498	0.4544	0.4442	0.4465	0.4476	0.4364
	Avg.	<u>0.5480</u>	0.5443	0.5466	0.5487	0.5418	0.5404	0.5437	0.5309
TSMixer	1D	0.6052	0.6538	0.5202	0.5722	0.6475	0.5493	0.6499	0.6525
	3D	0.5267	0.5571	0.4251	0.4290	0.5580	0.4402	0.5581	0.5596
	5D	0.4732	0.4959	0.3523	0.4067	0.4963	0.3709	0.4932	0.4973
	7D	0.4245	0.4487	0.3516	0.3703	0.4497	0.3206	0.4454	0.4474
	Avg.	0.5074	0.5389	0.4123	0.4446	0.5379	0.4202	0.5367	0.5392
NLinear	1D	0.6477	0.6485	0.6485	0.6481	0.6506	0.6529	0.6602	0.6561
	3D	0.5476	0.5556	0.5521	0.5472	0.5521	0.5566	0.5517	0.5582
	5D	0.4868	0.4928	0.4905	0.4865	0.4904	0.4936	0.4960	0.4950
	7D	0.4356	0.4355	0.4383	0.4345	0.4381	0.4440	0.4345	0.4450
	Avg.	0.5294	0.5331	0.5324	0.5291	0.5328	<u>0.5368</u>	0.5356	0.5386
TimesNet	1D	0.5936	0.6093	0.5855	0.5884	0.6067	0.5939	0.6086	0.6089
	3D	0.4772	0.5059	0.4986	0.4925	0.5117	0.4854	0.5093	0.5046
	5D	0.4304	0.4489	0.4264	0.4305	0.4460	0.4202	0.4456	0.4396
	7D	0.3878	0.3959	0.3696	0.3837	0.4095	0.3839	0.4091	0.4082
	Avg.	0.4723	0.4900	0.4700	0.4738	0.4935	0.4709	<u>0.4931</u>	0.4903

Table 45: MAE results of temporal information ablation study on S6.

Method	T	6H	12H	1D	2D	3D	4D	5D	6D
iTransformer	1D	0.1136	0.1123	0.1097	0.1123	0.1133	0.1198	0.1214	0.1209
	3D	0.1999	0.1978	0.1980	0.2025	0.2071	0.2121	0.2168	0.2137
	5D	0.2571	0.2537	0.2547	0.2586	0.2725	0.2776	0.2815	0.2779
	7D	0.3036	0.3007	0.3015	0.3059	0.3179	0.3242	0.3221	0.3261
	Avg.	0.2185	0.2161	0.2160	0.2198	0.2277	0.2334	0.2354	0.2346
PatchTST	1D	0.1266	0.1164	0.1135	0.1127	0.1094	0.1099	0.1106	0.1159
	3D	0.2123	0.2067	0.1988	0.1979	0.1963	0.1940	0.1979	0.2011
	5D	0.2693	0.2650	0.2566	0.2530	0.2554	0.2518	0.2522	0.2559
	7D	0.3143	0.3116	0.3039	0.2988	0.2984	0.2983	0.3011	0.3069
	Avg.	0.2306	0.2249	0.2182	0.2156	0.2149	0.2135	0.2155	0.2199
TSMixer	1D	0.1197	0.1216	0.1237	0.1269	0.1258	0.1269	0.1289	0.1266
	3D	0.2021	0.2033	0.2058	0.2087	0.2051	0.2073	0.2074	0.2052
	5D	0.2586	0.2592	0.2618	0.2639	0.2576	0.2604	0.2585	0.2582
	7D	0.3046	0.3049	0.3070	0.3092	0.3007	0.3029	0.2997	0.3009
	Avg.	0.2213	0.2222	0.2246	0.2272	0.2223	0.2244	0.2236	0.2227
NLinear	1D	0.1159	0.1180	0.1211	0.1462	0.1505	0.1508	0.1587	0.1586
	3D	0.1998	0.2011	0.2047	0.2220	0.2289	0.2305	0.2388	0.2382
	5D	0.2568	0.2577	0.2605	0.2743	0.2796	0.2870	0.2871	0.2925
	7D	0.3031	0.3036	0.3064	0.3186	0.3234	0.3293	0.3323	0.3355
	Avg.	0.2189	0.2201	0.2232	0.2403	0.2456	0.2494	0.2542	0.2562
TimesNet	1D	0.1175	0.1234	0.1357	0.1565	0.1750	0.1903	0.2066	0.2184
	3D	0.2017	0.2084	0.2235	0.2423	0.2618	0.2775	0.2880	0.3012
	5D	0.2602	0.2635	0.2757	0.2887	0.2989	0.3235	0.3386	0.3657
	7D	0.3086	0.3097	0.3254	0.3272	0.3570	0.3636	0.3804	0.4016
	Avg.	0.2220	0.2262	0.2401	0.2537	0.2732	0.2887	0.3034	0.3217

Table 46: MSE results of temporal information ablation study on S6.

Method	T	6H	12H	1D	2D	3D	4D	5D	6D
iTransformer	1D	0.0829	0.0815	0.0793	0.0825	0.0844	0.0878	0.0899	0.0899
	3D	0.2093	0.2068	0.2082	0.2152	0.2218	0.2281	0.2379	0.2306
	5D	0.3186	0.3162	0.3174	0.3219	0.3472	0.3551	0.3556	0.3594
	7D	0.4170	0.4149	0.4139	0.4220	0.4499	0.4614	0.4614	0.4757
	Avg.	0.2569	0.2549	0.2547	0.2604	0.2758	0.2831	0.2862	0.2889
PatchTST	1D	0.0888	0.0860	0.0851	0.0827	0.0795	0.0787	0.0788	0.0835
	3D	0.2151	0.2104	0.2107	0.2078	0.2064	0.2050	0.2072	0.2089
	5D	0.3243	0.3214	0.3210	0.3156	0.3130	0.3256	0.3408	0.3187
	7D	0.4236	0.4187	0.4270	0.4193	0.4275	0.4548	0.4255	0.4348
	Avg.	0.2630	0.2592	0.2609	0.2563	0.2566	0.2660	0.2631	0.2615
TSMixer	1D	0.0899	0.0899	0.0888	0.0887	0.0860	0.0853	0.0855	0.0837
	3D	0.2152	0.2145	0.2141	0.2121	0.2047	0.2042	0.2034	0.1992
	5D	0.3241	0.3228	0.3226	0.3170	0.3060	0.3050	0.3035	0.3024
	7D	0.4235	0.4222	0.4214	0.4142	0.3999	0.3982	0.3968	0.3985
	Avg.	0.2632	0.2623	0.2617	0.2580	0.2491	0.2482	0.2473	0.2460
NLinear	1D	0.0852	0.0858	0.0873	0.1111	0.1135	0.1122	0.1196	0.1176
	3D	0.2111	0.2115	0.2148	0.2370	0.2444	0.2436	0.2535	0.2509
	5D	0.3201	0.3206	0.3236	0.3439	0.3490	0.3587	0.3563	0.3629
	7D	0.4199	0.4200	0.4235	0.4427	0.4480	0.4555	0.4586	0.4623
	Avg.	0.2591	0.2595	0.2623	0.2836	0.2887	0.2925	0.2970	0.2984
TimesNet	1D	0.0866	0.0937	0.1104	0.1357	0.1603	0.1750	0.2039	0.2297
	3D	0.2177	0.2333	0.2594	0.2953	0.3156	0.3607	0.3811	0.4114
	5D	0.3316	0.3372	0.3781	0.3997	0.4169	0.4597	0.5154	0.5691
	7D	0.4363	0.4407	0.4918	0.4891	0.5910	0.5837	0.5953	0.6566
	Avg.	0.2681	0.2762	0.3100	0.3299	0.3710	0.3948	0.4239	0.4667

Table 47: SEDI(10%) results of temporal information ablation study on S6.

Method	T	6H	12H	1D	2D	3D	4D	5D	6D
iTransformer	1D	0.7825	0.7945	0.7941	0.7898	0.7703	0.7857	0.7728	0.7632
	3D	0.6792	0.6945	0.6987	0.6957	0.6847	0.6801	0.6691	0.6627
	5D	0.6220	0.6392	0.6420	0.6318	0.6201	0.6189	0.6178	0.6022
	7D	0.5797	0.5984	0.6006	0.5992	0.5867	0.5841	0.5738	0.5665
	Avg.	0.6659	0.6816	0.6838	0.6791	0.6654	0.6672	0.6584	0.6487
PatchTST	1D	0.7522	0.7946	0.8004	0.8033	0.8100	0.8118	0.8058	0.8099
	3D	0.6324	0.6425	0.6977	0.7010	0.7046	0.7030	0.7033	0.7032
	5D	0.5820	0.5796	0.6389	0.6427	0.6404	0.6467	0.6362	0.6326
	7D	0.5415	0.5314	0.5959	0.6024	0.5986	0.5957	0.5929	0.5982
	Avg.	0.6270	0.6370	0.6832	0.6874	0.6884	0.6893	0.6846	0.6860
TSMixer	1D	0.7812	0.7886	0.7954	0.7912	0.7931	0.7965	0.7905	0.7952
	3D	0.6861	0.6906	0.6971	0.7005	0.7016	0.7067	0.6946	0.6983
	5D	0.6307	0.6351	0.6397	0.6463	0.6445	0.6503	0.6400	0.6385
	7D	0.5917	0.5942	0.5984	0.6054	0.6003	0.6038	0.5959	0.5900
	Avg.	0.6724	0.6771	0.6827	0.6859	0.6849	0.6893	0.6803	0.6805
NLinear	1D	0.7915	0.8002	0.8029	0.7843	0.7891	0.7933	0.7929	0.7904
	3D	0.6956	0.6989	0.7019	0.6857	0.6958	0.6974	0.6925	0.6940
	5D	0.6359	0.6383	0.6447	0.6320	0.6329	0.6373	0.6422	0.6373
	7D	0.5944	0.5995	0.6016	0.5875	0.5889	0.5951	0.5973	0.5994
	Avg.	0.6794	0.6842	0.6878	0.6724	0.6767	0.6808	0.6812	0.6803
TimesNet	1D	0.7590	0.7610	0.7482	0.7130	0.6908	0.6854	0.6548	0.6392
	3D	0.6582	0.6606	0.6499	0.6148	0.5993	0.5889	0.5790	0.5731
	5D	0.5976	0.6096	0.5988	0.5779	0.5631	0.5461	0.5458	0.5246
	7D	0.5647	0.5692	0.5451	0.5447	0.5269	0.5263	0.5062	0.5060
	Avg.	0.6449	0.6501	0.6355	0.6126	0.5950	0.5867	0.5714	0.5607

Table 48: MAE results of spatial information ablation study on S6.

Method	T	MAE					MSE					SEDI(10%)				
		1.0R	1.2R	1.4R	1.6R	1.8R	1.0R	1.2R	1.4R	1.6R	1.8R	1.0R	1.2R	1.4R	1.6R	1.8R
iTransformer	1D	0.1123	0.1135	0.1125	0.1099	0.1119	0.0825	0.0825	0.0818	0.0822	0.0807	0.7898	0.7810	0.7894	0.7863	0.7939
	3D	0.2025	0.2038	0.2020	0.2024	0.2021	0.2152	0.2144	0.2117	0.2136	0.2119	0.6957	0.7011	0.6952	0.7022	0.7036
	5D	0.2586	0.2650	0.2584	0.2538	0.2513	0.3219	0.3314	0.3241	0.3257	0.3186	0.6318	0.6294	0.6329	0.6353	0.6361
	7D	0.3059	0.3040	0.3017	0.3006	0.2995	0.4220	0.4239	0.4181	0.4176	0.4197	0.5992	0.5973	0.5989	0.5972	0.5965
	Avg.	0.2198	0.2216	0.2187	0.2167	0.2162	0.2604	0.2631	0.2589	0.2598	0.2577	0.6791	0.6772	0.6791	0.6803	0.6825
PatchTST	1D	0.1127	0.1131	0.1139	0.1113	0.1118	0.0827	0.0833	0.0829	0.0825	0.0821	0.8033	0.8030	0.8066	0.8068	0.8095
	3D	0.1979	0.1967	0.1990	0.1976	0.1976	0.2078	0.2060	0.2087	0.2082	0.2083	0.7010	0.7036	0.7048	0.7040	0.7073
	5D	0.2530	0.2531	0.2548	0.2556	0.2550	0.3156	0.3133	0.3151	0.3182	0.3160	0.6427	0.6440	0.6433	0.6437	0.6483
	7D	0.2988	0.2971	0.2994	0.3000	0.2991	0.4193	0.4102	0.4102	0.4115	0.4097	0.6024	0.6035	0.6049	0.6056	0.6065
	Avg.	0.2156	0.2150	0.2168	0.2161	0.2159	0.2563	0.2532	0.2542	0.2551	0.2540	0.6874	0.6885	0.6899	0.6900	0.6929
TSMixer	1D	0.1269	0.1259	0.1255	0.1271	0.1256	0.0887	0.0882	0.0876	0.0888	0.0879	0.7912	0.7896	0.7918	0.7895	0.7891
	3D	0.2087	0.2082	0.2072	0.2075	0.2064	0.2121	0.2125	0.2096	0.2093	0.2096	0.7005	0.6996	0.7026	0.7011	0.7001
	5D	0.2639	0.2638	0.2621	0.2632	0.2625	0.3170	0.3182	0.3136	0.3147	0.3146	0.6463	0.6451	0.6503	0.6484	0.6492
	7D	0.3092	0.3086	0.3074	0.3082	0.3062	0.4142	0.4151	0.4117	0.4114	0.4097	0.6054	0.6042	0.6078	0.6066	0.6055
	Avg.	0.2272	0.2266	0.2255	0.2265	0.2252	0.2580	0.2585	0.2556	0.2560	0.2554	0.6859	0.6846	0.6881	0.6864	0.6860
NLinear	1D	0.1462	0.1482	0.1473	0.1464	0.1449	0.1111	0.1133	0.1123	0.1113	0.1097	0.7843	0.7816	0.7807	0.7811	0.7821
	3D	0.2220	0.2238	0.2227	0.2213	0.2205	0.2370	0.2397	0.2381	0.2358	0.2347	0.6857	0.6836	0.6850	0.6865	0.6876
	5D	0.2743	0.2756	0.2746	0.2744	0.2735	0.3439	0.3461	0.3443	0.3437	0.3423	0.6320	0.6300	0.6315	0.6317	0.6310
	7D	0.3186	0.3199	0.3191	0.3181	0.3173	0.4427	0.4450	0.4435	0.4415	0.4402	0.5875	0.5888	0.5904	0.5930	0.5932
	Avg.	0.2403	0.2419	0.2409	0.2400	0.2390	0.2836	0.2860	0.2845	0.2831	0.2817	0.6724	0.6710	0.6719	0.6731	0.6735
TimesNet	1D	0.1565	0.1582	0.1597	0.1655	0.1661	0.1357	0.1423	0.1392	0.1458	0.1492	0.7130	0.7099	0.7155	0.7093	0.7007
	3D	0.2423	0.2451	0.2408	0.2424	0.2454	0.2953	0.3075	0.2847	0.2893	0.3104	0.6148	0.6168	0.6174	0.6202	0.6131
	5D	0.2887	0.2935	0.3038	0.2924	0.2985	0.3997	0.4087	0.4531	0.4028	0.4239	0.5779	0.5812	0.5651	0.5824	0.5674
	7D	0.3272	0.3399	0.3380	0.3422	0.3413	0.4891	0.5223	0.5136	0.5219	0.5451	0.5447	0.5377	0.5359	0.5426	0.5410
	Avg.	0.2537	0.2592	0.2606	0.2606	0.2628	0.3299	0.3452	0.3476	0.3400	0.3571	0.6126	0.6114	0.6085	0.6136	0.6056

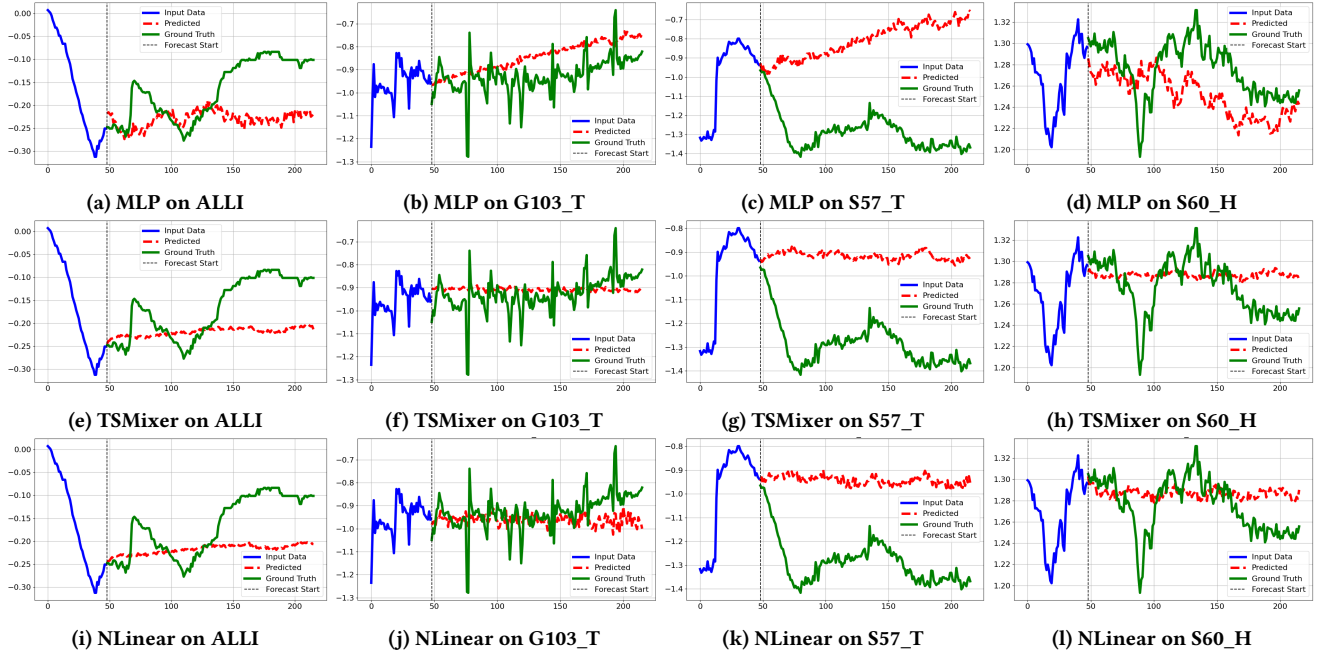


Figure 11: Cases visualization of MLP-based architecture methods on S7(Orlando area).

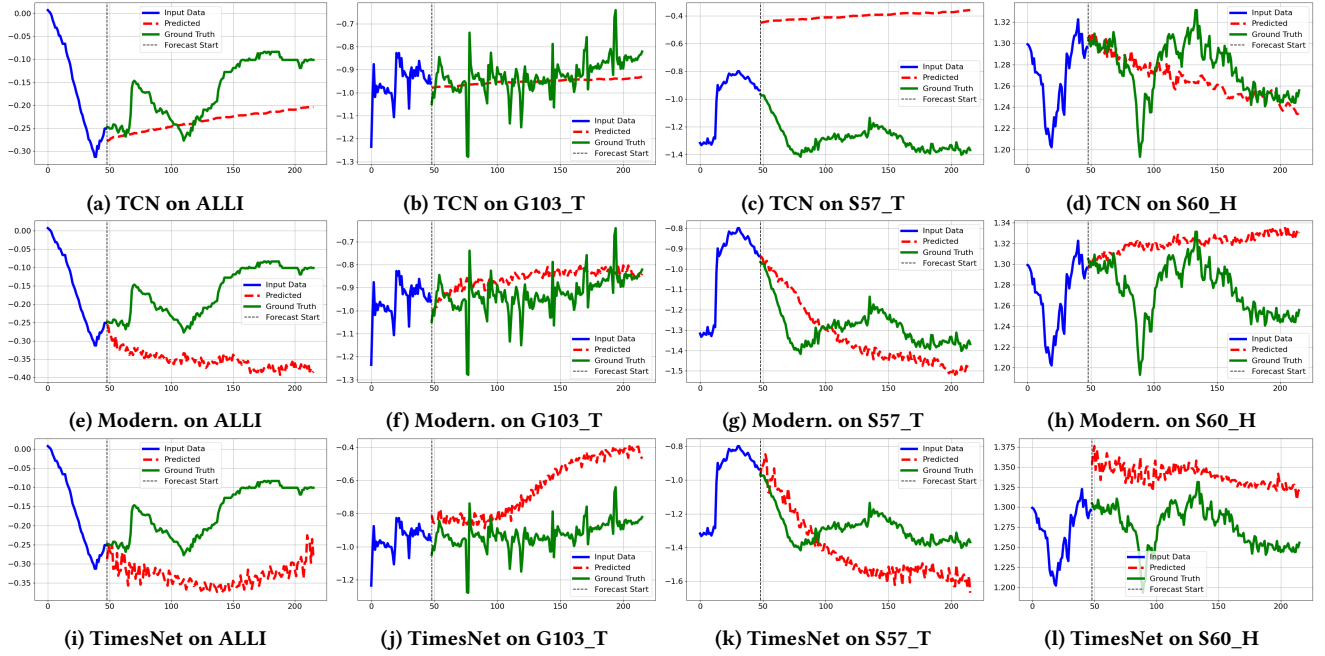


Figure 12: Cases visualization of CNN-based architecture methods on S7(Orlando area). Modern. means ModernTCN.

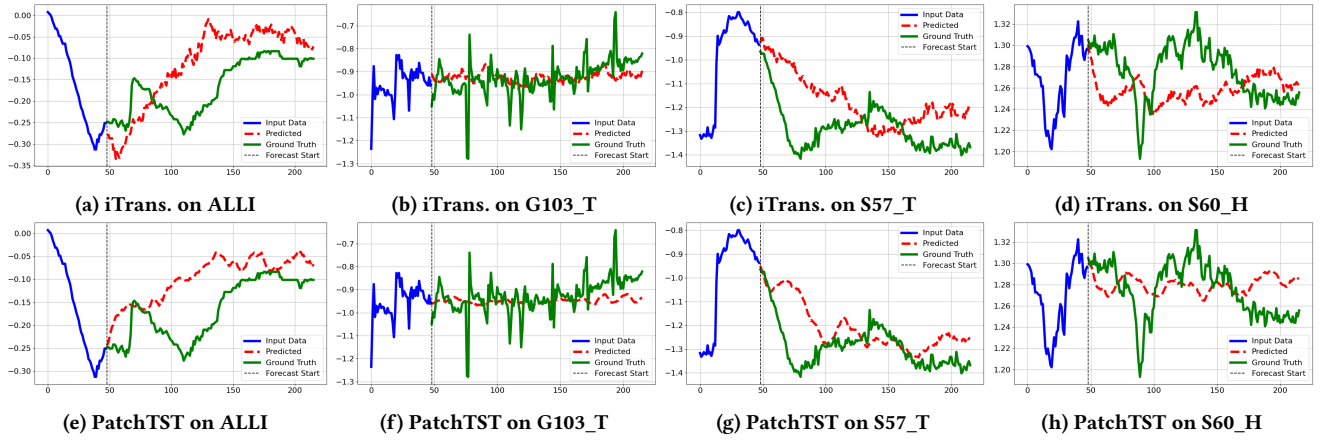


Figure 13: Cases visualization of Transformer-based architecture methods on S7(Orlando area). iTrans. means iTransformer.

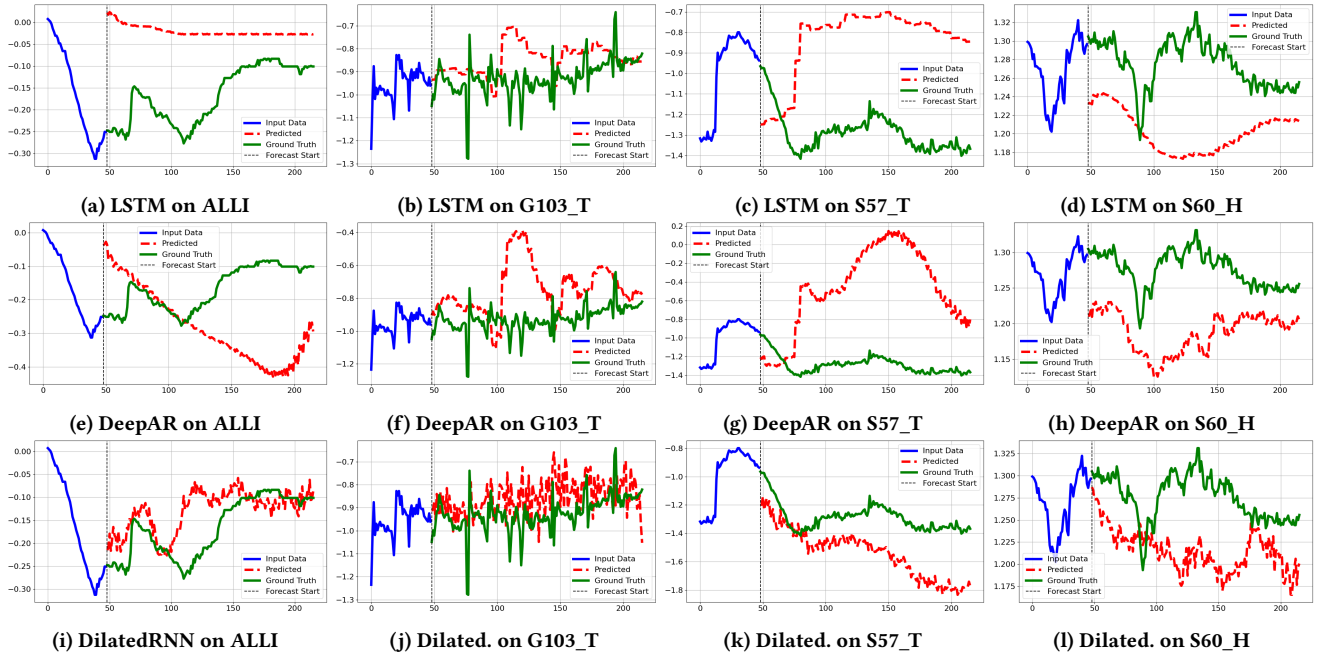


Figure 14: Cases visualization of RNN-based architecture methods on S7(Orlando area). Dilated. means DilatedRNN.

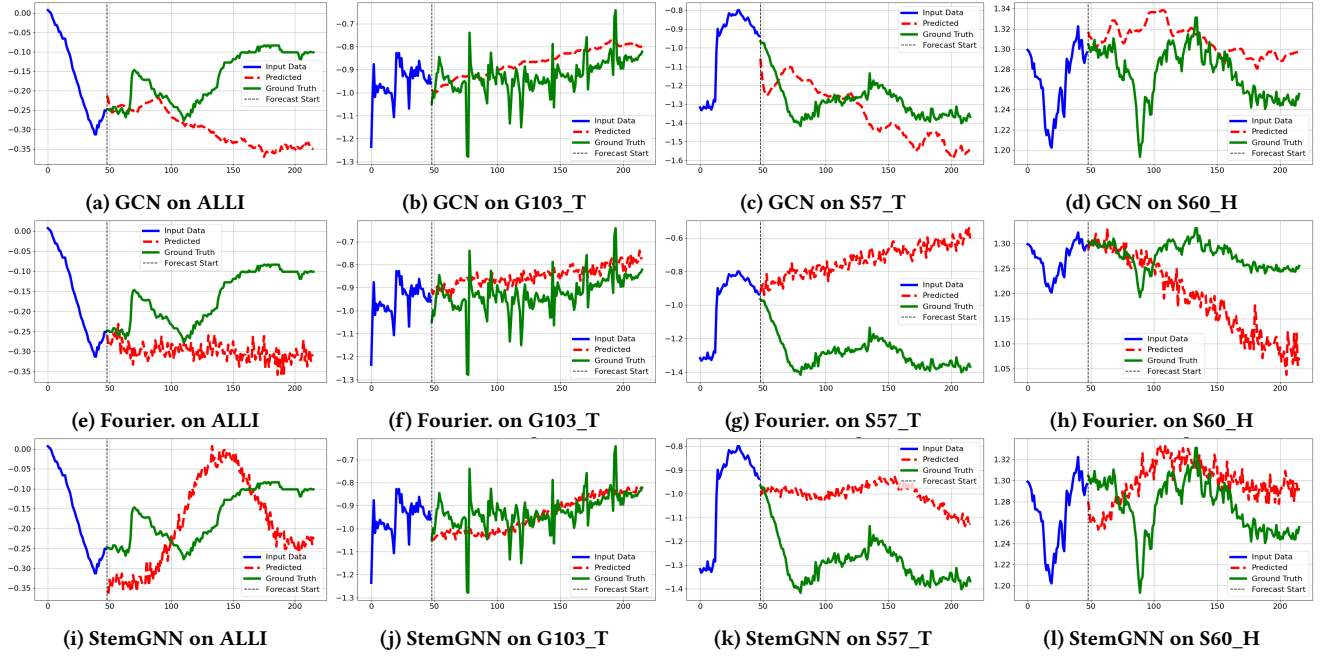


Figure 15: Cases visualization of GNN-based architecture methods on S7(Orlando area). Fourier. means FourierGNN.

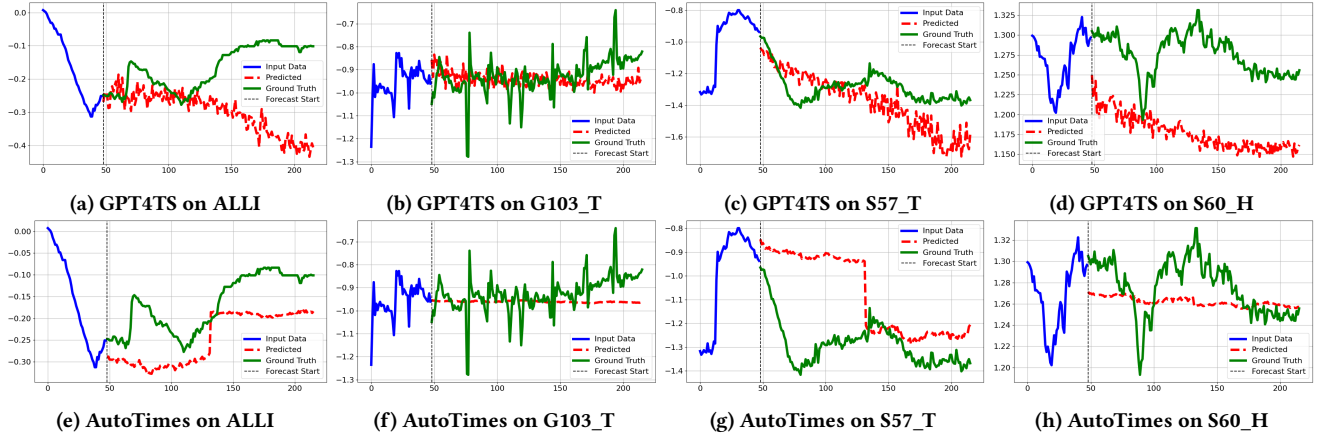


Figure 16: Cases visualization of LLM-based architecture methods on S7(Orlando area). Dilated. means DilatedRNN.